

# CHAPTER 1

## The Foundations: Logic and Proofs

### SECTION 1.1 Propositional Logic

2. Propositions must have clearly defined truth values, so a proposition must be a declarative sentence with no free variables.
- a) This is not a proposition; it's a command.
  - b) This is not a proposition; it's a question.
  - c) This is a proposition that is false, as anyone who has been to Maine knows.
  - d) This is not a proposition; its truth value depends on the value of  $x$ .
  - e) This is a proposition that is false.
  - f) This is not a proposition; its truth value depends on the value of  $n$ .
4. a) Jennifer and Teja are not friends.
- b) There are not 13 items in a baker's dozen. (Alternatively: The number of items in a baker's dozen is not equal to 13.)
  - c) Abby sent fewer than 101 text messages yesterday. Alternatively, Abby sent at most 100 text messages yesterday. Note: The first printing of this edition incorrectly rendered this exercise with "every day" in place of "yesterday." That makes it a much harder problem, because the days are quantified, and quantified propositions are not dealt with until a later section. It would be incorrect to say that the negation in that case is "Abby sent at most 100 text messages every day." Rather, a correct negation would be "There exists a day on which Abby sent at most 100 text messages." Saying "Abby did not send more than 100 text messages every day" is somewhat ambiguous—do we mean  $\neg \forall$  or do we mean  $\forall \neg$ ?
  - d) 121 is not a perfect square.
6. a) True, because  $288 > 256$  and  $288 > 128$ .
- b) True, because C has 5 MP resolution compared to B's 4 MP resolution. Note that only one of these conditions needs to be met because of the word or .
  - c) False, because its resolution is not higher (all of the statements would have to be true for the conjunction to be true).
  - d) False, because the hypothesis of this conditional statement is true and the conclusion is false.
  - e) False, because the first part of this biconditional statement is false and the second part is true.
8. a) I did not buy a lottery ticket this week.
- b) Either I bought a lottery ticket this week or [in the inclusive sense] I won the million dollar jackpot on Friday.
  - c) If I bought a lottery ticket this week, then I won the million dollar jackpot on Friday. d) I bought a lottery ticket this week and I won the million dollar jackpot on Friday.
  - e) I bought a lottery ticket this week if and only if I won the million dollar jackpot on Friday. f) If I did

not buy a lottery ticket this week, then I did not win the million dollar jackpot on Friday.

## 2 Chapter 1 The Foundations: Logic and Proofs

g) I did not buy a lottery ticket this week, and I did not win the million dollar jackpot on Friday. h) Either I did not buy a lottery ticket this week, or else I did buy one and won the million dollar jackpot on Friday.

10. a) The election is not decided.

b) The election is decided, or the votes have been counted.

c) The election is not decided, and the votes have been counted.

d) If the votes have been counted, then the election is decided.

e) If the votes have not been counted, then the election is not decided.

f) If the election is not decided, then the votes have not been counted.

g) The election is decided if and only if the votes have been counted.

h) Either the votes have not been counted, or else the election is not decided and the votes have been counted. Note that we were able to incorporate the parentheses by using the words either and else.

12. a) If you have the flu, then you miss the final exam.

b) You do not miss the final exam if and only if you pass the course.

c) If you miss the final exam, then you do not pass the course.

d) You have the flu, or miss the final exam, or pass the course.

e) It is either the case that if you have the flu then you do not pass the course or the case that if you miss the final exam then you do not pass the course (or both, it is understood).

f) Either you have the flu and miss the final exam, or you do not miss the final exam and do pass the

course. 14. a)  $r \wedge \neg q$  b)  $p \wedge q \wedge r$  c)  $r \rightarrow p$  d)  $p \wedge \neg q \wedge r$  e)  $(p \wedge q) \rightarrow r$  f)  $r \leftrightarrow (q \vee p)$

16. a) This is  $T \leftrightarrow T$ , which is true.

b) This is  $T \leftrightarrow F$ , which is false.

c) This is  $F \leftrightarrow F$ , which is true.

d) This is  $F \leftrightarrow T$ , which is false.

18. a) This is  $F \rightarrow F$ , which is true.

b) This is  $F \rightarrow F$ , which is true.

c) This is  $T \rightarrow F$ , which is false.

d) This is  $T \rightarrow T$ , which is true.

20. a) The employer making this request would be happy if the applicant knew both of these languages, so this is clearly an inclusive or .

b) The restaurant would probably charge extra if the diner wanted both of these items, so this is an exclusive or .

c) If a person happened to have both forms of identification, so much the better, so this is clearly an inclusive or .

d) This could be argued either way, but the inclusive interpretation seems more appropriate. This phrase means that faculty members who do not publish papers in research journals are likely to be fired from their jobs during the probationary period. On the other hand, it may happen that they will be fired even if they do publish (for example, if their teaching is poor).

22. a) The necessary condition is the conclusion: If you get promoted, then you wash the boss's car.

b) If the winds are from the south, then there will be a spring thaw.

### Section 1.1 Propositional Logic 3

c) The sufficient condition is the hypothesis: If you bought the computer less than a year ago, then the warranty is good.

d) If Willy cheats, then he gets caught.

e) The “only if” condition is the conclusion: If you access the website, then you must pay a subscription fee.

f) If you know the right people, then you will be elected.

g) If Carol is on a boat, then she gets seasick.

24. a) If I am to remember to send you the address, then you will have to send me an e-mail message. (This has been slightly reworded so that the tenses make more sense.)

b) If you were born in the United States, then you are a citizen of this country.

c) If you keep your textbook, then it will be a useful reference in your future courses. (The word “then” is understood in English, even if omitted.)

d) If their goaltender plays well, then the Red Wings will win the Stanley Cup.

e) If you get the job, then you had the best credentials.

f) If there is a storm, then the beach erodes.

g) If you log on to the server, then you have a valid password.

h) If you do not begin your climb too late, then you will reach the summit.

26. a) You will get an A in this course if and only if you learn how to solve discrete mathematics problems. b) You will be informed if and only if you read the newspaper every day. (It sounds better in this order; it would be logically equivalent to state this as “You read the newspaper every day if and only if you will be informed.”)

c) It rains if and only if it is a weekend day.

d) You can see the wizard if and only if he is not in.

28. a) Converse: If I stay home, then it will snow tonight. Contrapositive: If I do not stay at home, then it will not snow tonight. Inverse: If it does not snow tonight, then I will not stay home.

b) Converse: Whenever I go to the beach, it is a sunny summer day. Contrapositive: Whenever I do not go to the beach, it is not a sunny summer day. Inverse: Whenever it is not a sunny day, I do not go to the beach. c) Converse: If I sleep until noon, then I stayed up late. Contrapositive: If I do not sleep until noon, then I did not stay up late. Inverse: If I don’t stay up late, then I don’t sleep until noon.

30. A truth table will need  $2^n$  rows if there are  $n$  variables.

a)  $2^2 = 4$  b)  $2^3 = 8$  c)  $2^6 = 64$  d)  $2^5 = 32$

32. To construct the truth table for a compound proposition, we work from the inside out. In each case, we will show the intermediate steps. In part (d), for example, we first construct the truth tables for  $p \wedge q$  and for  $p \vee q$  and combine them to get the truth table for  $(p \wedge q) \rightarrow (p \vee q)$ . For parts (a) and (b) we have the following table (column three for part (a), column four for part (b)).

$p \quad \neg p \quad p \rightarrow \neg p \quad p \leftrightarrow \neg p$

T F F F  
F T T F

For parts (c) and (d) we have the following table.

$p \quad q \quad p \vee q \quad p \wedge q \quad p \oplus (p \vee q) \quad (p \wedge q) \rightarrow (p \vee q)$

T T T T F T

T F T F F T  
 F T T F T T  
 F F F F F T

#### 4 Chapter 1 The Foundations: Logic and Proofs

For part (e) we have the following table.

$p \quad q \quad \neg p \quad q \rightarrow \neg p \quad p \leftrightarrow q \quad (q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

T T F F T F  
 T F F T F F  
 F T T T F F  
 F F T T T T

For part (f) we have the following table.

$p \quad q \quad \neg q \quad p \leftrightarrow q \quad p \leftrightarrow \neg q \quad (p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

T T F T F T  
 T F T F T T  
 F T F F T T  
 F F T T F T

34. For parts (a) and (b) we have the following table (column two for part (a), column four for part (b)).  $p \quad p$

$\oplus \quad p \quad \neg p \quad p \oplus \neg p$

T F F T  
 F F T T

For parts (c) and (d) we have the following table (columns five and six).

$p \quad q \quad \neg p \quad \neg q \quad p \oplus \neg q \quad \neg p \oplus \neg q$

T T F F T F  
 T F F T F T  
 F T T F F T  
 F F T T T F

For parts (e) and (f) we have the following table (columns five and six). This time we have omitted the column explicitly showing the negation of  $q$ . Note that the first is a tautology and the second is a contradiction (see definitions in Section 1.3).

$p \quad q \quad p \oplus q \quad p \oplus \neg q \quad (p \oplus q) \vee (p \oplus \neg q) \quad (p \oplus q) \wedge (p \oplus \neg q)$

T T F T T F  
 T F T F T F  
 F T T F T F  
 F F T T T F

36. For parts (a) and (b), we have

$p \quad q \quad r \quad p \vee q \quad (p \vee q) \vee r \quad (p \vee q) \wedge r$

T T T T T T  
 T T F T T F  
 T F T T T T  
 T F F T T F  
 F T T T T T  
 F T F T T F

F F T F T F  
F F F F F F

For parts (c) and (d), we have  
Section 1.1 Propositional Logic 5

$p \ q \ r \ p \wedge q \ (p \wedge q) \vee r \ (p \wedge q) \wedge r$

T T T T T T  
T T F T T F  
T F T F T F  
T F F F F F  
F T T F T F  
F T F F F F  
F F T F T F  
F F F F F F

Finally, for parts (e) and (f) we have

$p \ q \ r \ \neg r \ p \vee q \ (p \vee q) \wedge \neg r \ p \wedge q \ (p \wedge q) \vee \neg r$

T T T F T F T T  
T T F T T T T T  
T F T F T F F F  
T F F T T T F T  
F T T F T F F F  
F T F T T T F T  
F F T F F F F F  
F F F T F F F T

38. This time the truth table needs  $2^4 = 16$  rows.

$p \ q \ r \ s \ p \rightarrow q \ (p \rightarrow q) \rightarrow r \ ((p \rightarrow q) \rightarrow r) \rightarrow s$

T T T T T T T  
T T T F T T F  
T T F T T F T  
T T F F T F T  
T F T T F T T  
T F T F F T F  
T F F T F T T  
T F F F T F F  
F T T T T T T  
F T T F T T F  
F T F T T F T  
F T F F T F T  
F F T T T T T  
F F T F T T F  
F F F T T F T  
F F F F T F T

40. This statement is true if and only if all three clauses,  $p \vee \neg q$ ,  $q \vee \neg r$ , and  $r \vee \neg p$  are true. Suppose  $p$ ,  $q$ , and  $r$  are all true. Because each clause has an unnegated variable, each clause is true. Similarly, if  $p$ ,  $q$ , and  $r$  are all false, then because each clause has a negated variable, each clause is true. On the other hand, if one of the variables is true and the other two false, then the clause containing the negation of that variable will be false, making the entire conjunction false; and similarly, if one of the variables is false and the other two true, then the clause containing that variable unnegated will be false, again making the entire conjunction false.

42. a) Since the condition is true, the statement is executed, so  $x$  is incremented and now has the value 2. b) Since the condition is false, the statement is not executed, so  $x$  is not incremented and now still has the value 1.  
c) Since the condition is true, the statement is executed, so  $x$  is incremented and now has the value 2. d) Since the condition is false, the statement is not executed, so  $x$  is not incremented and now still has the value 1.

## 6 Chapter 1 The Foundations: Logic and Proofs

- e) Since the condition is true when it is encountered (since  $x = 1$ ), the statement is executed, so  $x$  is incremented and now has the value 2. (It is irrelevant that the condition is now false.)
44. a)  $1\ 1000 \wedge (0\ 1011 \vee 1\ 1011) = 1\ 1000 \wedge 1\ 1011 = 1\ 1000$   
b)  $(0\ 1111 \wedge 1\ 0101) \vee 0\ 1000 = 0\ 0101 \vee 0\ 1000 = 0\ 1101$   
c)  $(0\ 1010 \oplus 1\ 1011) \oplus 0\ 1000 = 1\ 0001 \oplus 0\ 1000 = 1\ 1001$   
d)  $(1\ 1011 \vee 0\ 1010) \wedge (1\ 0001 \vee 1\ 1011) = 1\ 1011 \wedge 1\ 1011 = 1\ 1011$
46. The truth value of “Fred and John are happy” is  $\min(0.8, 0.4) = 0.4$ . The truth value of “Neither Fred nor John is happy” is  $\min(0.2, 0.6) = 0.2$ , since this statement means “Fred is not happy, and John is not happy,” and we computed the truth values of the two propositions in this conjunction in Exercise 45.
48. This cannot be a proposition, because it cannot have a truth value. Indeed, if it were true, then it would be truly asserting that it is false, a contradiction; on the other hand if it were false, then its assertion that it is false must be false, so that it would be true—again a contradiction. Thus this string of letters, while appearing to be a proposition, is in fact meaningless.
50. No. This is a classical paradox. (We will use the male pronoun in what follows, assuming that we are talking about males shaving their beards here, and assuming that all men have facial hair. If we restrict ourselves to beards and allow female barbers, then the barber could be female with no contradiction.) If such a barber existed, who would shave the barber? If the barber shaved himself, then he would be violating the rule that he shaves only those people who do not shave themselves. On the other hand, if he does not shave himself, then the rule says that he must shave himself. Neither is possible, so there can be no such barber.

## SECTION 1.2 Applications of Propositional Logic

2. Recall that  $p$  only if  $q$  means  $p \rightarrow q$ . In this case, if you can see the movie then you must have fulfilled one of the two requirements. Therefore the statement is  $m \rightarrow (e \vee p)$ . Notice that in everyday life one might actually say “You can see the movie if you meet one of these conditions,” but logically that is not what the rules really say.
4. The condition stated here is that if you use the network, then either you pay the fee or you are a subscriber. Therefore the proposition in symbols is  $w \rightarrow (d \vee s)$ .
6. This is similar to Exercise 2:  $u \rightarrow (b_{32} \wedge g_1 \wedge r_1 \wedge h_{16}) \vee (b_{64} \wedge g_2 \wedge r_2 \wedge h_{32})$ .
8. a) “But” means “and”:  $r \wedge \neg p$ .

b) “Whenever” means “if”:  $(r \wedge p) \rightarrow q$ .

c) Access being denied is the negation of  $q$ , so we have  $\neg r \rightarrow \neg q$ .

d) The hypothesis is a conjunction:  $(\neg p \wedge r) \rightarrow q$ .

10. We write these symbolically:  $u \rightarrow \neg a$ ,  $a \rightarrow s$ ,  $\neg s \rightarrow \neg u$ . Note that we can make all the conclusion true by making  $a$  false,  $s$  true, and  $u$  false. Therefore if the users cannot access the file system, they can save new files, and the system is not being upgraded, then all the conditional statements are true. Thus the system is consistent.

## Section 1.2 Applications of Propositional Logic 7

12. This system is consistent. We use  $L$ ,  $Q$ ,  $N$ , and  $B$  to stand for the basic propositions here, “The file system is locked,” “New messages will be queued,” “The system is functioning normally,” and “New messages will be sent to the message buffer,” respectively. Then the given specifications are  $\neg L \rightarrow Q$ ,  $\neg L \leftrightarrow N$ ,  $\neg Q \rightarrow B$ ,  $\neg L \rightarrow B$ , and  $\neg B$ . If we want consistency, then we had better have  $B$  false in order that  $\neg B$  be true. This requires that both  $L$  and  $Q$  be true, by the two conditional statements that have  $B$  as their consequence. The first conditional statement therefore is of the form  $F \rightarrow T$ , which is true. Finally, the biconditional  $\neg L \leftrightarrow N$  can be satisfied by taking  $N$  to be false. Thus this set of specifications is consistent. Note that there is just this one satisfying truth assignment.
14. This is similar to Example 6, about universities in New Mexico. To search for hiking in West Virginia, we could enter WEST AND VIRGINIA AND HIKING. If we enter (VIRGINIA AND HIKING) NOT WEST, then we’ll get websites about hiking in Virginia but not in West Virginia, except for sites that happen to use the word “west” in a different context (e.g., “Follow the stream west until you come to a clearing”).
16. a) If the explorer (a woman, so that our pronouns will not get confused here—the cannibals will be male) encounters a truth-teller, then he will honestly answer “no” to her question. If she encounters a liar, then the honest answer to her question is “yes,” so he will lie and answer “no.” Thus everybody will answer “no” to the question, and the explorer will have no way to determine which type of cannibal she is speaking to.  
b) There are several possible correct answers. One is the following question: “If I were to ask you if you always told the truth, would you say that you did?” Then if the cannibal is a truth teller, he will answer yes (truthfully), while if he is a liar, then, since in fact he would have said that he did tell the truth if questioned, he will now lie and answer no.
18. We will translate these conditions into statements in symbolic logic, using  $j$ ,  $s$ , and  $k$  for the propositions that Jasmine, Samir, and Kanti attend, respectively. The first statement is  $j \rightarrow \neg s$ . The second statement is  $s \rightarrow k$ . The last statement is  $\neg k \vee j$ , because “unless” means “or.” (We could also translate this as  $k \rightarrow j$ . From the comments following Definition 5 in the text, we know that  $p \rightarrow q$  is equivalent to “ $q$  unless  $\neg p$ .” In this case  $p$  is  $\neg j$  and  $q$  is  $\neg k$ .) First, suppose that  $s$  is true. Then the second statement tells us that  $k$  is also true, and then the last statement forces  $j$  to be true. But now the first statement forces  $s$  to be false. So we conclude that  $s$  must be false; Samir cannot attend. On the other hand, if  $s$  is false, then the first two statements are automatically true, no matter what the truth values of  $k$  and  $j$  are. If we look at the last statement, we see that it will be true as long as it is not the case that  $k$  is true and  $j$  is false. So the only combinations of friends that make everybody happy are Jasmine and Kanti, or Jasmine alone (or no one!).
20. If  $A$  is a knight, then his statement that both of them are knights is true, and both will be telling the truth. But that is impossible, because  $B$  is asserting otherwise (that  $A$  is a knave). If  $A$  is a knave, then  $B$ ’s assertion is true, so he must be a knight, and  $A$ ’s assertion is false, as it should be. Thus we conclude that  $A$  is a knave and  $B$  is a knight.

22. We can draw no conclusions. A knight will declare himself to be a knight, telling the truth. A knave will lie and assert that he is a knight. Since everyone will say "I am a knight," we can determine nothing.

24. Suppose that A is the knight. Then because he told the truth, C is the knave and therefore B is the spy. In this case both B and C are lying, which is consistent with their identities. To see that this is the only solution, first note that B cannot be the knight, because of his claim that A is the knight (which would then have to be a lie). Similarly, C cannot be the knight, because he would be lying when stating that he is the spy.

26. There is no solution, because neither a knight nor a knave would ever claim to be the knave.

#### 8 Chapter 1 The Foundations: Logic and Proofs

28. Suppose that A is the knight. Then B's statement is true, so he must be the spy, which means that C's statement is also true, but that is impossible because C would have to be the knave. Therefore A is not the knight. Next suppose that B is the knight. His true statement forces A to be the spy, which in turn forces C to be the knave; once more that is impossible because C said something true. The only other possibility is that C is the knight, which then forces B to be the spy and A the knave. This works out fine, because A is lying and B is telling the truth.

30. Neither A nor B can be the knave, because the knave cannot make the truthful statement that he is not the spy. Therefore C is the knave, and consequently A is not the spy. It follows that A is the knight and B is the spy. This works out fine, because A and B are then both telling the truth and C is lying.

32. a) We look at the three possibilities of who the innocent men might be. If Smith and Jones are innocent (and therefore telling the truth), then we get an immediate contradiction, since Smith said that Jones was a friend of Cooper, but Jones said that he did not even know Cooper. If Jones and Williams are the innocent truth-tellers, then we again get a contradiction, since Jones says that he did not know Cooper and was out of town, but Williams says he saw Jones with Cooper (presumably in town, and presumably if we was with him, then he knew him). Therefore it must be the case that Smith and Williams are telling the truth. Their statements do not contradict each other. Based on Williams' statement, we know that Jones is lying, since he said that he did not know Cooper when in fact he was with him. Therefore Jones is the murderer.

b) This is just like part (a), except that we are not told ahead of time that one of the men is guilty. Can none of them be guilty? If so, then they are all telling the truth, but this is impossible, because as we just saw, some of the statements are contradictory. Can more than one of them be guilty? If, for example, they are all guilty, then their statements give us no information. So that is certainly possible.

34. This information is enough to determine the entire system. Let each letter stand for the statement that the person whose name begins with that letter is chatting. Then the given information can be expressed symbolically as follows:  $\neg K \rightarrow H$ ,  $R \rightarrow \neg V$ ,  $\neg R \rightarrow V$ ,  $A \rightarrow R$ ,  $V \rightarrow K$ ,  $K \rightarrow V$ ,  $H \rightarrow A$ ,  $H \rightarrow K$ . Note that we were able to convert all of these statements into conditional statements. In what follows we will sometimes make use of the contrapositives of these conditional statements as well. First suppose that H is true. Then it follows that A and K are true, whence it follows that R and V are true. But R implies that V is false, so we get a contradiction. Therefore H must be false. From this it follows that K is true; whence V is true, and therefore R is false, as is A. We can now check that this assignment leads to a true value for each conditional statement. So we conclude that Kevin and Vijay are chatting but Heather, Randy, and Abby are not.

36. Note that Diana's statement is merely that she didn't do it.

a) John did it. There are four cases to consider. If Alice is the sole truth-teller, then Carlos did it; but this

means that John is telling the truth, a contradiction. If John is the sole truth-teller, then Diana must be lying, so she did it, but then Carlos is telling the truth, a contradiction. If Carlos is the sole truth-teller, then Diana did it, but that makes John truthful, again a contradiction. So the only possibility is that Diana is the sole truth-teller. This means that John is lying when he denied it, so he did it. Note that in this case both Alice and Carlos are indeed lying.

b) Again there are four cases to consider. Since Carlos and Diana are making contradictory statements, the liar must be one of them (we could have used this approach in part (a) as well). Therefore Alice is telling the truth, so Carlos did it. Note that John and Diana are telling the truth as well here, and it is Carlos who is lying.

38. This is often given as an exercise in constraint programming, and it is difficult to solve by hand. The following Section 1.3 Propositional Equivalences 9

table shows a solution consistent with all the clues, with the houses listed from left to right. Reportedly the solution is unique.

NATIONALITY Norwegian Italian Englishman Spaniard Japanese COLOR Yellow Blue Red White Green  
 PET Fox Horse Snail Dog Zebra JOB Diplomat Physician Photographer Violinist Painter DRINK Water Tea  
 Milk Juice Coffee In this solution the Japanese man owns the zebra, and the Norwegian drinks water. The logical reasoning needed to solve the problem is rather extensive, and the reader is referred to the following website containing the solution to a similar problem: [mathforum.org/library/drmath/view/55627.html](http://mathforum.org/library/drmath/view/55627.html).

40. a) Each of  $p$  and  $q$  is negated and fed to the OR gate. Therefore the output is  $(\neg p) \vee (\neg q)$ .

b)  $\neg(p \vee ((\neg p) \wedge q))$

42. We have the inputs come in from the left, in some cases passing through an inverter to form their negations. Certain pairs of them enter AND gates, and the outputs of these enter the final OR gate.

$p$

$r$

$q$

$r$

## SECTION 1.3 Propositional Equivalences

2. There are two cases. If  $p$  is true, then  $\neg(\neg p)$  is the negation of a false proposition, hence true. Similarly, if  $p$  is false, then  $\neg(\neg p)$  is also false. Therefore the two propositions are logically equivalent.

4. a) We construct the relevant truth table and note that the fifth and seventh columns are identical.  $p$

$$q \vee r \vee p \vee q \vee r \vee p \vee q \vee r$$

TTTTTTT  
 TTFTTTT  
 TFTTTTT  
 TFFTFTT  
 FTTTTTT  
 FTFTTTT  
 FFTFTTT  
 FFFFFFF

b) Again we construct the relevant truth table and note that the fifth and seventh columns are identical.

10 Chapter 1 The Foundations: Logic and Proofs

$$p \vee q \vee r \vee p \vee q \vee r \vee p \vee q \vee r$$

TTTTTTT  
 TTFTFFF  
 TFTFFFF  
 TFFFFFF  
 FTTFFTF  
 FTFFFFFF  
 FFTFFFF  
 FFFFFFF

6. We see that the fourth and seventh columns are identical.

$$p \vee q \vee r \vee p \vee q \vee r \vee p \vee q \vee r$$

TTTTFFF  
 TFFTFTT  
 FTFTTFT  
 FFFTFTT

8. We need to negate each part and swap “and” with “or.”

a) Kwame will not take a job in industry and will not go to graduate school.

b) Yoshiko does not know Java or does not know calculus.

c) James is not young, or he is not strong.

d) Rita will not move to Oregon and will not move to Washington.

10. We construct a truth table for each conditional statement and note that the relevant column contains only T's. For part (a) we have the following table.

$$p \vee q \vee r \vee p \vee q \vee r \vee p \vee q \vee r$$

TTFTFT  
 TFFTFT  
 FTTTTT  
 FFTFTT

For part (b) we have the following table. We omit the columns showing  $p \rightarrow q$  and  $q \rightarrow r$  so that the table will fit on the page.

$$p \vee q \vee r \vee p \vee q \vee r \vee p \vee q \vee r$$

TTTTTT  
 TFFTFT  
 TFTTTF  
 TFFFFFF  
 FTTTTT  
 FTFTTF  
 FFTTTF

F F F T T T

For part (c) we have the following table.

| $p$ | $q$ | $p \rightarrow q$ | $p \wedge (p \rightarrow q)$ | $[p \wedge (p \rightarrow q)] \rightarrow q$ |
|-----|-----|-------------------|------------------------------|----------------------------------------------|
| T   | T   | T                 | T                            | T                                            |
| T   | F   | F                 | F                            | T                                            |
| F   | T   | T                 | F                            | T                                            |
| F   | F   | T                 | F                            | T                                            |

For part (d) we have the following table. We have omitted some of the intermediate steps to make the table fit.

### Section 1.3 Propositional Equivalences 11

| $p$ | $q$ | $r$ | $(p \vee q) \wedge (p \rightarrow r) \wedge (p \rightarrow r)$ | $[(p \vee q) \wedge (p \rightarrow r) \wedge (p \rightarrow r)] \rightarrow r$ |
|-----|-----|-----|----------------------------------------------------------------|--------------------------------------------------------------------------------|
| T   | T   | T   | T                                                              | T                                                                              |
| T   | T   | F   | F                                                              | T                                                                              |
| T   | F   | T   | F                                                              | T                                                                              |
| T   | F   | F   | F                                                              | T                                                                              |
| F   | T   | T   | F                                                              | T                                                                              |
| F   | T   | F   | F                                                              | T                                                                              |
| F   | F   | T   | F                                                              | T                                                                              |
| F   | F   | F   | F                                                              | T                                                                              |

12. We argue directly by showing that if the hypothesis is true, then so is the conclusion. An alternative approach, which we show only for part (a), is to use the equivalences listed in the section and work symbolically. a) Assume the hypothesis is true. Then  $p$  is false. Since  $p \vee q$  is true, we conclude that  $q$  must be true. Here is a more “algebraic” solution:  $[\neg p \wedge (p \vee q)] \rightarrow q \equiv \neg[\neg p \wedge (p \vee q)] \vee q \equiv \neg\neg p \vee \neg(p \vee q) \vee q \equiv p \vee \neg(p \vee q) \vee q \equiv (p \vee q) \vee \neg(p \vee q) \equiv T$ . The reasons for these logical equivalences are, respectively, Table 7, line 1; De Morgan’s law; double negation; commutative and associative laws; negation law. b) We want to show that if the entire hypothesis is true, then the conclusion  $p \rightarrow r$  is true. To do this, we need only show that if  $p$  is true, then  $r$  is true. Suppose  $p$  is true. Then by the first part of the hypothesis, we conclude that  $q$  is true. It now follows from the second part of the hypothesis that  $r$  is true, as desired. c) Assume the hypothesis is true. Then  $p$  is true, and since the second part of the hypothesis is true, we conclude that  $q$  is also true, as desired. d) Assume the hypothesis is true. Since the first part of the hypothesis is true, we know that either  $p$  or  $q$  is true. If  $p$  is true, then the second part of the hypothesis tells us that  $r$  is true; similarly, if  $q$  is true, then the third part of the hypothesis tells us that  $r$  is true. Thus in either case we conclude that  $r$  is true.
14. This is not a tautology. It is saying that knowing that the hypothesis of an conditional statement is false allows us to conclude that the conclusion is also false, and we know that this is not valid reasoning. To show that it is not a tautology, we need to find truth assignments for  $p$  and  $q$  that make the entire proposition false. Since this is possible only if the conclusion is false, we want to let  $q$  be true; and since we want the hypothesis to be true, we must also let  $p$  be false. It is easy to check that if, indeed,  $p$  is false and  $q$  is true, then the conditional statement is false. Therefore it is not a tautology.
16. The first of these propositions is true if and only if  $p$  and  $q$  have the same truth value. The second is true if and only if either  $p$  and  $q$  are both true, or  $p$  and  $q$  are both false. Clearly these two conditions are saying the same thing.
18. It is easy to see from the definitions of conditional statement and negation that each of these propositions is false in the case in which  $p$  is true and  $q$  is false, and true in the other three cases. Therefore the two

propositions are logically equivalent.

20. It is easy to see from the definitions of the logical operations involved here that each of these propositions is true in the cases in which  $p$  and  $q$  have the same truth value, and false in the cases in which  $p$  and  $q$  have opposite truth values. Therefore the two propositions are logically equivalent.
22. Suppose that  $(p \rightarrow q) \wedge (p \rightarrow r)$  is true. We want to show that  $p \rightarrow (q \wedge r)$  is true, which means that we want to show that  $q \wedge r$  is true whenever  $p$  is true. If  $p$  is true, since we know that both  $p \rightarrow q$  and  $p \rightarrow r$  are true from our assumption, we can conclude that  $q$  is true and that  $r$  is true. Therefore  $q \wedge r$  is true, as desired. Conversely, suppose that  $p \rightarrow (q \wedge r)$  is true. We need to show that  $p \rightarrow q$  is true and that  $p \rightarrow r$  is true, which means that if  $p$  is true, then so are  $q$  and  $r$ . But this follows from  $p \rightarrow (q \wedge r)$ .

## 12 Chapter 1 The Foundations: Logic and Proofs

24. We determine exactly which rows of the truth table will have T as their entries. Now  $(p \rightarrow q) \vee (p \rightarrow r)$  will be true when either of the conditional statements is true. The conditional statement will be true if  $p$  is false, or if  $q$  in one case or  $r$  in the other case is true, i.e., when  $q \vee r$  is true, which is precisely when  $p \rightarrow (q \vee r)$  is true. Since the two propositions are true in exactly the same situations, they are logically equivalent.
26. Applying the third and first equivalences in Table 7, we have  $\neg p \rightarrow (q \rightarrow r) \equiv p \vee (q \rightarrow r) \equiv p \vee \neg q \vee r$ . Applying the first equivalence in Table 7 to  $q \rightarrow (p \vee r)$  shows that  $\neg q \vee p \vee r$  is equivalent to it. But these are equivalent by the commutative and associative laws.
28. We know that  $p \leftrightarrow q$  is true precisely when  $p$  and  $q$  have the same truth value. But this happens precisely when  $\neg p$  and  $\neg q$  have the same truth value, that is,  $\neg p \leftrightarrow \neg q$ .
30. The conclusion  $q \vee r$  will be true in every case except when  $q$  and  $r$  are both false. But if  $q$  and  $r$  are both false, then one of  $p \vee q$  or  $\neg p \vee r$  is false, because one of  $p$  or  $\neg p$  is false. Thus in this case the hypothesis  $(p \vee q) \wedge (\neg p \vee r)$  is false. An conditional statement in which the conclusion is true or the hypothesis is false is true, and that completes the argument.
32. We just need to find an assignment of truth values that makes one of these propositions true and the other false. We can let  $p$  be true and the other two variables be false. Then the first statement will be  $F \rightarrow F$ , which is true, but the second will be  $F \wedge T$ , which is false.
34. We apply the rules stated in the preamble.  
a)  $p \wedge \neg q$  b)  $p \vee (q \wedge (r \vee F))$  c)  $(p \vee \neg q) \wedge (q \vee T)$
36. If  $s$  has any occurrences of  $\wedge$ ,  $\vee$ ,  $T$ , or  $F$ , then the process of forming the dual will change it. Therefore  $s^* = s$  if and only if  $s$  is simply one propositional variable (like  $p$ ). A more difficult question is to determine when  $s^*$  will be logically equivalent to  $s$ . For example,  $p \vee F$  is logically equivalent to its dual  $p \wedge T$ , because both are logically equivalent to  $p$ .
38. The table is in fact displayed so as to exhibit the duality. The two identity laws are duals of each other, the two domination laws are duals of each other, etc. The only law not listed with another, the double negation law, is its own dual, since there are no occurrences of  $\wedge$ ,  $\vee$ ,  $T$ , or  $F$  to replace.
40. Following the hint, we easily see that the answer is  $p \wedge q \wedge \neg r$ .
42. The statement of the problem is really the solution. Each line of the truth table corresponds to exactly one combination of truth values for the  $n$  atomic propositions involved. We can write down a conjunction that is true precisely in this case, namely the conjunction of all the atomic propositions that are true and the negations of all the atomic propositions that are false. If we do this for each line of the truth table for which

the value of the compound proposition is to be true, and take the disjunction of the resulting propositions, then we have the desired proposition in its disjunctive normal form.

44. Given a compound proposition  $p$ , we can, by Exercise 43, write down a proposition  $q$  that is logically equivalent to  $p$  and uses only  $\neg$ ,  $\wedge$ , and  $\vee$ . Now by De Morgan's law we can get rid of all the  $\vee$ 's by replacing each occurrence of  $p_1 \vee p_2 \vee \cdots \vee p_n$  with  $\neg(\neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n)$ .

46. We write down the truth table corresponding to the definition.

| $p$ | $q$ | $p \mid q$ |
|-----|-----|------------|
| T   | T   | F          |
| T   | F   | T          |
| F   | T   | T          |
| F   | F   | T          |

### Section 1.3 Propositional Equivalences 13

48. We write down the truth table corresponding to the definition.

| $p$ | $q$ | $p \downarrow q$ |
|-----|-----|------------------|
| T   | T   | F                |
| T   | F   | F                |
| F   | T   | F                |
| F   | F   | T                |

50. a) From the definition (or as seen in the truth table constructed in Exercise 48),  $p \downarrow p$  is false when  $p$  is true and true when  $p$  is false, exactly as  $\neg p$  is; thus the two are logically equivalent.
- b) The proposition  $(p \downarrow q) \downarrow (p \downarrow q)$  is equivalent, by part (a), to  $\neg(p \downarrow q)$ , which from the definition (or truth table or Exercise 49) is clearly equivalent to  $p \vee q$ .
- c) By Exercise 45, every compound proposition is logically equivalent to one that uses only  $\neg$  and  $\vee$ . But by parts (a) and (b) of the present exercise, we can get rid of all the negations and disjunctions by using NOR's. Thus every compound proposition can be converted into a logically equivalent compound proposition involving only NOR's.
52. This exercise is similar to Exercise 50. First we can see from the truth tables that  $(p \mid p) \equiv (\neg p)$  and that  $((p \mid p) \mid (q \mid q)) \equiv (p \vee q)$ . Then we argue exactly as in part (c) of Exercise 50: by Exercise 45, every compound proposition is logically equivalent to one that uses only  $\neg$  and  $\vee$ . But by our observations at the beginning of the present exercise, we can get rid of all the negations and disjunctions by using NAND's. Thus every compound proposition can be converted into a logically equivalent compound proposition involving only NAND's.
54. To show that these are not logically equivalent, we need only find one assignment of truth values to  $p$ ,  $q$ , and  $r$  for which the truth values of  $p \mid (q \mid r)$  and  $(p \mid q) \mid r$  differ. One such assignment is T for  $p$  and F for  $q$  and  $r$ . Then computing from the truth tables (or definitions), we see that  $p \mid (q \mid r)$  is false and  $(p \mid q) \mid r$  is true.
56. To say that  $p$  and  $q$  are logically equivalent is to say that the truth tables for  $p$  and  $q$  are identical; similarly, to say that  $q$  and  $r$  are logically equivalent is to say that the truth tables for  $q$  and  $r$  are identical. Clearly if the truth tables for  $p$  and  $q$  are identical, and the truth tables for  $q$  and  $r$  are identical, then the truth tables for  $p$  and  $r$  are identical (this is a fundamental axiom of the notion of equality). Therefore  $p$  and  $r$  are logically equivalent. (We are assuming—and there is no loss of generality in doing so—that the same atomic variables appear in all three propositions.)

58. If we want the first two of these to be true, then  $p$  and  $q$  must have the same truth value. If  $q$  is true, then the third and fourth expressions will be true, and if  $r$  is false, the last expression will be true. So all five of these disjunctions will be true if we set  $p$  and  $q$  to be true, and  $r$  to be false.

60. These follow directly from the definitions. An unsatisfiable compound proposition is one that is true for no assignment of truth values to its variables, which is the same as saying that it is false for every assignment of truth values, which is the same as saying that its negation is true for every assignment of truth values. That is the definition of a tautology. Conversely, the negation of a tautology (i.e., a proposition that is true for every assignment of truth values to its variables) will be false for every assignment of truth values, and therefore will be unsatisfiable.

62. In each case we hunt for truth assignments that make all the disjunctions true.

a) Since  $p$  occurs in four of the five disjunctions, we can make  $p$  true, and then make  $q$  false (and make  $r$  and  $s$  anything we please). Thus this proposition is satisfiable.

14 Chapter 1 The Foundations: Logic and Proofs

b) This is satisfiable by, for example, setting  $p$  to be false (that takes care of the first, second, and fourth disjunctions),  $s$  to be false (for the third and sixth disjunctions),  $q$  to be true (for the fifth disjunction), and  $r$  to be anything.

c) It is not hard to find a satisfying truth assignment, such as  $p, q$ , and  $s$  true, and  $r$  false.

64. Recall that  $p(i, j, n)$  asserts that the cell in row  $i$ , column  $j$  contains the number  $n$ . Thus  $\bigvee_{n=1}^9 p(i, j, n)$  asserts that this cell contains at least one number. To assert that every cell contains at least one number, we take the conjunction of these statements over all cells:  $\bigwedge_{i=1}^9 \bigwedge_{j=1}^9 \bigvee_{n=1}^9 p(i, j, n)$ .

66. There are nine blocks, in three rows and three columns. Let  $r$  and  $s$  index the row and column of the block, respectively, where we start counting at 0, so that  $0 \leq r \leq 2$  and  $0 \leq s \leq 2$ . (For example,  $r = 0, s = 1$  corresponds to the block in the first row of blocks and second column of blocks.) The key point is to notice that the block corresponding to the pair  $(r, s)$  contains the cells that are in rows  $3r + 1, 3r + 2$ , and  $3r + 3$  and columns  $3s + 1, 3s + 2$ , and  $3s + 3$ . Therefore  $p(3r + i, 3s + j, n)$  asserts that a particular cell in this block contains the number  $n$ , where  $1 \leq i \leq 3$  and  $1 \leq j \leq 3$ . If we take the disjunction over all these values of  $i$  and  $j$ , then we obtain  $\bigvee_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$ , asserting that some cell in this block contains the number  $n$ . Because we want this to be true for every number and for every block, we form the triply-indexed conjunction given in the text.

## SECTION 1.4 Predicates and Quantifiers

2. a) This is true, since there is an  $a$  in orange. b) This is false, since there is no  $a$  in lemon. c) This is false, since there is no  $a$  in true. d) This is true, since there is an  $a$  in false.

4. a) Here  $x$  is still equal to 0, since the condition is false.

b) Here  $x$  is still equal to 1, since the condition is false.

c) This time  $x$  is equal to 1 at the end, since the condition is true, so the statement  $x := 1$  is executed.

6. The answers given here are not unique, but care must be taken not to confuse nonequivalent sentences. Parts (c) and (f) are equivalent; and parts (d) and (e) are equivalent. But these two pairs are not equivalent to each other.
- a) Some student in the school has visited North Dakota. (Alternatively, there exists a student in the school who has visited North Dakota.)
  - b) Every student in the school has visited North Dakota. (Alternatively, all students in the school have visited North Dakota.)
  - c) This is the negation of part (a): No student in the school has visited North Dakota. (Alternatively, there does not exist a student in the school who has visited North Dakota.)
  - d) Some student in the school has not visited North Dakota. (Alternatively, there exists a student in the school who has not visited North Dakota.)
  - e) This is the negation of part (b): It is not true that every student in the school has visited North Dakota. (Alternatively, not all students in the school have visited North Dakota.)
  - f) All students in the school have not visited North Dakota. (This is technically the correct answer, although common English usage takes this sentence to mean—incorrectly—the answer to part (e). To be perfectly clear, one could say that every student in this school has failed to visit North Dakota, or simply that no student has visited North Dakota.)

#### Section 1.4 Predicates and Quantifiers 15

8. Note that part (b) and part (c) are not the sorts of things one would normally say. a) If an animal is a rabbit, then that animal hops. (Alternatively, every rabbit hops.)
- b) Every animal is a rabbit and hops.
  - c) There exists an animal such that if it is a rabbit, then it hops. (Note that this is trivially true, satisfied, for example, by lions, so it is not the sort of thing one would say.)
  - d) There exists an animal that is a rabbit and hops. (Alternatively, some rabbits hop. Alternatively, some hopping animals are rabbits.)
10. a) We assume that this means that one student has all three animals:  $\exists x(C(x) \wedge D(x) \wedge F(x))$ .  
 b)  $\forall x(C(x) \vee D(x) \vee F(x))$  c)  $\exists x(C(x) \wedge F(x) \wedge \neg D(x))$   
 d) This is the negation of part (a):  $\neg \exists x(C(x) \wedge D(x) \wedge F(x))$ .  
 e) Here the owners of these pets can be different:  $(\exists x C(x)) \wedge (\exists x D(x)) \wedge (\exists x F(x))$ . There is no harm in using the same dummy variable, but this could also be written, for example, as  $(\exists x C(x)) \wedge (\exists y D(y)) \wedge (\exists z F(z))$ .
12. a) Since  $0 + 1 > 2 \cdot 0$ , we know that  $Q(0)$  is true.  
 b) Since  $(-1) + 1 > 2 \cdot (-1)$ , we know that  $Q(-1)$  is true.  
 c) Since  $1 + 1 > 2 \cdot 1$ , we know that  $Q(1)$  is false.  
 d) From part (a) we know that there is at least one  $x$  that makes  $Q(x)$  true, so  $\exists x Q(x)$  is true. e)  
 From part (c) we know that there is at least one  $x$  that makes  $Q(x)$  false, so  $\forall x Q(x)$  is false. f)  
 From part (c) we know that there is at least one  $x$  that makes  $Q(x)$  false, so  $\exists x \neg Q(x)$  is true. g)  
 From part (a) we know that there is at least one  $x$  that makes  $Q(x)$  true, so  $\forall x \neg Q(x)$  is false.
14. a) Since  $(-1)^3 = -1$ , this is true.  
 b) Since  $(\frac{1}{2})^4 < (\frac{1}{2})^2$ , this is true.  
 c) Since  $(-x)^2 = ((-1)x)^2 = (-1)^2 x^2 = x^2$ , we know that  $\forall x((-x)^2 = x^2)$  is true.  
 d) Twice a positive number is larger than the number, but this inequality is not true for negative numbers or

0. Therefore  $\forall x(2x > x)$  is false.

16. a) true ( $x = \sqrt{2}$ ) b) false ( $\sqrt{-1}$  is not a real number)

c) true (the left-hand side is always at least 2) d) false (not true for  $x = 1$  or  $x = 0$ )

18. Existential quantifiers are like disjunctions, and universal quantifiers are like conjunctions. See Examples 11 and 16.

a) We want to assert that  $P(x)$  is true for some  $x$  in the domain, so either  $P(-2)$  is true or  $P(-1)$  is true or  $P(0)$  is true or  $P(1)$  is true or  $P(2)$  is true. Thus the answer is  $P(-2) \vee P(-1) \vee P(0) \vee P(1) \vee P(2)$ . The other parts of this exercise are similar. Note that by De Morgan's laws, the expression in part (c) is logically equivalent to the expression in part (f), and the expression in part (d) is logically equivalent to the expression in part (e).

b)  $P(-2) \wedge P(-1) \wedge P(0) \wedge P(1) \wedge P(2)$

c)  $\neg P(-2) \vee \neg P(-1) \vee \neg P(0) \vee \neg P(1) \vee \neg P(2)$

d)  $\neg P(-2) \wedge \neg P(-1) \wedge \neg P(0) \wedge \neg P(1) \wedge \neg P(2)$

e) This is just the negation of part (a):  $\neg(P(-2) \vee P(-1) \vee P(0) \vee P(1) \vee P(2))$

f) This is just the negation of part (b):  $\neg(P(-2) \wedge P(-1) \wedge P(0) \wedge P(1) \wedge P(2))$

20. Existential quantifiers are like disjunctions, and universal quantifiers are like conjunctions. See Examples 11 and 16.

16 Chapter 1 The Foundations: Logic and Proofs

a) We want to assert that  $P(x)$  is true for some  $x$  in the domain, so either  $P(-5)$  is true or  $P(-3)$  is true or  $P(-1)$  is true or  $P(1)$  is true or  $P(3)$  is true or  $P(5)$  is true. Thus the answer is  $P(-5) \vee P(-3) \vee P(-1) \vee P(1) \vee P(3) \vee P(5)$ .

b)  $P(-5) \wedge P(-3) \wedge P(-1) \wedge P(1) \wedge P(3) \wedge P(5)$

c) The formal translation is as follows:  $((-5 = 1) \rightarrow P(-5)) \wedge ((-3 = 1) \rightarrow P(-3)) \wedge ((-1 = 1) \rightarrow P(-1)) \wedge ((1 = 1) \rightarrow P(1)) \wedge ((3 = 1) \rightarrow P(3)) \wedge ((5 = 1) \rightarrow P(5))$ . However, since the hypothesis  $x = 1$  is false when  $x$  is 1 and true when  $x$  is anything other than 1, we have more simply  $P(-5) \wedge P(-3) \wedge P(-1) \wedge P(3) \wedge P(5)$ . d) The formal translation is as follows:  $((-5 \geq 0) \wedge P(-5)) \vee ((-3 \geq 0) \wedge P(-3)) \vee ((-1 \geq 0) \wedge P(-1)) \vee ((1 \geq 0) \wedge P(1)) \vee ((3 \geq 0) \wedge P(3)) \vee ((5 \geq 0) \wedge P(5))$ . Since only three of the  $x$ 's in the domain meet the condition, the answer is equivalent to  $P(1) \vee P(3) \vee P(5)$ .

e) For the second part we again restrict the domain:  $(\neg P(-5) \vee \neg P(-3) \vee \neg P(-1) \vee \neg P(1) \vee \neg P(3) \vee \neg P(5)) \wedge (P(-1) \wedge P(-3) \wedge P(-5))$ . This is equivalent to  $(\neg P(1) \vee \neg P(3) \vee \neg P(5)) \wedge (P(-1) \wedge P(-3) \wedge P(-5))$ .

22. Many answer are possible in each case.

a) A domain consisting of a few adults in certain parts of India would make this true. If the domain were all residents of the United States, then this is certainly false.

b) If the domain is all residents of the United States, then this is true. If the domain is the set of pupils in a first grade class, it is false.

c) If the domain consists of all the United States Presidents whose last name is Bush, then the statement is true. If the domain consists of all United States Presidents, then the statement is false. d) If the domain were all residents of the United States, then this is certainly true. If the domain consists of all babies born in the last five minutes, one would expect the statement to be false (it's not even clear that these babies "know" their mothers yet).

24. In order to do the translation the second way, we let  $C(x)$  be the propositional function " $x$  is in your class." Note that for the second way, we always want to use conditional statements with universal quantifiers and

conjunctions with existential quantifiers.

a) Let  $P(x)$  be “x has a cellular phone.” Then we have  $\forall x P(x)$  the first way, or  $\forall x(C(x) \rightarrow P(x))$  the second way.

b) Let  $F(x)$  be “x has seen a foreign movie.” Then we have  $\exists x F(x)$  the first way, or  $\exists x(C(x) \wedge F(x))$  the second way.

c) Let  $S(x)$  be “x can swim.” Then we have  $\exists x \neg S(x)$  the first way, or  $\exists x(C(x) \wedge \neg S(x))$  the second way. d) Let  $Q(x)$  be “x can solve quadratic equations.” Then we have  $\forall x Q(x)$  the first way, or  $\forall x(C(x) \rightarrow Q(x))$  the second way.

e) Let  $R(x)$  be “x wants to be rich.” Then we have  $\exists x \neg R(x)$  the first way, or  $\exists x(C(x) \wedge \neg R(x))$  the second way.

26. In all of these, we will let  $Y(x)$  be the propositional function that x is in your school or class, as appropriate.

a) If we let  $U(x)$  be “x has visited Uzbekistan,” then we have  $\exists x U(x)$  if the domain is just your schoolmates, or  $\exists x(Y(x) \wedge U(x))$  if the domain is all people. If we let  $V(x, y)$  mean that person x has visited country y, then we can rewrite this last one as  $\exists x(Y(x) \wedge V(x, \text{Uzbekistan}))$ .

b) If we let  $C(x)$  and  $P(x)$  be the propositional functions asserting that x has studied calculus and C++, respectively, then we have  $\forall x(C(x) \wedge P(x))$  if the domain is just your schoolmates, or  $\forall x(Y(x) \rightarrow (C(x) \wedge P(x)))$  if the domain is all people. If we let  $S(x, y)$  mean that person x has studied subject y, then we can rewrite this last one as  $\forall x(Y(x) \rightarrow (S(x, \text{calculus}) \wedge S(x, \text{C++})))$ .

c) If we let  $B(x)$  and  $M(x)$  be the propositional functions asserting that x owns a bicycle and a motorcycle, respectively, then we have  $\forall x(\neg(B(x) \wedge M(x)))$  if the domain is just your schoolmates, or  $\forall x(Y(x) \rightarrow \neg(B(x) \wedge M(x)))$  if the domain is all people. Note that “no one” became “for all . . . not.”

d) If we let  $H(x)$  be “x is happy,” then we have  $\exists x \neg H(x)$  if the domain is just your schoolmates, or  $\exists x(Y(x) \wedge \neg H(x))$  if the domain is all people. If we let  $E(x, y)$  mean that person x is in mental state y, then we can rewrite this last one as  $\exists x(Y(x) \wedge \neg E(x, \text{happy}))$ .

e) If we let  $T(x)$  be “x was born in the twentieth century,” then we have  $\forall x T(x)$  if the domain is just your schoolmates, or  $\forall x(Y(x) \rightarrow T(x))$  if the domain is all people. If we let  $B(x, y)$  mean that person x was born in the y<sup>th</sup> century, then we can rewrite this last one as  $\forall x(Y(x) \rightarrow B(x, 20))$ .

28. Let  $R(x)$  be “x is in the correct place”; let  $E(x)$  be “x is in excellent condition”; let  $T(x)$  be “x is a [or your] tool”; and let the domain of discourse be all things.

a) There exists something not in the correct place:  $\exists x \neg R(x)$ .

b) If something is a tool, then it is in the correct place and in excellent condition:  $\forall x (T(x) \rightarrow (R(x) \wedge E(x)))$ .

c)  $\forall x (R(x) \wedge E(x))$

d) This is saying that everything fails to satisfy the condition:  $\forall x \neg (R(x) \wedge E(x))$ .

e) There exists a tool with this property:  $\exists x (T(x) \wedge \neg R(x) \wedge E(x))$ .

30. a)  $P(1, 3) \vee P(2, 3) \vee P(3, 3)$  b)  $P(1, 1) \wedge P(1, 2) \wedge P(1, 3)$

c)  $\neg P(2, 1) \vee \neg P(2, 2) \vee \neg P(2, 3)$  d)  $\neg P(1, 2) \wedge \neg P(2, 2) \wedge \neg P(3, 2)$

32. In each case we need to specify some propositional functions (predicates) and identify the domain of discourse.

a) Let  $F(x)$  be “x has fleas,” and let the domain of discourse be dogs. Our original statement is  $\forall x F(x)$ . Its negation is  $\exists x \neg F(x)$ . In English this reads “There is a dog that does not have fleas.” b) Let  $H(x)$

be “x can add,” where the domain of discourse is horses. Then our original statement is  $\exists x H(x)$ . Its negation is  $\forall x \neg H(x)$ . In English this is rendered most simply as “No horse can add.” c) Let  $C(x)$  be “x can climb,” and let the domain of discourse be koalas. Our original statement is  $\forall x C(x)$ . Its negation is  $\exists x \neg C(x)$ . In English this reads “There is a koala that cannot climb.”

d) Let  $F(x)$  be “x can speak French,” and let the domain of discourse be monkeys. Our original statement is  $\neg \exists x F(x)$  or  $\forall x \neg F(x)$ . Its negation is  $\exists x F(x)$ . In English this reads “There is a monkey that can speak French.”

e) Let  $S(x)$  be “x can swim” and let  $C(x)$  be “x can catch fish,” where the domain of discourse is pigs. Then our original statement is  $\exists x (S(x) \wedge C(x))$ . Its negation is  $\forall x \neg (S(x) \wedge C(x))$ , which could also be written  $\forall x (\neg S(x) \vee \neg C(x))$  by De Morgan’s law. In English this is “No pig can both swim and catch fish,” or “Every pig either is unable to swim or is unable to catch fish.”

34. a) Let  $S(x)$  be “x obeys the speed limit,” where the domain of discourse is drivers. The original statement is  $\exists x \neg S(x)$ , the negation is  $\forall x S(x)$ , “All drivers obey the speed limit.”

b) Let  $S(x)$  be “x is serious,” where the domain of discourse is Swedish movies. The original statement is  $\forall x S(x)$ , the negation is  $\exists x \neg S(x)$ , “Some Swedish movies are not serious.”

c) Let  $S(x)$  be “x can keep a secret,” where the domain of discourse is people. The original statement is  $\neg \exists x S(x)$ , the negation is  $\exists x S(x)$ , “Some people can keep a secret.”

d) Let  $A(x)$  be “x has a good attitude,” where the domain of discourse is people in this class. The original statement is  $\exists x \neg A(x)$ , the negation is  $\forall x A(x)$ , “Everyone in this class has a good attitude.”

36. a) Since  $1^2 = 1$ , this statement is false;  $x = 1$  is a counterexample. So is  $x = 0$  (these are the only two counterexamples).

18 Chapter 1 The Foundations: Logic and Proofs b) There are two counterexamples:  $x = \sqrt{2}$  and  $x = -\sqrt{2}$ .

c) There is one counterexample:  $x = 0$ .

38. a) Some system is open. b) Every system is either malfunctioning or in a diagnostic state. c) Some system is open, or some system is in a diagnostic state. d) Some system is unavailable. e) No system is working. (We could also say “Every system is not working,” as long as we understood that this is different from “Not every system is working.”)

40. There are many ways to write these, depending on what we use for predicates.

a) Let  $F(x)$  be “There is less than x megabytes free on the hard disk,” with the domain of discourse being positive numbers, and let  $W(x)$  be “User x is sent a warning message.” Then we have  $F(30) \rightarrow \forall x W(x)$ . b) Let  $O(x)$  be “Directory x can be opened,” let  $C(x)$  be “File x can be closed,” and let  $E$  be the proposition “System errors have been detected.” Then we have  $E \rightarrow ((\forall x \neg O(x)) \wedge (\forall x \neg C(x)))$ .

c) Let  $B$  be the proposition “The file system can be backed up,” and let  $L(x)$  be “User x is currently logged on.” Then we have  $(\exists x L(x)) \rightarrow \neg B$ .

d) Let  $D(x)$  be “Product x can be delivered,” and let  $M(x)$  be “There are at least x megabytes of memory available” and  $S(x)$  be “The connection speed is at least x kilobits per second,” where the domain of discourse for the last two propositional functions are positive numbers. Then we have  $(M(8) \wedge S(56)) \rightarrow D(\text{video on demand})$ .

42. There are many ways to write these, depending on what we use for predicates.

a) Let  $A(x)$  be “User x has access to an electronic mailbox.” Then we have  $\forall x A(x)$ . b) Let  $A(x, y)$  be “Group member x can access resource y,” and let  $S(x, y)$  be “System x is in state y.” Then we have  $S(\text{file system, locked}) \rightarrow \forall x A(x, \text{system mailbox})$ .

c) Let  $S(x, y)$  be "System  $x$  is in state  $y$ ." Recalling that "only if" indicates a necessary condition, we have  $S(\text{firewall, diagnostic}) \rightarrow S(\text{proxy server, diagnostic})$ .

d) Let  $T(x)$  be "The throughput is at least  $x$  kbps," where the domain of discourse is positive numbers, let  $M(x, y)$  be "Resource  $x$  is in mode  $y$ ," and let  $S(x, y)$  be "Router  $x$  is in state  $y$ ." Then we have  $(T(100) \wedge \neg T(500) \wedge \neg M(\text{proxy server, diagnostic})) \rightarrow \exists x S(x, \text{normal})$ .

44. We want propositional functions  $P$  and  $Q$  that are sometimes, but not always, true (so that the second biconditional is  $F \leftrightarrow F$  and hence true), but such that there is an  $x$  making one true and the other false. For example, we can take  $P(x)$  to mean that  $x$  is an even number (a multiple of 2) and  $Q(x)$  to mean that  $x$  is a multiple of 3. Then an example like  $x = 4$  or  $x = 9$  shows that  $\forall x(P(x) \leftrightarrow Q(x))$  is false.

46. a) There are two cases. If  $A$  is true, then  $(\forall x P(x)) \vee A$  is true, and since  $P(x) \vee A$  is true for all  $x$ ,  $\forall x(P(x) \vee A)$  is also true. Thus both sides of the logical equivalence are true (hence equivalent). Now suppose that  $A$  is false. If  $P(x)$  is true for all  $x$ , then the left-hand side is true. Furthermore, the right-hand side is also true (since  $P(x) \vee A$  is true for all  $x$ ). On the other hand, if  $P(x)$  is false for some  $x$ , then both sides are false. Therefore again the two sides are logically equivalent.

b) There are two cases. If  $A$  is true, then  $(\exists x P(x)) \vee A$  is true, and since  $P(x) \vee A$  is true for some (really all)  $x$ ,  $\exists x(P(x) \vee A)$  is also true. Thus both sides of the logical equivalence are true (hence equivalent). Now suppose that  $A$  is false. If  $P(x)$  is true for at least one  $x$ , then the left-hand side is true. Furthermore, the right-hand side is also true (since  $P(x) \vee A$  is true for that  $x$ ). On the other hand, if  $P(x)$  is false for all  $x$ , then both sides are false. Therefore again the two sides are logically equivalent.

48. a) There are two cases. If  $A$  is false, then both sides of the equivalence are true, because a conditional statement with a false hypothesis is true. If  $A$  is true, then  $A \rightarrow P(x)$  is equivalent to  $P(x)$  for each  $x$ , so the left-hand side is equivalent to  $\forall x P(x)$ , which is equivalent to the right-hand side.

#### Section 1.4 Predicates and Quantifiers 19

b) There are two cases. If  $A$  is false, then both sides of the equivalence are true, because a conditional statement with a false hypothesis is true (and we are assuming that the domain is nonempty). If  $A$  is true, then  $A \rightarrow P(x)$  is equivalent to  $P(x)$  for each  $x$ , so the left-hand side is equivalent to  $\exists x P(x)$ , which is equivalent to the right-hand side.

50. It is enough to find a counterexample. It is intuitively clear that the first proposition is asserting much more than the second. It is saying that one of the two predicates,  $P$  or  $Q$ , is universally true; whereas the second proposition is simply saying that for every  $x$  either  $P(x)$  or  $Q(x)$  holds, but which it is may well depend on  $x$ . As a simple counterexample, let  $P(x)$  be the statement that  $x$  is odd, and let  $Q(x)$  be the statement that  $x$  is even. Let the domain of discourse be the positive integers. The second proposition is true, since every positive integer is either odd or even. But the first proposition is false, since it is neither the case that all positive integers are odd nor the case that all of them are even.

52. a) This is false, since there are many values of  $x$  that make  $x > 1$  true.

b) This is false, since there are two values of  $x$  that make  $x^2 = 1$  true.

c) This is true, since by algebra we see that the unique solution to the equation is  $x = 3$ .

d) This is false, since there are no values of  $x$  that make  $x = x + 1$  true.

54. There are only three cases in which  $\exists x!P(x)$  is true, so we form the disjunction of these three cases. The answer is thus  $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (\neg P(1) \wedge P(2) \wedge \neg P(3)) \vee (\neg P(1) \wedge \neg P(2) \wedge P(3))$ .

56. A Prolog query returns a yes/no answer if there are no variables in the query, and it returns the values that make the query true if there are.
- None of the facts was that Kevin was enrolled in EE 222. So the response is no.
  - One of the facts was that Kiko was enrolled in Math 273. So the response is yes.
  - Prolog returns the names of the courses for which Grossman is the instructor, namely just cs301.
  - Prolog returns the names of the instructor for CS 301, namely grossman.
  - Prolog returns the names of the instructors teaching any course that Kevin is enrolled in, namely chan, since Chan is the instructor in Math 273, the only course Kevin is enrolled in.
58. Following the idea and syntax of Example 28, we have the following rule:  
 $\text{grandfather}(X,Y) :- \text{father}(X,Z), \text{father}(Z,Y); \text{father}(X,Z), \text{mother}(Z,Y).$   
 Note that we used the comma to mean “and” and the semicolon to mean “or.” For X to be the grandfather of Y, X must be either Y’s father’s father or Y’s mother’s father.
60. a)  $\forall x(P(x) \rightarrow Q(x))$  b)  $\exists x(R(x) \wedge \neg Q(x))$  c)  $\exists x(R(x) \wedge \neg P(x))$   
 d) Yes. The unsatisfactory excuse guaranteed by part (b) cannot be a clear explanation by part (a).
62. a)  $\forall x(P(x) \rightarrow \neg S(x))$  b)  $\forall x(R(x) \rightarrow S(x))$  c)  $\forall x(Q(x) \rightarrow P(x))$  d)  $\forall x(Q(x) \rightarrow \neg R(x))$  e) Yes. If x is one of my poultry, then he is a duck (by part (c)), hence not willing to waltz (part (a)). Since officers are always willing to waltz (part (b)), x is not an officer.
- 20 Chapter 1 The Foundations: Logic and Proofs

## SECTION 1.5 Nested Quantifiers

- There exists a real number x such that for every real number y ,  $xy = y$  . This is asserting the existence of a multiplicative identity for the real numbers, and the statement is true, since we can take  $x = 1$ .
  - For every real number x and real number y , if x is nonnegative and y is negative, then the difference  $x - y$  is positive. Or, more simply, a nonnegative number minus a negative number is positive (which is true).
  - For every real number x and real number y , there exists a real number z such that  $x = y + z$  . This is a true statement, since we can take  $z = x - y$  in each case.
- Some student in your class has taken some computer science course.
  - There is a student in your class who has taken every computer science course.
  - Every student in your class has taken at least one computer science course.
  - There is a computer science course that every student in your class has taken.
  - Every computer science course has been taken by at least one student in your class.
  - Every student in your class has taken every computer science course.
- Randy Goldberg is enrolled in CS 252.
  - Someone is enrolled in Math 695.
  - Carol Sitea is enrolled in some course.
  - Some student is enrolled simultaneously in Math 222 and CS 252.
  - There exist two distinct people, the second of whom is enrolled in every course that the first is enrolled in.
  - There exist two distinct people enrolled in exactly the same courses.
- $\exists x \exists y Q(x, y)$

b) This is the negation of part (a), and so could be written either  $\neg \exists x \exists y Q(x, y)$  or  $\forall x \forall y \neg Q(x, y)$ . c) We assume from the wording that the statement means that the same person appeared on both shows:

$\exists x (Q(x, \text{Jeopardy}) \wedge Q(x, \text{Wheel of Fortune}))$

d)  $\forall y \exists x Q(x, y)$  e)  $\exists x_1 \exists x_2 (Q(x_1, \text{Jeopardy}) \wedge Q(x_2, \text{Jeopardy}) \wedge x_1 \neq x_2)$

10. a)  $\forall x F(x, \text{Fred})$  b)  $\forall y F(\text{Evelyn}, y)$  c)  $\forall x \exists y F(x, y)$  d)  $\neg \exists x \forall y F(x, y)$  e)  $\forall y \exists x F(x, y)$  f)  $\neg \exists x (F(x, \text{Fred}) \wedge F(x, \text{Jerry}))$

g)  $\exists y_1 \exists y_2 (F(\text{Nancy}, y_1) \wedge F(\text{Nancy}, y_2) \wedge y_1 \neq y_2 \wedge \forall y (F(\text{Nancy}, y) \rightarrow (y = y_1 \vee y = y_2)))$  h)  $\exists y (\forall x F(x, y) \wedge \forall z (\forall x F(x, z) \rightarrow z = y))$  i)  $\neg \exists x F(x, x)$

j)  $\exists x \exists y (x \neq y \wedge F(x, y) \wedge \forall z ((F(x, z) \wedge z \neq x) \rightarrow z = y))$  (We do not assume that this sentence is asserting that this person can or cannot fool her/himself.)

12. The answers to this exercise are not unique; there are many ways of expressing the same propositions symbolically. Note that  $C(x, y)$  and  $C(y, x)$  say the same thing.

a)  $\neg I(\text{Jerry})$  b)  $\neg C(\text{Rachel}, \text{Chelsea})$  c)  $\neg C(\text{Jan}, \text{Sharon})$  d)  $\neg \exists x C(x, \text{Bob})$  e)  $\forall x (x \neq \text{Joseph} \leftrightarrow C(x, \text{Sanjay}))$  f)  $\exists x \neg I(x)$  g)  $\neg \forall x I(x)$  (same as (f))

h)  $\exists x \forall y (x = y \leftrightarrow I(y))$  i)  $\exists x \forall y (x \neq y \leftrightarrow I(y))$  j)  $\forall x (I(x) \rightarrow \exists y (x \neq y \wedge C(x, y)))$  k)  $\exists x (I(x) \wedge \forall y (x \neq y \rightarrow \neg C(x, y)))$  l)  $\exists x \exists y (x \neq y \wedge \neg C(x, y))$  m)  $\exists x \forall y C(x, y)$  n)  $\exists x \exists y (x \neq y \wedge \forall z (\neg C(x, z) \wedge C(y, z)))$  o)  $\exists x \exists y (x \neq y \wedge \forall z (C(x, z) \vee C(y, z)))$

14. The answers to this exercise are not unique; there are many ways of expressing the same propositions symbolically. Our domain of discourse for persons here consists of people in this class. We need to make up a predicate in each case.

#### Section 1.5 Nested Quantifiers 21

a) Let  $S(x, y)$  mean that person  $x$  can speak language  $y$ . Then our statement is  $\exists x S(x, \text{Hindi})$ . b) Let  $P(x, y)$  mean that person  $x$  plays sport  $y$ . Then our statement is  $\forall x \exists y P(x, y)$ . c) Let  $V(x, y)$  mean that person  $x$  has visited state  $y$ . Then our statement is  $\exists x (V(x, \text{Alaska}) \wedge \neg V(x, \text{Hawaii}))$ .

d) Let  $L(x, y)$  mean that person  $x$  has learned programming language  $y$ . Then our statement is  $\forall x \exists y L(x, y)$ . e) Let  $T(x, y)$  mean that person  $x$  has taken course  $y$ , and let  $O(y, z)$  mean that course  $y$  is offered by department  $z$ . Then our statement is  $\exists x \exists z \forall y (O(y, z) \rightarrow T(x, y))$ .

f) Let  $G(x, y)$  mean that persons  $x$  and  $y$  grew up in the same town. Then our statement is  $\exists x \exists y (x \neq y \wedge G(x, y) \wedge \forall z (G(x, z) \rightarrow (x = y \vee x = z)))$ .

g) Let  $C(x, y, z)$  mean that persons  $x$  and  $y$  have chatted with each other in chat group  $z$ . Then our statement is  $\forall x \exists y \exists z (x \neq y \wedge C(x, y, z))$ .

16. We let  $P(s, c, m)$  be the statement that student  $s$  has class standing  $c$  and is majoring in  $m$ . The variable  $s$  ranges over students in the class, the variable  $c$  ranges over the four class standings, and the variable  $m$  ranges over all possible majors.

a) The proposition is  $\exists s \exists m P(s, \text{junior}, m)$ . It is true from the given information.

b) The proposition is  $\forall s \exists c P(s, c, \text{computer science})$ . This is false, since there are some mathematics majors. c) The proposition is  $\exists s \exists c \exists m P(s, c, m) \wedge (c \neq \text{junior}) \wedge (m \neq \text{mathematics})$ . This is true, since there is a sophomore majoring in computer science.

d) The proposition is  $\forall s \exists c P(s, c, \text{computer science}) \vee \exists m P(s, \text{sophomore}, m)$ . This is false, since there is a freshman mathematics major.

e) The proposition is  $\exists m \forall c \exists s P(s, c, m)$ . This is false. It cannot be that  $m$  is mathematics, since there is

no senior mathematics major, and it cannot be that  $m$  is computer science, since there is no freshman computer science major. Nor, of course, can  $m$  be any other major.

18. a)  $\forall f (H(f) \rightarrow \exists c A(c))$ , where  $A(x)$  means that console  $x$  is accessible, and  $H(x)$  means that fault condition  $x$  is happening  
 b)  $(\forall u \exists m (A(m) \wedge S(u, m))) \rightarrow \forall u R(u)$ , where  $A(x)$  means that the archive contains message  $x$ ,  $S(x, y)$  means that user  $x$  sent message  $y$ , and  $R(x)$  means that the e-mail address of user  $x$  can be retrieved  
 c)  $(\forall b \exists m D(m, b)) \leftrightarrow \exists p \neg C(p)$ , where  $D(x, y)$  means that mechanism  $x$  can detect breach  $y$ , and  $C(x)$  means that process  $x$  has been compromised  
 d)  $\forall x \forall y (x \neq y \rightarrow \exists p \exists q (p \neq q \wedge C(p, x, y) \wedge C(q, x, y)))$ , where  $C(p, x, y)$  means that path  $p$  connects endpoint  $x$  to endpoint  $y$   
 e)  $\forall x ((\forall u K(x, u)) \leftrightarrow x = \text{SysAdm})$ , where  $K(x, y)$  means that person  $x$  knows the password of user  $y$
20. a)  $\forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (xy > 0))$  b)  $\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow ((x + y)/2 > 0))$  c) What does “necessarily” mean in this context? The best explanation is to assert that a certain universal conditional statement is not true. So we have  $\neg \forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (x - y < 0))$ . Note that we do not want to put the negation symbol inside (it is not true that the difference of two negative integers is never negative), nor do we want to negate just the conclusion (it is not true that the sum is always nonnegative). We could rewrite our solution by passing the negation inside, obtaining  $\exists x \exists y ((x < 0) \wedge (y < 0) \wedge (x - y \geq 0))$ . d)  $\forall x \forall y (|x + y| \leq |x| + |y|)$
22.  $\exists x \forall a \forall b \forall c ((x > 0) \wedge x = a^2 + b^2 + c^2)$ , where the domain of discourse consists of all integers
24. a) There exists an additive identity for the real numbers—a number that when added to every number does not change its value.
- 22 Chapter 1 The Foundations: Logic and Proofs
- b) A nonnegative number minus a negative number is positive.  
 c) The difference of two nonpositive numbers is not necessarily nonpositive.  
 d) The product of two numbers is nonzero if and only if both factors are nonzero.
26. a) This is false, since  $1 + 1 \neq 1 - 1$ . b) This is true, since  $2 + 0 = 2 - 0$ .  
 c) This is false, since there are many values of  $y$  for which  $1 + y \neq 1 - y$ .  
 d) This is false, since the equation  $x + 2 = x - 2$  has no solution.  
 e) This is true, since we can take  $x = y = 0$ . f) This is true, since we can take  $y = 0$  for each  $x$ . g) This is true, since we can take  $y = 0$ . h) This is false, since part (d) was false. i) This is certainly false.
28. a) true (let  $y = x^2$ ) b) false (no such  $y$  exists if  $x$  is negative) c) true (let  $x = 0$ ) d) false (the commutative law for addition always holds) e) true (let  $y = 1/x$ )  
 f) false (the reciprocal of  $y$  depends on  $y$ —there is not one  $x$  that works for all  $y$ ) g) true (let  $y = 1 - x$ ) h) false (this system of equations is inconsistent)  
 i) false (this system has only one solution; if  $x = 0$ , for example, then no  $y$  satisfies  $y = 2 \wedge -y = 1$ ) j) true (let  $z = (x + y)/2$ )
30. We need to use the transformations shown in Table 2 of Section 1.4, replacing  $\neg \forall$  by  $\exists \neg$ , and replacing  $\neg \exists$  by  $\forall \neg$ . In other words, we push all the negation symbols inside the quantifiers, changing the sense of the quantifiers as we do so, because of the equivalences in Table 2 of Section 1.4. In addition, we need to use De Morgan’s laws (Section 1.3) to change the negation of a conjunction to the disjunction of the negations

and to change the negation of a disjunction to the conjunction of the negations. We also use the fact that  $\neg\neg p \equiv p$ . a)  $\forall y \forall x \neg P(x, y)$  b)  $\exists x \forall y \neg P(x, y)$  c)  $\forall y (\neg Q(y) \vee \exists x R(x, y))$   
d)  $\forall y (\forall x \neg R(x, y) \wedge \exists x \neg S(x, y))$  e)  $\forall y (\exists x \forall z \neg T(x, y, z) \wedge \forall x \exists z \neg U(x, y, z))$

32. As we push the negation symbol toward the inside, each quantifier it passes must change its type. For logical connectives we either use De Morgan's laws or recall that  $\neg(p \rightarrow q) \equiv p \wedge \neg q$  (Table 7 in Section 1.3) and that  $\neg(p \leftrightarrow q) \equiv \neg p \leftrightarrow q$  (Exercise 21 in Section 1.3).

a)  $\neg \exists z \forall y \forall x T(x, y, z) \equiv \forall z \neg \forall y \forall x T(x, y, z)$   
 $\equiv \forall z \exists y \neg \forall x T(x, y, z)$   
 $\equiv \forall z \exists y \exists x \neg T(x, y, z)$

b)  $\neg (\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)) \equiv \neg \exists x \exists y P(x, y) \vee \neg \forall x \forall y Q(x, y) \equiv \forall x \neg \exists y$   
 $P(x, y) \vee \exists x \neg \forall y Q(x, y)$   
 $\equiv \forall x \forall y \neg P(x, y) \vee \exists x \exists y \neg Q(x, y)$

c)  $\neg \exists x \exists y (Q(x, y) \leftrightarrow Q(y, x)) \equiv \forall x \neg \exists y (Q(x, y) \leftrightarrow Q(y, x))$   
 $\equiv \forall x \forall y \neg (Q(x, y) \leftrightarrow Q(y, x))$   
 $\equiv \forall x \forall y (\neg Q(x, y) \leftrightarrow Q(y, x))$

d)  $\neg \forall y \exists x \exists z (T(x, y, z) \vee Q(x, y)) \equiv \exists y \neg \exists x \exists z (T(x, y, z) \vee Q(x, y))$   
 $\equiv \exists y \forall x \neg \exists z (T(x, y, z) \vee Q(x, y))$   
 $\equiv \exists y \forall x \forall z \neg (T(x, y, z) \vee Q(x, y))$   
 $\equiv \exists y \forall x \forall z (\neg T(x, y, z) \wedge \neg Q(x, y))$

## Section 1.5 Nested Quantifiers 23

34. The logical expression is asserting that the domain consists of at most two members. (It is saying that whenever you have two unequal objects, any object has to be one of those two. Note that this is vacuously true for domains with one element.) Therefore any domain having one or two members will make it true (such as the female members of the United States Supreme Court in 2005), and any domain with more than two members will make it false (such as all members of the United States Supreme Court in 2005).
36. In each case we need to specify some predicates and identify the domain of discourse. a) Let  $L(x, y)$  mean that person  $x$  has lost  $y$  dollars playing the lottery. The original statement is then  $\neg \exists x \exists y (y > 1000 \wedge L(x, y))$ . Its negation of course is  $\exists x \exists y (y > 1000 \wedge L(x, y))$ ; someone has lost more than \$1000 playing the lottery.
- b) Let  $C(x, y)$  mean that person  $x$  has chatted with person  $y$ . The given statement is  $\exists x \exists y (y \neq x \wedge \forall z (z \neq x \rightarrow (z = y \leftrightarrow C(x, z))))$ . The negation is therefore  $\forall x \forall y (y \neq x \rightarrow \exists z (z \neq x \wedge \neg (z = y \leftrightarrow C(x, z))))$ . In English, everybody in this class has either chatted with no one else or has chatted with two or more others.
- c) Let  $E(x, y)$  mean that person  $x$  has sent e-mail to person  $y$ . The given statement is  $\neg \exists x \exists y \exists z (y \neq z \wedge x \neq y \wedge x \neq z \wedge \forall w (w \neq x \rightarrow (E(x, w) \leftrightarrow (w = y \vee w = z))))$ . The negation is obviously  $\exists x \exists y \exists z (y \neq z \wedge x \neq y \wedge x \neq z \wedge \forall w (w \neq x \rightarrow (E(x, w) \leftrightarrow (w = y \vee w = z))))$ . In English, some student in this class has sent e-mail to exactly two other students in this class.
- d) Let  $S(x, y)$  mean that student  $x$  has solved exercise  $y$ . The statement is  $\exists x \forall y S(x, y)$ . The negation is  $\forall x \exists y \neg S(x, y)$ . In English, for every student in this class, there is some exercise that he or she has not solved. (One could also interpret the given statement as asserting that for every exercise, there exists a

student— perhaps a different one for each exercise—who has solved it. In that case the order of the quantifiers would be reversed. Word order in English sometimes makes for a little ambiguity.)

e) Let  $S(x, y)$  mean that student  $x$  has solved exercise  $y$ , and let  $B(y, z)$  mean that exercise  $y$  is in section  $z$  of the book. The statement is  $\neg \exists x \forall z \exists y (B(y, z) \wedge S(x, y))$ . The negation is of course  $\exists x \forall z \exists y (B(y, z) \wedge S(x, y))$ . In English, some student has solved at least one exercise in every section of this book.

38. a) In English, the negation is “Some student in this class does not like mathematics.” With the obvious propositional function, this is  $\exists x \neg L(x)$ .

b) In English, the negation is “Every student in this class has seen a computer.” With the obvious propositional function, this is  $\forall x S(x)$ .

c) In English, the negation is “For every student in this class, there is a mathematics course that this student has not taken.” With the obvious propositional function, this is  $\forall x \exists c \neg T(x, c)$ .

d) As in Exercise 15f, let  $P(z, y)$  be “Room  $z$  is in building  $y$ ,” and let  $Q(x, z)$  be “Student  $x$  has been in room  $z$ .” Then the original statement is  $\exists x \forall y \exists z P(z, y) \wedge Q(x, z)$ . To form the negation, we change all the quantifiers and put the negation on the inside, then apply De Morgan’s law. The negation is therefore  $\forall x \exists y \forall z \neg P(z, y) \vee \neg Q(x, z)$ , which is also equivalent to  $\forall x \exists y \forall z P(z, y) \rightarrow \neg Q(x, z)$ . In English, this could be read, “For every student there is a building such that for every room in that building, the student has not been in that room.”

40. a) There are many counterexamples. If  $x = 2$ , then there is no  $y$  among the integers such that  $2 = 1/y$ , since the only solution of this equation is  $y = 1/2$ . Even if we were working in the domain of real numbers,  $x = 0$  would provide a counterexample, since  $0 = 1/y$  for no real number  $y$ .

b) We can rewrite  $y^2 - x < 100$  as  $y^2 < 100 + x$ . Since squares can never be negative, no such  $y$  exists if  $x$  is, say,  $-200$ . This  $x$  provides a counterexample.

c) This is not true, since sixth powers are both squares and cubes. Trivial counterexamples would include  $x = y = 0$  and  $x = y = 1$ , but we can also take something like  $x = 27$  and  $y = 9$ , since  $27^2 = 3^6 = 9^3$ .

## 24 Chapter 1 The Foundations: Logic and Proofs

42. The distributive law is just the statement that  $x(y+z) = xy+xz$  for all real numbers. Therefore the expression we want is  $\forall x \forall y \forall z (x(y+z) = xy+xz)$ , where the quantifiers are assumed to range over (i.e., the domain of discourse is) the real numbers.

44. We want to say that for each triple of coefficients (the  $a$ ,  $b$ , and  $c$  in the expression  $ax^2 + bx + c$ , where we insist that  $a \neq 0$  so that this actually is quadratic), there are at most two values of  $x$  making that expression equal to 0. The domain here is all real numbers. We write  $\forall a \forall b \forall c (a \neq 0 \rightarrow \forall x_1 \forall x_2 \forall x_3 (ax_1^2 + bx_1 + c = 0 \wedge ax_2^2 + bx_2 + c = 0 \wedge ax_3^2 + bx_3 + c = 0) \rightarrow (x_1 = x_2 \vee x_1 = x_3 \vee x_2 = x_3))$ .

46. This statement says that there is a number that is less than or equal to all squares. a) This is false, since no matter how small a positive number  $x$  we might choose, if we let  $y = \sqrt{x}/2$ , then  $x = 4y^2$ , and it will not be true that  $x \leq y^2$ .

b) This is true, since we can take  $x = -1$ , for example.

c) This is true, since we can take  $x = -1$ , for example.

48. We need to show that each of these propositions implies the other. Suppose that  $\forall x P(x) \vee \forall x Q(x)$  is true. We want to show that  $\forall x \forall y (P(x) \vee Q(y))$  is true. By our hypothesis, one of two things must be true. Either

$P$  is universally true, or  $Q$  is universally true. In the first case,  $\forall x \forall y (P(x) \vee Q(y))$  is true, since the first expression in the disjunction is true, no matter what  $x$  and  $y$  are; and in the second case,  $\forall x \forall y (P(x) \vee Q(y))$  is also true, since now the second expression in the disjunction is true, no matter what  $x$  and  $y$  are. Next we need to prove the converse. So suppose that  $\forall x \forall y (P(x) \vee Q(y))$  is true. We want to show that  $\forall x P(x) \vee \forall x Q(x)$  is true. If  $\forall x P(x)$  is true, then we are done. Otherwise,  $P(x_0)$  must be false for some  $x_0$  in the domain of discourse. For this  $x_0$ , then, the hypothesis tells us that  $P(x_0) \vee Q(y)$  is true, no matter what  $y$  is. Since  $P(x_0)$  is false, it must be the case that  $Q(y)$  is true for each  $y$ . In other words,  $\forall y Q(y)$  is true, or, to change the name of the meaningless quantified variable,  $\forall x Q(x)$  is true. This certainly implies that  $\forall x P(x) \vee \forall x Q(x)$  is true, as desired.

50. a) By Exercises 45 and 46b in Section 1.4, we can simply bring the existential quantifier outside:  $\exists x (P(x) \vee Q(x) \vee A)$ .

b) By Exercise 48 of the current section, the expression inside the parentheses is logically equivalent to  $\forall x \forall y (P(x) \vee Q(y))$ . Applying the negation operation, we obtain  $\exists x \exists y \neg (P(x) \vee Q(y))$ . c) First we rewrite this using Table 7 in Section 1.3 as  $\exists x Q(x) \vee \neg \exists x P(x)$ , which is equivalent to  $\exists x Q(x) \vee \forall x \neg P(x)$ . To combine the existential and universal statements we use Exercise 49b of the current section, obtaining  $\forall x \exists y (\neg P(x) \vee Q(y))$ , which is in prenex normal form.

52. We simply want to say that there exists an  $x$  such that  $P(x)$  holds, and that every  $y$  such that  $P(y)$  holds must be this same  $x$ . Thus we write  $\exists x P(x) \wedge \forall y (P(y) \rightarrow y = x)$ . Even more compactly, we can write  $\exists x \forall y (P(y) \leftrightarrow y = x)$ .

Section 1.6 Rules of Inference 25

## SECTION 1.6 Rules of Inference

2. This is modus tollens. The first statement is  $p \rightarrow q$ , where  $p$  is "George does not have eight legs" and  $q$  is "George is not a spider." The second statement is  $\neg q$ . The third is  $\neg p$ . Modus tollens is valid. We can therefore conclude that the conclusion of the argument (third statement) is true, given that the hypotheses (the first two statements) are true.

4. a) We have taken the conjunction of two propositions and asserted one of them. This is, according to Table 1, simplification.

b) We have taken the disjunction of two propositions and the negation of one of them, and asserted the other. This is, according to Table 1, disjunctive syllogism. See Table 1 for the other parts of this exercise as well. c) modus ponens d) addition e) hypothetical syllogism

6. Let  $r$  be the proposition "It rains," let  $f$  be the proposition "It is foggy," let  $s$  be the proposition "The sailing race will be held," let  $l$  be the proposition "The life saving demonstration will go on," and let  $t$  be the proposition "The trophy will be awarded." We are given premises  $(\neg r \vee \neg f) \rightarrow (s \wedge l)$ ,  $s \rightarrow t$ , and  $\neg t$ . We want to conclude  $r$ . We set up the proof in two columns, with reasons, as in Example 6. Note that it is valid to replace subexpressions by other expressions logically equivalent to them.

Step Reason

1.  $\neg t$  Hypothesis

2.  $s \rightarrow t$  Hypothesis

3.  $\neg s$  Modus tollens using (1) and (2)

4.  $(\neg r \vee \neg f) \rightarrow (s \wedge l)$  Hypothesis

5.  $(\neg(s \wedge l)) \rightarrow \neg(\neg r \vee \neg f)$  Contrapositive of (4)

6.  $(\neg s \vee \neg I) \rightarrow (r \wedge f)$  De Morgan's law and double negative
7.  $\neg s \vee \neg I$  Addition, using (3)
8.  $r \wedge f$  Modus ponens using (6) and (7)
9.  $r$  Simplification using (8)

8. First we use universal instantiation to conclude from "For all  $x$ , if  $x$  is a man, then  $x$  is not an island" the special case of interest, "If Manhattan is a man, then Manhattan is not an island." Then we form the contrapositive (using also double negative): "If Manhattan is an island, then Manhattan is not a man." Finally we use modus ponens to conclude that Manhattan is not a man. Alternatively, we could apply modus tollens.

10. a) If we use modus tollens starting from the back, then we conclude that I am not sore. Another application of modus tollens then tells us that I did not play hockey.  
 b) We really can't conclude anything specific here.  
 c) By universal instantiation, we conclude from the first conditional statement by modus ponens that dragon flies have six legs, and we conclude by modus tollens that spiders are not insects. We could say using existential generalization that, for example, there exists a non-six-legged creature that eats a six-legged creature, and that there exists a non-insect that eats an insect.  
 d) We can apply universal instantiation to the conditional statement and conclude that if Homer (respectively, Maggie) is a student, then he (she) has an Internet account. Now modus tollens tells us that Homer is not a student. There are no conclusions to be drawn about Maggie.  
 e) The first conditional statement is that if  $x$  is healthy to eat, then  $x$  does not taste good. Universal instantiation and modus ponens therefore tell us that tofu does not taste good. The third sentence says that if you eat  $x$ , then  $x$  tastes good. Therefore the fourth hypothesis already follows (by modus tollens) from the first three. No conclusions can be drawn about cheeseburgers from these statements.

## 26 Chapter 1 The Foundations: Logic and Proofs

f) By disjunctive syllogism, the first two hypotheses allow us to conclude that I am hallucinating. Therefore by modus ponens we know that I see elephants running down the road.

12. Applying Exercise 11, we want to show that the conclusion  $r$  follows from the five premises  $(p \wedge t) \rightarrow (r \vee s)$ ,  $q \rightarrow (u \wedge t)$ ,  $u \rightarrow p$ ,  $\neg s$ , and  $q$ . From  $q$  and  $q \rightarrow (u \wedge t)$  we get  $u \wedge t$  by modus ponens. From there we get both  $u$  and  $t$  by simplification (and the commutative law). From  $u$  and  $u \rightarrow p$  we get  $p$  by modus ponens. From  $p$  and  $t$  we get  $p \wedge t$  by conjunction. From that and  $(p \wedge t) \rightarrow (r \vee s)$  we get  $r \vee s$  by modus ponens. From that and  $\neg s$  we finally get  $r$  by disjunctive syllogism.
14. In each case we set up the proof in two columns, with reasons, as in Example 6.  
 a) Let  $c(x)$  be " $x$  is in this class," let  $r(x)$  be " $x$  owns a red convertible," and let  $t(x)$  be " $x$  has gotten a speeding ticket." We are given premises  $c(\text{Linda})$ ,  $r(\text{Linda})$ ,  $\forall x(r(x) \rightarrow t(x))$ , and we want to conclude  $\exists x(c(x) \wedge t(x))$ .

Step Reason

1.  $\forall x(r(x) \rightarrow t(x))$  Hypothesis
2.  $r(\text{Linda}) \rightarrow t(\text{Linda})$  Universal instantiation using (1)
3.  $r(\text{Linda})$  Hypothesis
4.  $t(\text{Linda})$  Modus ponens using (2) and (3)
5.  $c(\text{Linda})$  Hypothesis
6.  $c(\text{Linda}) \wedge t(\text{Linda})$  Conjunction using (4) and (5)
7.  $\exists x(c(x) \wedge t(x))$  Existential generalization using (6)

b) Let  $r(x)$  be “ $x$  is one of the five roommates listed,” let  $d(x)$  be “ $x$  has taken a course in discrete mathematics,” and let  $a(x)$  be “ $x$  can take a course in algorithms.” We are given premises  $\forall x(r(x) \rightarrow d(x))$  and  $\forall x(d(x) \rightarrow a(x))$ , and we want to conclude  $\forall x(r(x) \rightarrow a(x))$ . In what follows  $y$  represents an arbitrary person.

Step Reason

1.  $\forall x(r(x) \rightarrow d(x))$  Hypothesis
2.  $r(y) \rightarrow d(y)$  Universal instantiation using (1)
3.  $\forall x(d(x) \rightarrow a(x))$  Hypothesis
4.  $d(y) \rightarrow a(y)$  Universal instantiation using (3)
5.  $r(y) \rightarrow a(y)$  Hypothetical syllogism using (2) and (4)
6.  $\forall x(r(x) \rightarrow a(x))$  Universal generalization using (5)

c) Let  $s(x)$  be “ $x$  is a movie produced by Sayles,” let  $c(x)$  be “ $x$  is a movie about coal miners,” and let  $w(x)$  be “movie  $x$  is wonderful.” We are given premises  $\forall x(s(x) \rightarrow w(x))$  and  $\exists x(s(x) \wedge c(x))$ , and we want to conclude  $\exists x(c(x) \wedge w(x))$ . In our proof,  $y$  represents an unspecified particular movie. Step Reason

1.  $\exists x(s(x) \wedge c(x))$  Hypothesis
2.  $s(y) \wedge c(y)$  Existential instantiation using (1)
3.  $s(y)$  Simplification using (2)
4.  $\forall x(s(x) \rightarrow w(x))$  Hypothesis
5.  $s(y) \rightarrow w(y)$  Universal instantiation using (4)
6.  $w(y)$  Modus ponens using (3) and (5)
7.  $c(y)$  Simplification using (2)
8.  $w(y) \wedge c(y)$  Conjunction using (6) and (7)
9.  $\exists x(c(x) \wedge w(x))$  Existential generalization using (8)

d) Let  $c(x)$  be “ $x$  is in this class,” let  $f(x)$  be “ $x$  has been to France,” and let  $l(x)$  be “ $x$  has visited the Louvre.” We are given premises  $\exists x(c(x) \wedge f(x))$ ,  $\forall x(f(x) \rightarrow l(x))$ , and we want to conclude  $\exists x(c(x) \wedge l(x))$ .

#### Section 1.6 Rules of Inference 27

In our proof,  $y$  represents an unspecified particular person.

Step Reason

1.  $\exists x(c(x) \wedge f(x))$  Hypothesis
2.  $c(y) \wedge f(y)$  Existential instantiation using (1)
3.  $f(y)$  Simplification using (2)
4.  $c(y)$  Simplification using (2)
5.  $\forall x(f(x) \rightarrow l(x))$  Hypothesis
6.  $f(y) \rightarrow l(y)$  Universal instantiation using (5)
7.  $l(y)$  Modus ponens using (3) and (6)
8.  $c(y) \wedge l(y)$  Conjunction using (4) and (7)
9.  $\exists x(c(x) \wedge l(x))$  Existential generalization using (8)

16. a) This is correct, using universal instantiation and modus tollens.

b) This is not correct. After applying universal instantiation, it contains the fallacy of denying the hypothesis.

c) After applying universal instantiation, it contains the fallacy of affirming the conclusion. d) This is correct, using universal instantiation and modus ponens.

18. We know that some  $s$  exists that makes  $S(s, \text{Max})$  true, but we cannot conclude that Max is one such  $s$ . Therefore this first step is invalid.

20. a) This is invalid. It is the fallacy of affirming the conclusion. Letting  $a = -2$  provides a counterexample. b)

This is valid; it is modus ponens.

22. We will give an argument establishing the conclusion. We want to show that all hummingbirds are small. Let Tweety be an arbitrary hummingbird. We must show that Tweety is small. The first premise implies that if Tweety is a hummingbird, then Tweety is richly colored. Therefore by (universal) modus ponens we can conclude that Tweety is richly colored. The third premise implies that if Tweety does not live on honey, then Tweety is not richly colored. Therefore by (universal) modus tollens we can now conclude that Tweety does live on honey. Finally, the second premise implies that if Tweety is a large bird, then Tweety does not live on honey. Therefore again by (universal) modus tollens we can now conclude that Tweety is not a large bird, i.e., that Tweety is small, as desired. Notice that we invoke universal generalization as the last step.

24. Steps 3 and 5 are incorrect; simplification applies to conjunctions, not disjunctions.

26. We want to show that the conditional statement  $P(a) \rightarrow R(a)$  is true for all  $a$  in the domain; the desired conclusion then follows by universal generalization. Thus we want to show that if  $P(a)$  is true for a particular  $a$ , then  $R(a)$  is also true. For such an  $a$ , by universal modus ponens from the first premise we have  $Q(a)$ , and then by universal modus ponens from the second premise we have  $R(a)$ , as desired.

28. We want to show that the conditional statement  $\neg R(a) \rightarrow P(a)$  is true for all  $a$  in the domain; the desired conclusion then follows by universal generalization. Thus we want to show that if  $\neg R(a)$  is true for a particular  $a$ , then  $P(a)$  is also true. For such an  $a$ , universal modus tollens applied to the second premise gives us  $\neg(\neg P(a) \wedge Q(a))$ . By rules from propositional logic, this gives us  $P(a) \vee \neg Q(a)$ . By universal generalization from the first premise, we have  $P(a) \vee Q(a)$ . Now by resolution we can conclude  $P(a) \vee P(a)$ , which is logically equivalent to  $P(a)$ , as desired.

30. Let  $a$  be "Allen is a good boy"; let  $h$  be "Hillary is a good girl"; let  $d$  be "David is happy." Then our assumptions are  $\neg a \vee h$  and  $a \vee d$ . Using resolution gives us  $h \vee d$ , as desired.

## 28 Chapter 1 The Foundations: Logic and Proofs

32. We apply resolution to give the tautology  $(p \vee F) \wedge (\neg p \vee F) \rightarrow (F \vee F)$ . The left-hand side is equivalent to  $p \wedge \neg p$ , since  $p \vee F$  is equivalent to  $p$ , and  $\neg p \vee F$  is equivalent to  $\neg p$ . The right-hand side is equivalent to  $F$ . Since the conditional statement is true, and the conclusion is false, it follows that the hypothesis,  $p \wedge \neg p$ , is false, as desired.

34. Let us use the following letters to stand for the relevant propositions:  $d$  for "logic is difficult";  $s$  for "many students like logic"; and  $e$  for "mathematics is easy." Then the assumptions are  $d \vee \neg s$  and  $e \rightarrow \neg d$ . Note that the first of these is equivalent to  $s \rightarrow d$ , since both forms are false if and only if  $s$  is true and  $d$  is false. In addition, let us note that the second assumption is equivalent to its contrapositive,  $d \rightarrow \neg e$ . And finally, by combining these two conditional statements, we see that  $s \rightarrow \neg e$  also follows from our assumptions.

a) Here we are asked whether we can conclude that  $s \rightarrow \neg e$ . As we noted above, the answer is yes, this conclusion is valid.

b) The question concerns  $\neg e \rightarrow \neg s$ . This is equivalent to its contrapositive,  $s \rightarrow e$ . That doesn't seem to follow from our assumptions, so let's find a case in which the assumptions hold but this conditional statement does not. This conditional statement fails in the case in which  $s$  is true and  $e$  is false. If we take  $d$  to be true as well, then both of our assumptions are true. Therefore this conclusion is not valid.

c) The issue is  $\neg e \vee d$ , which is equivalent to the conditional statement  $e \rightarrow d$ . This does not follow from our assumptions. If we take  $d$  to be false,  $e$  to be true, and  $s$  to be false, then this proposition is false but our assumptions are true.

d) The issue is  $\neg d \vee \neg e$ , which is equivalent to the conditional statement  $d \rightarrow \neg e$ . We noted above that this validly follows from our assumptions.

e) This sentence says  $\neg s \rightarrow (\neg e \vee \neg d)$ . The only case in which this is false is when  $s$  is false and both  $e$  and  $d$  are true. But in this case, our assumption  $e \rightarrow \neg d$  is also violated. Therefore, in all cases in which the assumptions hold, this statement holds as well, so it is a valid conclusion.

## SECTION 1.7 Introduction to Proofs

2. We must show that whenever we have two even integers, their sum is even. Suppose that  $a$  and  $b$  are two even integers. Then there exist integers  $s$  and  $t$  such that  $a = 2s$  and  $b = 2t$ . Adding, we obtain  $a + b = 2s + 2t = 2(s + t)$ . Since this represents  $a + b$  as 2 times the integer  $s + t$ , we conclude that  $a + b$  is even, as desired.
4. We must show that whenever we have an even integer, its negative is even. Suppose that  $a$  is an even integer. Then there exists an integer  $s$  such that  $a = 2s$ . Its additive inverse is  $-2s$ , which by rules of arithmetic and algebra (see Appendix 1) equals  $2(-s)$ . Since this is 2 times the integer  $-s$ , it is even, as desired.
6. An odd number is one of the form  $2n + 1$ , where  $n$  is an integer. We are given two odd numbers, say  $2a + 1$  and  $2b + 1$ . Their product is  $(2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$ . This last expression shows that the product is odd, since it is of the form  $2n + 1$ , with  $n = 2ab + a + b$ .
8. Let  $n = m^2$ . If  $m = 0$ , then  $n + 2 = 2$ , which is not a perfect square, so we can assume that  $m \geq 1$ . The smallest perfect square greater than  $n$  is  $(m + 1)^2$ , and we have  $(m + 1)^2 = m^2 + 2m + 1 = n + 2m + 1 > n + 2 \cdot 1 + 1 > n + 2$ . Therefore  $n + 2$  cannot be a perfect square.
10. A rational number is a number that can be written in the form  $x/y$  where  $x$  and  $y$  are integers and  $y \neq 0$ . Suppose that we have two rational numbers, say  $a/b$  and  $c/d$ . Then their product is, by the usual rules for multiplication of fractions,  $(ac)/(bd)$ . Note that both the numerator and the denominator are integers, and that  $bd \neq 0$  since  $b$  and  $d$  were both nonzero. Therefore the product is, by definition, a rational number.

### Section 1.7 Introduction to Proofs 29

12. This is true. Suppose that  $a/b$  is a nonzero rational number and that  $x$  is an irrational number. We must prove that the product  $xa/b$  is also irrational. We give a proof by contradiction. Suppose that  $xa/b$  were rational. Since  $a/b \neq 0$ , we know that  $a \neq 0$ , so  $b/a$  is also a rational number. Let us multiply this rational number  $b/a$  by the assumed rational number  $xa/b$ . By Exercise 26, the product is rational. But the product is  $(b/a)(xa/b) = x$ , which is irrational by hypothesis. This is a contradiction, so in fact  $xa/b$  must be irrational, as desired.
14. If  $x$  is rational and not zero, then by definition we can write  $x = p/q$ , where  $p$  and  $q$  are nonzero integers. Since  $1/x$  is then  $q/p$  and  $p \neq 0$ , we can conclude that  $1/x$  is rational.
16. We give a proof by contraposition. If it is not true that  $m$  is even or  $n$  is even, then  $m$  and  $n$  are both odd. By Exercise 6, this tells us that  $mn$  is odd, and our proof is complete.
18. a) We must prove the contrapositive: If  $n$  is odd, then  $3n + 2$  is odd. Assume that  $n$  is odd. Then we can write  $n = 2k + 1$  for some integer  $k$ . Then  $3n + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1$ . Thus  $3n + 2$  is two times some integer plus 1, so it is odd.  
 b) Suppose that  $3n + 2$  is even and that  $n$  is odd. Since  $3n + 2$  is even, so is  $3n$ . If we add subtract an odd

number from an even number, we get an odd number, so  $3n - n = 2n$  is odd. But this is obviously not true. Therefore our supposition was wrong, and the proof by contradiction is complete.

20. We need to prove the proposition "If 1 is a positive integer, then  $1^2 \geq 1$ ." The conclusion is the true statement  $1 \geq 1$ . Therefore the conditional statement is true. This is an example of a trivial proof, since we merely showed that the conclusion was true.
22. We give a proof by contradiction. Suppose that we don't get a pair of blue socks or a pair of black socks. Then we drew at most one of each color. This accounts for only two socks. But we are drawing three socks. Therefore our supposition that we did not get a pair of blue socks or a pair of black socks is incorrect, and our proof is complete.
24. We give a proof by contradiction. If there were at most two days falling in the same month, then we could have at most  $2 \cdot 12 = 24$  days, since there are 12 months. Since we have chosen 25 days, at least three of them must fall in the same month.
26. We need to prove two things, since this is an "if and only if" statement. First let us prove directly that if  $n$  is even then  $7n + 4$  is even. Since  $n$  is even, it can be written as  $2k$  for some integer  $k$ . Then  $7n + 4 = 14k + 4 = 2(7k + 2)$ . This is 2 times an integer, so it is even, as desired. Next we give a proof by contraposition that if  $7n + 4$  is even then  $n$  is even. So suppose that  $n$  is not even, i.e., that  $n$  is odd. Then  $n$  can be written as  $2k + 1$  for some integer  $k$ . Thus  $7n + 4 = 14k + 11 = 2(7k + 5) + 1$ . This is 1 more than 2 times an integer, so it is odd. That completes the proof by contraposition.
28. There are two things to prove. For the "if" part, there are two cases. If  $m = n$ , then of course  $m^2 = n^2$ ; if  $m = -n$ , then  $m^2 = (-n)^2 = (-1)^2 n^2 = n^2$ . For the "only if" part, we suppose that  $m^2 = n^2$ . Putting everything on the left and factoring, we have  $(m + n)(m - n) = 0$ . Now the only way that a product of two numbers can be zero is if one of them is zero. Therefore we conclude that either  $m + n = 0$  (in which case  $m = -n$ ), or else  $m - n = 0$  (in which case  $m = n$ ), and our proof is complete.
30. We write these in symbols:  $a < b$ ,  $(a + b)/2 > a$ , and  $(a + b)/2 < b$ . The latter two are equivalent to  $a + b > 2a$  and  $a + b < 2b$ , respectively, and these are in turn equivalent to  $b > a$  and  $a < b$ , respectively. It is now clear that all three statements are equivalent.

### 30 Chapter 1 The Foundations: Logic and Proofs

32. We give direct proofs that (i) implies (ii), that (ii) implies (iii), and that (iii) implies (i). That will suffice. For the first, suppose that  $x = p/q$  where  $p$  and  $q$  are integers with  $q \neq 0$ . Then  $x/2 = p/(2q)$ , and this is rational, since  $p$  and  $2q$  are integers with  $2q \neq 0$ . For the second, suppose that  $x/2 = p/q$  where  $p$  and  $q$  are integers with  $q \neq 0$ . Then  $x = (2p)/q$ , so  $3x - 1 = (6p)/q - 1 = (6p - q)/q$  and this is rational, since  $6p - q$  and  $q$  are integers with  $q \neq 0$ . For the last, suppose that  $3x - 1 = p/q$  where  $p$  and  $q$  are integers with  $q \neq 0$ . Then  $x = (p/q + 1)/3 = (p + q)/(3q)$ , and this is rational, since  $p + q$  and  $3q$  are integers with  $3q \neq 0$ .
34. No. This line of reasoning shows that if  $\sqrt{2x^2 - 1} = x$ , then we must have  $x = 1$  or  $x = -1$ . These are therefore the only possible solutions, but we have no guarantee that they are solutions, since not all of our steps were reversible (in particular, squaring both sides). Therefore we must substitute these values back into the original equation to determine whether they do indeed satisfy it.
36. The only conditional statements not shown directly are  $p_1 \leftrightarrow p_2$ ,  $p_2 \leftrightarrow p_4$ , and  $p_3 \leftrightarrow p_4$ . But these each follow with one or more intermediate steps:  $p_1 \leftrightarrow p_2$ , since  $p_1 \leftrightarrow p_3$  and  $p_3 \leftrightarrow p_2$ ;  $p_2 \leftrightarrow p_4$ , since  $p_2 \leftrightarrow p_1$  (just established) and  $p_1 \leftrightarrow p_4$ ; and  $p_3 \leftrightarrow p_4$ , since  $p_3 \leftrightarrow p_1$  and  $p_1 \leftrightarrow p_4$ .

38. We must find a number that cannot be written as the sum of the squares of three integers. We claim that 7 is such a number (in fact, it is the smallest such number). The only squares that can be used to contribute to the sum are 0, 1, and 4. We cannot use two 4's, because their sum exceeds 7. Therefore we can use at most one 4, which means that we must get 3 using just 0's and 1's. Clearly three 1's are required for this, bringing the total number of squares used to four. Thus 7 cannot be written as the sum of three squares.
40. Suppose that we look at the ten groups of integers in three consecutive locations around the circle (first second-third, second-third-fourth, . . . , eighth-ninth-tenth, ninth-tenth-first, and tenth-first-second). Since each number from 1 to 10 gets used three times in these groups, the sum of the sums of the ten groups must equal three times the sum of the numbers from 1 to 10, namely  $3 \cdot 55 = 165$ . Therefore the average sum is  $165/10 = 16.5$ . By Exercise 39, at least one of the sums must be greater than or equal to 16.5, and since the sums are whole numbers, this means that at least one of the sums must be greater than or equal to 17.
42. We show that each of these is equivalent to the statement (v)  $n$  is odd, say  $n = 2k+1$ . Example 1 showed that (v) implies (i), and Example 8 showed that (i) implies (v). For  $(v) \rightarrow (ii)$  we see that  $1 - n = 1 - (2k + 1) = 2(-k)$  is even. Conversely, if  $n$  were even, say  $n = 2m$ , then we would have  $1 - n = 1 - 2m = 2(-m) + 1$ , so  $1 - n$  would be odd, and this completes the proof by contraposition that  $(ii) \rightarrow (v)$ . For  $(v) \rightarrow (iii)$ , we see that  $n^3 = (2k+1)^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1$  is odd. Conversely, if  $n$  were even, say  $n = 2m$ , then we would have  $n^3 = 2(4m^3)$ , so  $n^3$  would be even, and this completes the proof by contraposition that  $(iii) \rightarrow (v)$ . Finally, for  $(v) \rightarrow (iv)$ , we see that  $n^2 + 1 = (2k + 1)^2 + 1 = 4k^2 + 4k + 2 = 2(2k^2 + 2k + 1)$  is even. Conversely, if  $n$  were even, say  $n = 2m$ , then we would have  $n^2 + 1 = 2(2m^2) + 1$ , so  $n^2 + 1$  would be odd, and this completes the proof by contraposition that  $(iv) \rightarrow (v)$ .

#### Section 1.8 Proof Methods and Strategy 31

### SECTION 1.8 Proof Methods and Strategy

2. The cubes that might go into the sum are 1, 8, 27, 64, 125, 216, 343, 512, and 729. We must show that no two of these sum to a number on this list. If we try the 45 combinations  $(1 + 1, 1 + 8, \dots, 1 + 729, 8 + 8, 8 + 27, \dots, 8 + 729, \dots, 729 + 729)$ , we see that none of them works. Having exhausted the possibilities, we conclude that no cube less than 1000 is the sum of two cubes.
4. There are three main cases, depending on which of the three numbers is smallest. If  $a$  is smallest (or tied for smallest), then clearly  $a \leq \min(b, c)$ , and so the left-hand side equals  $a$ . On the other hand, for the right-hand side we have  $\min(a, c) = a$  as well. In the second case,  $b$  is smallest (or tied for smallest). The same reasoning shows us that the right-hand side equals  $b$ ; and the left-hand side is  $\min(a, b) = b$  as well. In the final case, in which  $c$  is smallest (or tied for smallest), the left-hand side is  $\min(a, c) = c$ , whereas the right-hand side is clearly also  $c$ . Since one of the three has to be smallest we have taken care of all the cases.
6. Because  $x$  and  $y$  are of opposite parities, we can assume, without loss of generality, that  $x$  is even and  $y$  is odd. This tells us that  $x = 2m$  for some integer  $m$  and  $y = 2n + 1$  for some integer  $n$ . Then  $5x+5y = 5(2m)+5(2n+1) = 10m + 10n+1 = 10(m+n)+1 = 2 \cdot 5(m+n)+1$ , which satisfies the definition of being an odd number.
8. The number 1 has this property, since the only positive integer not exceeding 1 is 1 itself, and therefore the sum is 1. This is a constructive proof.

10. The only perfect squares that differ by 1 are 0 and 1. Therefore these two consecutive integers cannot both be perfect squares. This is a nonconstructive proof—we do not know which of them meets the requirement. (In fact, a computer algebra system will tell us that neither of them is a perfect square.)
12. Of these three numbers, at least two must have the same sign (both positive or both negative), since there are only two signs. (It is conceivable that some of them are zero, but we view zero as positive for the purposes of this problem.) The product of two with the same sign is nonnegative. This was a nonconstructive proof, since we have not identified which product is nonnegative. (In fact, a computer algebra system will tell us that all three are positive, so all three products are positive.)
14. An assertion like this one is implicitly universally quantified—it means that for all rational numbers  $a$  and  $b$ ,  $a^b$  is rational. To disprove such a statement it suffices to provide one counterexample. Take  $a = 2$  and  $b = 1/2$ . Then  $a^b = 2^{1/2} = \sqrt{2}$ , and we know from Example 10 in Section 1.7 that  $\sqrt{2}$  is not rational.
16. We know from algebra that the following equations are equivalent:  $ax + b = c$ ,  $ax = c - b$ ,  $x = (c - b)/a$ . This shows, constructively, what the unique solution of the given equation is.
18. Given  $r$ , let  $a$  be the closest integer to  $r$  less than  $r$ , and let  $b$  be the closest integer to  $r$  greater than  $r$ . In the notation to be introduced in Section 2.3,  $a = \lfloor r \rfloor$  and  $b = \lceil r \rceil$ . In fact,  $b = a + 1$ . Clearly the distance between  $r$  and any integer other than  $a$  or  $b$  is greater than 1 so cannot be less than  $1/2$ . Furthermore, since  $r$  is irrational, it cannot be exactly half-way between  $a$  and  $b$ , so exactly one of  $r - a < 1/2$  and  $b - r < 1/2$  holds.
20. Given  $x$ , let  $n$  be the greatest integer less than or equal to  $x$ , and let  $\{x\} = x - n$ . In the notation to be introduced in Section 2.3,  $n = \lfloor x \rfloor$ . Clearly  $0 \leq \{x\} < 1$ , and  $\{x\}$  is unique for this  $n$ . Any other choice of  $n$  would cause the required  $\{x\}$  to be less than 0 or greater than or equal to 1, so  $n$  is unique as well.

### 32 Chapter 1 The Foundations: Logic and Proofs

22. We follow the hint. The square of every real number is nonnegative, so  $(x - 1/x)^2 \geq 0$ . Multiplying this out and simplifying, we obtain  $x^2 - 2 + 1/x^2 \geq 0$ , so  $x^2 + 1/x^2 \geq 2$ , as desired.
24. Let  $x = 1$  and  $y = 10$ . Then their arithmetic mean is 5.5 and their quadratic mean is  $\sqrt{50.5} \approx 7.11$ . Similarly, if  $x = 5$  and  $y = 8$ , then the arithmetic mean is  $(5+8)/2 = 6.5$  and the quadratic mean is  $\sqrt{(5^2 + 8^2)/2} \approx 6.67$ . So we conjecture that the quadratic mean is always greater than or equal to the arithmetic mean. Thus we want to prove that  $\sqrt{x^2 + y^2} \geq \frac{x+y}{2}$ .

$$\sqrt{x^2 + y^2} \geq \frac{x+y}{2}$$

for all positive real numbers  $x$  and  $y$ . Doing some algebra, we find that this inequality is equivalent to the true statement that  $(x - y)^2 \geq 0$ :

$$\begin{aligned} \sqrt{x^2 + y^2} &\geq \frac{x+y}{2} \\ 2\sqrt{x^2 + y^2} &\geq x+y \\ 2x^2 + 2y^2 &\geq x^2 + 2xy + y^2 \\ x^2 - 2xy + y^2 &\geq 0 \\ (x - y)^2 &\geq 0 \end{aligned}$$

In fact, our argument also shows that equality holds if and only if  $x = y$ .

26. If we were to end up with nine 0's, then in the step before this we must have had either nine 0's or nine 1's, since each adjacent pair of bits must have been equal and therefore all the bits must have been the same. Thus if we are to start with something other than nine 0's and yet end up with nine 0's, we must have had nine 1's at some point. But in the step before that each adjacent pair of bits must have been different; in other words, they must have alternated 0, 1, 0, 1, and so on. This is impossible with an odd number of bits. This contradiction shows that we can never get nine 0's.
28. Clearly only the last two digits of  $n$  contribute to the last two digits of  $n^2$ . So we can compute  $0^2, 1^2, 2^2, 3^2, \dots, 99^2$ , and record the last two digits, omitting repetitions. We obtain 00, 01, 04, 09, 16, 25, 36, 49, 64, 81, 21, 44, 69, 96, 56, 89, 24, 61, 41, 84, 29, 76. From that point on, the list repeats in reverse order (as we take the squares from  $25^2$  to  $49^2$ , and then it all repeats again as we take the squares from  $50^2$  to  $99^2$ ). The reason for these last two statements are that  $(50 - n)^2 = 2500 - 100n + n^2$ , so  $(50 - n)^2$  and  $n^2$  have the same two final digits, and  $(50 + n)^2 = 2500 + 100n + n^2$ , so  $(50 + n)^2$  and  $n^2$  have the same two final digits. Thus our list (which contains 22 numbers) is complete.
30. If  $|y| \geq 2$ , then  $2x^2 + 5y^2 \geq 2x^2 + 20 \geq 20$ , so the only possible values of  $y$  to try are 0 and  $\pm 1$ . In the former case we would be looking for solutions to  $2x^2 = 14$  and in the latter case to  $2x^2 = 9$ . Clearly there are no integer solutions to these equations, so there are no solutions to the original equation.
32. Following the hint, we let  $x = m^2 - n^2$ ,  $y = 2mn$ , and  $z = m^2 + n^2$ . Then  $x^2 + y^2 = (m^2 - n^2)^2 + (2mn)^2 = m^4 - 2m^2n^2 + n^4 + 4m^2n^2 = m^4 + 2m^2n^2 + n^4 = (m^2 + n^2)^2 = z^2$ . Thus we have found infinitely many solutions, since  $m$  and  $n$  can be arbitrarily large.
34. One proof that  $\sqrt[3]{2}$  is irrational is similar to the proof that  $\sqrt{2}$  is irrational, given in Example 10 in Section 1.7. It is a proof by contradiction. Suppose that  $2^{1/3}$  (or  $\sqrt[3]{2}$ , which is the same thing) is the rational number  $p/q$ , where  $p$  and  $q$  are positive integers with no common factors (the fraction is in lowest terms). Cubing, we see that  $2 = p^3/q^3$ , or, equivalently,  $p^3 = 2q^3$ . Thus  $p^3$  is even. Since the product of odd numbers is odd, this means that  $p$  is even, so we can write  $p = 2s$ . Substituting into the equation  $p^3 = 2q^3$ , we obtain  $8s^3 = 2q^3$ , which simplifies to  $4s^3 = q^3$ .

#### Section 1.8 Proof Methods and Strategy 33

Now we play the same game with  $q$ . Since  $q^3$  is even,  $q$  must be even. We have now concluded that  $p$  and  $q$  are both even, that is, that 2 is a common divisor of  $p$  and  $q$ . This contradicts the choice of  $p/q$  to be in lowest terms. Therefore our original assumption—that  $\sqrt[3]{2}$  is rational—is in error, so we have proved that  $\sqrt[3]{2}$  is irrational.

36. The average of two different numbers is certainly always between the two numbers. Furthermore, the average  $a$  of rational number  $x$  and irrational number  $y$  must be irrational, because the equation  $a = (x + y)/2$  leads to  $y = 2a - x$ , which would be rational if  $a$  were rational.
38. The solution is not unique, but here is one way to measure out four gallons. Fill the 5-gallon jug from the 8-gallon jug, leaving the contents (3, 5, 0), where we are using the ordered triple to record the amount of water in the 8-gallon jug, the 5-gallon jug, and the 3-gallon jug, respectively. Next fill the 3-gallon jug from the 5-gallon jug, leaving (3, 2, 3). Pour the contents of the 3-gallon jug back into the 8-gallon jug, leaving (6, 2, 0). Empty the 5-gallon jug's contents into the 3-gallon jug, leaving (6, 0, 2), and then fill the 5-gallon jug from the 8-gallon jug, producing (1, 5, 2). Finally, top off the 3-gallon jug from the 5-gallon jug, and we'll

have (1, 4, 3), with four gallons in the 5-gallon jug.

40. a)  $16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

b)  $11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

c)  $35 \rightarrow 106 \rightarrow 53 \rightarrow 160 \rightarrow 80 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

d)  $113 \rightarrow 340 \rightarrow 170 \rightarrow 85 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

42. This is easily done, by laying the dominoes horizontally, three in the first and last rows and four in each of the other six rows.

44. Without loss of generality, we number the squares from 1 to 25, starting in the top row and proceeding left to right in each row; and we assume that squares 5 (upper right corner), 21 (lower left corner), and 25 (lower right corner) are the missing ones. We argue that there is no way to cover the remaining squares with dominoes.

By symmetry we can assume that there is a domino placed in 1-2 (using the obvious notation). If square 3 is covered by 3-8, then the following dominoes are forced in turn: 4-9, 10-15, 19-20, 23-24, 17-22, and 13-18, and now no domino can cover square 14. Therefore we must use 3-4 along with 1-2. If we use all of 17-22, 18-23, and 19-24, then we are again quickly forced into a sequence of placements that lead to a contradiction. Therefore without loss of generality, we can assume that we use 22-23, which then forces 19-24, 15-20, 9-10, 13-14, 7-8, 6-11, and 12-17, and we are stuck once again. This completes the proof by contradiction that no placement is possible.

46. The barriers shown in the diagram split the board into one continuous closed path of 64 squares, each adjacent to the next (for example, start at the upper left corner, go all the way to the right, then all the way down, then all the way to the left, and then weave your way back up to the starting point). Because each square in the path is adjacent to its neighbors, the colors alternate. Therefore, if we remove one black square and one white square, this closed path decomposes into two paths, each of which starts in one color and ends in the other color (and therefore has even length). Clearly each such path can be covered by dominoes by starting at one end. This completes the proof.

48. If we study Figure 7, we see that by rotating or reflecting the board, we can make any square we wish nonwhite, with the exception of the squares with coordinates (3, 3), (3, 6), (6, 3), and (6, 6). Therefore the same argument as was used in Example 22 shows that we cannot tile the board using straight triominoes if

34 Chapter 1 The Foundations: Logic and Proofs

any one of those other 60 squares is removed. The following drawing (rotated as necessary) shows that we can tile the board using straight triominoes if one of those four squares is removed.

50. We will use a coloring of the  $10 \times 10$  board with four colors as the basis for a proof by contradiction showing that no such tiling exists. Assume that 25 straight tetrominoes can cover the board. Some will be placed horizontally and some vertically. Because there is an odd number of tiles, the number placed horizontally and the number placed vertically cannot both be odd, so assume without loss of generality that an even number of tiles are placed horizontally. Color the squares in order using the colors red, blue, green, yellow in that order repeatedly, starting in the upper left corner and proceeding row by row, from left to right in each row. Then it is clear that every horizontally placed tile covers one square of each color and each vertically placed tile covers either zero or two squares of each color. It follows that in this tiling an even number of squares of each color are covered. But this contradicts the fact that there are 25 squares of each color. Therefore no such coloring exists.

## SUPPLEMENTARY EXERCISES FOR CHAPTER 1

2. The truth table is as follows.

| $p$ | $q$ | $r$ | $p \vee q$ | $p \wedge q$ | $\neg r$ | $(p \vee q) \rightarrow (p \wedge \neg r)$ |
|-----|-----|-----|------------|--------------|----------|--------------------------------------------|
| T   | T   | T   | T          | T            | F        | F                                          |
| T   | T   | F   | T          | T            | T        | T                                          |
| T   | F   | T   | T          | F            | F        | F                                          |
| T   | F   | F   | T          | F            | T        | T                                          |
| F   | T   | T   | T          | F            | F        | F                                          |
| F   | T   | F   | T          | F            | T        | T                                          |
| F   | F   | T   | F          | F            | F        | T                                          |
| F   | F   | F   | F          | F            | T        | T                                          |

4. a) The converse is "If I drive to work today, then it will rain." The contrapositive is "If I do not drive to work today, then it will not rain." The inverse is "If it does not rain today, then I will not drive to work." b) The converse is "If  $x \geq 0$  then  $|x| = x$ ." The contrapositive is "If  $x < 0$  then  $|x| \neq x$ ." The inverse is "If  $|x| \neq x$ , then  $x < 0$ ." c) The converse is "If  $n^2$  is greater than 9, then  $n$  is greater than 3." The contrapositive is "If  $n^2$  is not greater than 9, then  $n$  is not greater than 3." The inverse is "If  $n$  is not greater than 3, then  $n^2$  is not greater than 9."
6. The inverse of  $p \rightarrow q$  is  $\neg p \rightarrow \neg q$ . Therefore the inverse of the inverse is  $\neg\neg p \rightarrow \neg\neg q$ , which is equivalent to  $p \rightarrow q$  (the original proposition). The converse of  $p \rightarrow q$  is  $q \rightarrow p$ . Therefore the inverse of the converse is  $\neg q \rightarrow \neg p$ , which is the contrapositive of the original proposition. The inverse of the contrapositive is  $q \rightarrow p$ , which is the same as the converse of the original statement.

Supplementary Exercises 35

8. Let  $t$  be "Sergei takes the job offer"; let  $b$  be "Sergei gets a signing bonus"; and let  $h$  be "Sergei will receive a higher salary." The given statements are  $t \rightarrow b$ ,  $t \rightarrow h$ ,  $b \rightarrow \neg h$ , and  $t$ . By modus ponens we can conclude  $b$

and  $h$  from the first two conditional statements, and therefore we can conclude  $\neg h$  from the third conditional statement. We now have the contradiction  $h \wedge \neg h$ , so these statements are inconsistent.

10. We make a table of the eight possibilities for  $p$ ,  $q$ , and  $r$ , showing the truth values of the three propositions.

$$p \quad q \quad r \quad p \rightarrow q \quad \neg(p \vee r) \vee q \quad q$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| T | T | T | T | T | T |
| T | T | F | T | T | T |
| T | F | T | F | F | T |
| T | F | F | F | F | T |
| F | T | T | T | T | F |
| F | T | F | T | T | F |
| F | F | T | T | F | F |
| F | F | F | T | T | F |

If we look at the first row of the table, we see that if the student accepts all three propositions, then the resulting commitments are consistent, because the propositions are all true in this case in which  $p$ ,  $q$ , and  $r$  are all true. Similarly, looking at the sixth row of the table, where  $p$  and  $r$  are false but  $q$  is true, we see that a student who accepts the first two propositions and rejects the third also wins. Scanning the entire table, we see that the winning answers are accept-accept-accept, reject-reject-accept, accept-accept-reject, and accept-reject-reject.

12. As we saw from the examples in the previous exercises, one winning strategy is just to assume that all the variables are true and answer “accept” or “reject” according to whether the given proposition is true or false.

14. A knight would never claim that she is a knave, so we know that Anita is a knave. Because she is lying and the first part of her conjunction is true, it must be the second part that is false, and so Bohan must be a knave. If Carmen were a knight, then Bohan’s statement would be true; because Bohan is a knave, we know that that cannot be, so we conclude that Carmen is also a knave.

16. If  $S$  is a proposition, then it is either true or false. If  $S$  is false, then the statement “If  $S$  is true, then unicorns live” is vacuously true; but this statement is  $S$ , so we would have a contradiction. Therefore  $S$  is true, so the statement “If  $S$  is true, then unicorns live” is true and has a true hypothesis. Hence it has a true conclusion (modus ponens), and so unicorns live. But we know that unicorns do not live. It follows that  $S$  cannot be a proposition.

18. From the given information we know that  $p_1, p_3, p_5, \dots$  are true and  $p_2, p_4, p_6, \dots$  are false. Therefore  $p_i$

$\bigwedge_{i=1}^{100} p_{i+1}$  is always false, and so the disjunction  $\bigvee_{i=1}^{100} p_{i+1}$  is always true, and so the conjunction  $\bigwedge_{i=1}^{100} (p_i \wedge p_{i+1})$  is also false. On the other hand,  $\bigvee_{i=1}^{100} (p_i \vee p_{i+1})$  is also true.

20. a) The answer is  $\exists x P(x)$  if we do not read any significance into the use of the plural, and  $\exists x \exists y (P(x) \wedge P(y) \wedge x \neq y)$  if we do.  
 b)  $\neg \forall x P(x)$ , or, equivalently,  $\exists x \neg P(x)$  c)  $\forall y Q(y)$   
 d)  $\forall x P(x)$  (the class has nothing to do with it) e)  $\exists y \neg Q(y)$

22. The given statement tells us that there are exactly two elements in the domain. Therefore the statement will be true as long as we choose the domain to be anything with size 2, such as the United States presidents named Bush.

24. We want to say that for every  $y$ , there do not exist four different people each of whom is the grandmother of  $y$ . Thus we have  $\forall x \neg \exists a \exists b \exists c \exists d (a \neq b \wedge a \neq c \wedge a \neq d \wedge b \neq c \wedge b \neq d \wedge c \neq d \wedge G(a, y) \wedge G(b, y) \wedge G(c, y) \wedge G(d, y))$ .
26. a) Since there is no real number whose square is  $-1$ , it is true that there exist exactly 0 values of  $x$  such that  $x^2 = -1$ .  
 b) This is true, because 0 is the one and only value of  $x$  such that  $|x| = 0$ .  
 c) This is true, because  $\sqrt{2}$  and  $-\sqrt{2}$  are the only values of  $x$  such that  $x^2 = 2$ .  
 d) This is false, because there are more than three values of  $x$  such that  $x = |x|$ , namely all positive real numbers.
28. Let us assume the hypothesis. This means that there is some  $x_0$  such that  $P(x_0, y)$  holds for all  $y$ . Then it is certainly true that for all  $y$  there exists an  $x$  such that  $P(x, y)$  is true, since in each case we can take  $x = x_0$ . Note that the converse is not always a tautology, since the  $x$  in  $\forall y \exists x P(x, y)$  can depend on  $y$ .
30. No. Here is an example. Let  $P(x, y)$  be  $x > y$ , where we are talking about integers. Then for every  $y$  there does exist an  $x$  such that  $x > y$ ; we could take  $x = y + 1$ , for example. However, there does not exist an  $x$  such that for every  $y$ ,  $x > y$ ; in other words, there is no superlarge integer (if for no other reason than that no integer can be larger than itself).
32. a) It will snow today, but I will not go skiing tomorrow.  
 b) Some person in this class does not understand mathematical induction.  
 c) All students in this class like discrete mathematics.  
 d) There is some mathematics class in which all the students stay awake during lectures.
34. Let  $W(r)$  means that room  $r$  is painted white. Let  $I(r, b)$  mean that room  $r$  is in building  $b$ . Let  $L(b, u)$  mean that building  $b$  is on the campus of United States university  $u$ . Then the statement is that there is some university  $u$  and some building on the campus of  $u$  such that every room in  $b$  is painted white. In symbols this is  $\exists u \exists b (L(b, u) \wedge \forall r (I(r, b) \rightarrow W(r)))$ .
36. To say that there are exactly two elements that make the statement true is to say that two elements exist that make the statement true, and that every element that makes the statement true is one of these two elements. More compactly, we can phrase the last part by saying that an element makes the statement true if and only if it is one of these two elements. In symbols this is  $\exists x \exists y (x \neq y \wedge \forall z (P(z) \leftrightarrow (z = x \vee z = y)))$ . In English we might express the rule as follows. The hypotheses are that  $P(x)$  and  $P(y)$  are both true, that  $x \neq y$ , and that every  $z$  that satisfies  $P(z)$  must be either  $x$  or  $y$ . The conclusion is that there are exactly two elements that make  $P$  true.
38. We give a proof by contraposition. If  $x$  is rational, then  $x = p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Then  $x^3 = p^3/q^3$ , and we have expressed  $x^3$  as the quotient of two integers, the second of which is not zero. This by definition means that  $x^3$  is rational, and that completes the proof of the contrapositive of the original statement.
40. Let  $m$  be the square root of  $n$ , rounded down if it is not a whole number. (In the notation to be introduced in Section 2.3, we are letting  $m = \lfloor \sqrt{n} \rfloor$ .) We can see that this is the unique solution in a couple of ways. First, clearly the different choices of  $m$  correspond to a partition of  $N$ , namely into  $\{0\}$ ,  $\{1, 2, 3\}$ ,  $\{4, 5, 6, 7, 8\}$ ,  $\{9,$

10, 11, 12, 13, 14, 15}, . . . . So every  $n$  is in exactly one of these sets. Alternatively, take the square root of the given inequalities to give  $m \leq \sqrt{n} < m + 1$ . That  $m$  is then the floor of  $\sqrt{n}$  (and that  $m$  is unique) follows from statement (1a) of Table 1 in Section 2.3.

#### Supplementary Exercises 37

42. A constructive proof seems indicated. We can look for examples by hand or with a computer program. The smallest ones to be found are  $50 = 5^2 + 5^2 = 1^2 + 7^2$  and  $65 = 4^2 + 7^2 = 1^2 + 8^2$ .
44. We claim that the number 7 is not the sum of at most two squares and a cube. The first two positive squares are 1 and 4, and the first positive cube is 1, and these are the only numbers that could be used in forming the sum. Clearly no sum of three or fewer of these is 7. This counterexample disproves the statement.
46. We give a proof by contradiction. If  $\sqrt{2} + \sqrt{3}$  were rational, then so would be its square, which is  $5 + 2\sqrt{6}$ . Subtracting 5 and dividing by 2 then shows that  $\sqrt{6}$  is rational, but this contradicts the theorem we are told to assume.

#### 38 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

## CHAPTER 2

### Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

#### SECTION 2.1 Sets

2. There are of course an infinite number of correct answers.
- $\{3n \mid n = 0, 1, 2, 3, 4\}$  or  $\{x \mid x \text{ is a multiple of } 3 \wedge 0 \leq x \leq 12\}$ .
  - $\{x \mid -3 \leq x \leq 3\}$ , where we are assuming that the domain (universe of discourse) is the set of integers.
  - $\{x \mid x \text{ is a letter of the word monopoly other than l or y}\}$ .
4. Recall that one set is a subset of another set if every element of the first set is also an element of the second.
- The second condition imposes an extra requirement, so clearly the second set is a subset of the first, but not vice versa.
  - Again the second condition imposes an extra requirement, so the second set is a subset of the first, but not vice versa.
  - There could well be students studying discrete mathematics but not data structures (for example, pure math majors) and students studying data structure but not discrete mathematics (at least not this semester— one could argue that the knowing the latter is necessary to really understand the former!), so neither set is a subset of the other.
6. Each of the sets is a subset of itself. Aside from that, the only relations are  $B \subseteq A$ ,  $C \subseteq A$ , and  $C \subseteq D$ .
8. a) Since the set contains only integers and  $\{2\}$  is a set, not an integer,  $\{2\}$  is not an element. b) Since the set contains only integers and  $\{2\}$  is a set, not an integer,  $\{2\}$  is not an element. c) The set has two elements. One of them is patently  $\{2\}$ .

- d) The set has two elements. One of them is patently  $\{2\}$ .
- e) The set has two elements. One of them is patently  $\{2\}$ .
- f) The set has only one element,  $\{\{2\}\}$ ; since this is not the same as  $\{2\}$  (the former is a set containing a set, whereas the latter is a set containing a number),  $\{2\}$  is not an element of  $\{\{\{2\}\}\}$ .

10. a) true b) true c) false—see part (a) d) true

- e) true—the one element in the set on the left is an element of the set on the right, and the sets are not equal
- f) true—similar to part (e) g) false—the two sets are equal

12. The numbers 1, 3, 5, 7, and 9 form a subset of the set of all ten positive integers under discussion, as shown here.

Section 2.1 Sets 39 14. We put the subsets inside the supersets. Thus the answer is as shown.

16. We allow B and C to overlap, because we are told nothing about their relationship. The set A must be a subset of each of them, and that forces it to be positioned as shown. We cannot actually show the properness of the subset relationships in the diagram, because we don't know where the elements in B and C that are not in A are located—there might be only one (which is in both B and C), or they might be located in portions of B and/or C outside the other. Thus the answer is as shown, but with the added condition that there must be at least one element of B not in A and one element of C not in A.

18. Since the empty set is a subset of every set, we just need to take a set B that contains  $\emptyset$  as an element. Thus we can let  $A = \emptyset$  and  $B = \{\emptyset\}$  as the simplest example.

20. The cardinality of a set is the number of elements it has.

- a) The empty set has no elements, so its cardinality is 0.
- b) This set has one element (the empty set), so its cardinality is 1.
- c) This set has two elements, so its cardinality is 2.
- d) This set has three elements, so its cardinality is 3.

22. The union of all the sets in the power set of a set X must be exactly X. In other words, we can recover X from its power set, uniquely. Therefore the answer is yes.

- 24. a) The power set of every set includes at least the empty set, so the power set cannot be empty. Thus  $\emptyset$  is not the power set of any set.
- b) This is the power set of  $\{a\}$ .
- c) This set has three elements. Since 3 is not a power of 2, this set cannot be the power set of any set.
- d) This is the power set of  $\{a, b\}$ .

26. We need to show that every element of  $A \times B$  is also an element of  $C \times D$ . By definition, a typical element of  $A \times B$  is a pair  $(a, b)$  where  $a \in A$  and  $b \in B$ . Because  $A \subseteq C$ , we know that  $a \in C$ ; similarly,  $b \in D$ . Therefore  $(a, b) \in C \times D$ .

28. By definition it is the set of all ordered pairs  $(c, p)$  such that  $c$  is a course and  $p$  is a professor. The elements of this set are the possible teaching assignments for the mathematics department.

30. We can conclude that  $A = \emptyset$  or  $B = \emptyset$ . To prove this, suppose that neither  $A$  nor  $B$  were empty. Then there would be elements  $a \in A$  and  $b \in B$ . This would give at least one element, namely  $(a, b)$ , in  $A \times B$ , so  $A \times B$  would not be the empty set. This contradiction shows that either  $A$  or  $B$  (or both, it goes without saying) is empty.

#### 40 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

32. In each case the answer is a set of 3-tuples.

a)  $\{(a, x, 0), (a, x, 1), (a, y, 0), (a, y, 1), (b, x, 0), (b, x, 1), (b, y, 0), (b, y, 1), (c, x, 0), (c, x, 1), (c, y, 0), (c, y, 1)\}$

b)  $\{(0, x, a), (0, x, b), (0, x, c), (0, y, a), (0, y, b), (0, y, c), (1, x, a), (1, x, b), (1, x, c), (1, y, a), (1, y, b), (1, y, c)\}$

c)  $\{(0, a, x), (0, a, y), (0, b, x), (0, b, y), (0, c, x), (0, c, y), (1, a, x), (1, a, y), (1, b, x), (1, b, y), (1, c, x), (1, c, y)\}$

d)  $\{(x, x, x), (x, x, y), (x, y, x), (x, y, y), (y, x, x), (y, x, y), (y, y, x), (y, y, y)\}$

34. Recall that  $A^3$  consists of all the ordered triples  $(x, y, z)$  of elements of  $A$ .

a)  $\{(a, a, a)\}$  b)  $\{(0, 0, 0), (0, 0, a), (0, a, 0), (0, a, a), (a, 0, 0), (a, 0, a), (a, a, 0), (a, a, a)\}$

36. The set  $A \times B \times C$  consists of ordered triples  $(a, b, c)$  with  $a \in A$ ,  $b \in B$ , and  $c \in C$ . There are  $m$  choices for the first coordinate. For each of these, there are  $n$  choices for the second coordinate, giving us  $mn$  choices for the first two coordinates. For each of these, there are  $p$  choices for the third coordinate, giving us  $mnp$  choices in all. Therefore  $A \times B \times C$  has  $mnp$  elements. This is an application of the product rule (see Chapter 6).

38. Suppose  $A \neq B$  and neither  $A$  nor  $B$  is empty. We must prove that  $A \times B \neq B \times A$ . Since  $A \neq B$ , either we can find an element  $x$  that is in  $A$  but not  $B$ , or vice versa. The two cases are similar, so without loss of generality, let us assume that  $x$  is in  $A$  but not  $B$ . Also, since  $B$  is not empty, there is some element  $y \in B$ . Then  $(x, y)$  is in  $A \times B$  by definition, but it is not in  $B \times A$  since  $x \notin B$ . Therefore  $A \times B \neq B \times A$ .

40. The only difference between  $(A \times B) \times (C \times D)$  and  $A \times (B \times C) \times D$  is parentheses, so for all practical purposes one can think of them as essentially the same thing. By Definition 8, the elements of  $(A \times B) \times (C \times D)$  consist of ordered pairs  $(x, y)$ , where  $x \in A \times B$  and  $y \in C \times D$ , so the typical element of  $(A \times B) \times (C \times D)$  looks like  $((a, b), (c, d))$ . By Definition 9, the elements of  $A \times (B \times C) \times D$  consist of 3-tuples  $(a, x, d)$ , where  $a \in A$ ,  $d \in D$ , and  $x \in B \times C$ , so the typical element of  $A \times (B \times C) \times D$  looks like  $(a, (b, c), d)$ . The structures  $((a, b), (c, d))$  and  $(a, (b, c), d)$  are different, even if they convey exactly the same information (the first is a pair, and the second is a 3-tuple). To be more precise, there is a natural one-to-one correspondence between  $(A \times B) \times (C \times D)$  and  $A \times (B \times C) \times D$  given by  $((a, b), (c, d)) \leftrightarrow (a, (b, c), d)$ .

42. a) There is a real number whose cube is  $-1$ . This is true, since  $x = -1$  is a solution. b) There is an integer such that the number obtained by adding 1 to it is greater than the integer. This is true—in fact, every integer satisfies this statement.

c) For every integer, the number obtained by subtracting 1 is again an integer. This is true.

d) The square of every integer is an integer. This is true.

44. In each case we want the set of all values of  $x$  in the domain (the set of integers) that satisfy the given equation or inequality.

- a) It is exactly the positive integers that satisfy this inequality. Therefore the truth set is  $\{x \in \mathbb{Z} \mid x^3 \geq 1\} = \{x \in \mathbb{Z} \mid x \geq 1\} = \{1, 2, 3, \dots\}$ .
- b) The square roots of 2 are not integers, so the truth set is the empty set,  $\emptyset$ .
- c) Negative integers certainly satisfy this inequality, as do all positive integers greater than 1. However,  $0^2 < 1^2$  and  $1^2 < 1^2$ . Thus the truth set is  $\{x \in \mathbb{Z} \mid x < x^2\} = \{x \in \mathbb{Z} \mid x \neq 0 \wedge x \neq 1\} = \{\dots, -3, -2, -1, 2, 3, \dots\}$ .

46. a) If  $S \in S$ , then by the defining condition for  $S$  we conclude that  $S \notin S$ , a contradiction. b) If  $S \notin S$ , then by the defining condition for  $S$  we conclude that it is not the case that  $S \notin S$  (otherwise  $S$  would be an element of  $S$ ), again a contradiction.

## Section 2.2 Set Operations 41

### SECTION 2.2 Set Operations

2. a)  $A \cap B$  b)  $A \cap B$ , which is the same as  $A - B$  c)  $A \cup B$  d)  $A \cup B$

4. Note that  $A \subseteq B$ .

- a)  $\{a, b, c, d, e, f, g, h\} = B$  b)  $\{a, b, c, d, e\} = A$
- c) There are no elements in  $A$  that are not in  $B$ , so the answer is  $\emptyset$ . d)  $\{f, g, h\}$

6. a)  $A \cup \emptyset = \{x \mid x \in A \vee x \in \emptyset\} = \{x \mid x \in A \vee F\} = \{x \mid x \in A\} = A$

b)  $A \cap U = \{x \mid x \in A \wedge x \in U\} = \{x \mid x \in A \wedge T\} = \{x \mid x \in A\} = A$

8. a)  $A \cup A = \{x \mid x \in A \vee x \in A\} = \{x \mid x \in A\} = A$

b)  $A \cap A = \{x \mid x \in A \wedge x \in A\} = \{x \mid x \in A\} = A$

10. a)  $A - \emptyset = \{x \mid x \in A \wedge x \notin \emptyset\} = \{x \mid x \in A \wedge T\} = \{x \mid x \in A\} = A$

b)  $\emptyset - A = \{x \mid x \in \emptyset \wedge x \notin A\} = \{x \mid F \wedge x \notin A\} = \{x \mid F\} = \emptyset$

12. We will show that these two sets are equal by showing that each is a subset of the other. Suppose  $x \in A \cup (A \cap B)$ . Then  $x \in A$  or  $x \in A \cap B$  by the definition of union. In the former case, we have  $x \in A$ , and in the latter case we have  $x \in A$  and  $x \in B$  by the definition of intersection; thus in any event,  $x \in A$ , so we have proved that the left-hand side is a subset of the right-hand side. Conversely, let  $x \in A$ . Then by the definition of union,  $x \in A \cup (A \cap B)$  as well. Thus we have shown that the right-hand side is a subset of the left-hand side.

14. Since  $A = (A - B) \cup (A \cap B)$ , we conclude that  $A = \{1, 5, 7, 8\} \cup \{3, 6, 9\} = \{1, 3, 5, 6, 7, 8, 9\}$ . Similarly  $B = (B - A) \cup (A \cap B) = \{2, 10\} \cup \{3, 6, 9\} = \{2, 3, 6, 9, 10\}$ .

16. a) If  $x$  is in  $A \cap B$ , then perforce it is in  $A$  (by definition of intersection).

b) If  $x$  is in  $A$ , then perforce it is in  $A \cup B$  (by definition of union).

c) If  $x$  is in  $A - B$ , then perforce it is in  $A$  (by definition of difference).

d) If  $x \in A$  then  $x \notin B - A$ . Therefore there can be no elements in  $A \cap (B - A)$ , so  $A \cap (B - A) = \emptyset$ . e) The left-hand side consists precisely of those things that are either elements of  $A$  or else elements of  $B$  but not  $A$ , in other words, things that are elements of either  $A$  or  $B$  (or, of course, both). This is precisely the definition of the right-hand side.

18. a) Suppose that  $x \in A \cup B$ . Then either  $x \in A$  or  $x \in B$ . In either case, certainly  $x \in A \cup B \cup C$ . This establishes the desired inclusion.
- b) Suppose that  $x \in A \cap B \cap C$ . Then  $x$  is in all three of these sets. In particular, it is in both  $A$  and  $B$  and therefore in  $A \cap B$ , as desired.
- c) Suppose that  $x \in (A - B) - C$ . Then  $x$  is in  $A - B$  but not in  $C$ . Since  $x \in A - B$ , we know that  $x \in A$  (we also know that  $x \notin B$ , but that won't be used here). Since we have established that  $x \in A$  but  $x \notin C$ , we have proved that  $x \in A - C$ .
- d) To show that the set given on the left-hand side is empty, it suffices to assume that  $x$  is some element in that set and derive a contradiction, thereby showing that no such  $x$  exists. So suppose that  $x \in (A - C) \cap (C - B)$ . Then  $x \in A - C$  and  $x \in C - B$ . The first of these statements implies by definition that  $x \notin C$ , while the second implies that  $x \in C$ . This is impossible, so our proof by contradiction is complete.
- e) To establish the equality, we need to prove inclusion in both directions. To prove that  $(B - A) \cup (C - A) \subseteq (B \cup C) - A$ , suppose that  $x \in (B - A) \cup (C - A)$ . Then either  $x \in (B - A)$  or  $x \in (C - A)$ . Without loss of
- 42 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

generality, assume the former (the proof in the latter case is exactly parallel.) Then  $x \in B$  and  $x \notin A$ . From the first of these assertions, it follows that  $x \in B \cup C$ . Thus we can conclude that  $x \in (B \cup C) - A$ , as desired. For the converse, that is, to show that  $(B \cup C) - A \subseteq (B - A) \cup (C - A)$ , suppose that  $x \in (B \cup C) - A$ . This means that  $x \in (B \cup C)$  and  $x \notin A$ . The first of these assertions tells us that either  $x \in B$  or  $x \in C$ . Thus either  $x \in B - A$  or  $x \in C - A$ . In either case,  $x \in (B - A) \cup (C - A)$ . (An alternative proof could be given by using Venn diagrams, showing that both sides represent the same region.)

20. a) It is always the case that  $B \subseteq A \cup B$ , so it remains to show that  $A \cup B \subseteq B$ . But this is clear because if  $x \in A \cup B$ , then either  $x \in A$ , in which case  $x \in B$  (because we are given  $A \subseteq B$ ) or  $x \in B$ ; in either case  $x \in B$ .
- b) It is always the case that  $A \cap B \subseteq A$ , so it remains to show that  $A \subseteq A \cap B$ . But this is clear because if  $x \in A$ , then  $x \in B$  as well (because we are given  $A \subseteq B$ ), so  $x \in A \cap B$ .
22. First we show that every element of the left-hand side must be in the right-hand side as well. If  $x \in A \cap (B \cap C)$ , then  $x$  must be in  $A$  and also in  $B \cap C$ . Hence  $x$  must be in  $A$  and also in  $B$  and in  $C$ . Since  $x$  is in both  $A$  and  $B$ , we conclude that  $x \in A \cap B$ . This, together with the fact that  $x \in C$  tells us that  $x \in (A \cap B) \cap C$ , as desired. The argument in the other direction (if  $x \in (A \cap B) \cap C$  then  $x$  must be in  $A \cap (B \cap C)$ ) is nearly identical.
24. First suppose  $x$  is in the left-hand side. Then  $x$  must be in  $A$  but in neither  $B$  nor  $C$ . Thus  $x \in A - C$ , but  $x \notin B - C$ , so  $x$  is in the right-hand side. Next suppose that  $x$  is in the right-hand side. Thus  $x$  must be in  $A - C$  and not in  $B - C$ . The first of these implies that  $x \in A$  and  $x \notin C$ . But now it must also be the case that  $x \notin B$ , since otherwise we would have  $x \in B - C$ . Thus we have shown that  $x$  is in  $A$  but in neither  $B$  nor  $C$ , which implies that  $x$  is in the left-hand side.

26. The set is shaded in each case.

28. Here is a Venn diagram that can be used for four sets. Notice that sets A and B are not convex in this picture. We have shaded set A. Notice that each of the 16 different combinations are represented by a region.

We can now shade in the appropriate regions for each of the expressions in this exercise.

Section 2.2 Set Operations 43

30. a) We cannot conclude that  $A = B$ . For instance, if A and B are both subsets of C, then this equation will always hold, and A need not equal B.
- b) We cannot conclude that  $A = B$ ; let  $C = \emptyset$ , for example.
- c) By putting the two conditions together, we can now conclude that  $A = B$ . By symmetry, it suffices to prove that  $A \subseteq B$ . Suppose that  $x \in A$ . There are two cases. If  $x \in C$ , then  $x \in A \cap C = B \cap C$ , which forces  $x \in B$ . On the other hand, if  $x \notin C$ , then because  $x \in A \cup C = B \cup C$ , we must have  $x \in B$ .

32. This is the set of elements in exactly one of these sets, namely  $\{2, 5\}$ .

34. The figure is as shown; we shade that portion of  $A$  that is not in  $B$  and that portion of  $B$  that is not in  $A$ .

36. There are precisely two ways that an item can be in either  $A$  or  $B$  but not both. It can be in  $A$  but not  $B$  (which is equivalent to saying that it is in  $A - B$ ), or it can be in  $B$  but not  $A$  (which is equivalent to saying that it is in  $B - A$ ). Thus an element is in  $A \oplus B$  if and only if it is in  $(A - B) \cup (B - A)$ .

38. a) This is clear from the symmetry (between  $A$  and  $B$ ) in the definition of symmetric difference. b) We prove two things. To show that  $A \subseteq (A \oplus B) \oplus B$ , suppose  $x \in A$ . If  $x \in B$ , then  $x \notin A \oplus B$ , so  $x$  is an element of the right-hand side. On the other hand if  $x \notin B$ , then  $x \in A \oplus B$ , so again  $x$  is in the right-hand side. Conversely, suppose  $x$  is an element of the right-hand side. There are two cases. If  $x \notin B$ , then necessarily  $x \in A \oplus B$ , whence  $x \in A$ . If  $x \in B$ , then necessarily  $x \notin A \oplus B$ , and the only way for that to happen (since  $x \in B$ ) is for  $x$  to be in  $A$ .

44 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices 40. This is an identity; each side

consists of those things that are in an odd number of the sets  $A$ ,  $B$ , and  $C$ . 42. This is an identity; each side

consists of those things that are in an odd number of the sets  $A$ ,  $B$ ,  $C$ , and  $D$ .

44. A finite set is a set with  $k$  elements for some natural number  $k$ . Suppose that  $A$  has  $n$  elements and  $B$  has  $m$  elements. Then the number of elements in  $A \cup B$  is at most  $n + m$  (it might be less because  $A \cap B$  might be nonempty). Therefore by definition,  $A \cup B$  is finite.

46. To count the elements of  $A \cup B \cup C$  we proceed as follows. First we count the elements in each of the sets and add. This certainly gives us all the elements in the union, but we have overcounted. Each element in  $A \cap B$ ,  $A \cap C$ , and  $B \cap C$  has been counted twice. Therefore we subtract the cardinalities of these intersections to make up for the overcount. Finally, we have compensated a bit too much, since the elements of  $A \cap B \cap C$  have now been counted three times and subtracted three times. We adjust by adding back the cardinality of  $A \cap B \cap C$ .

48. We note that these sets are increasing, that is,  $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$ . Therefore, the union of any collection of these sets is just the one with the largest subscript, and the intersection is just the one with the smallest subscript.

a)  $A_n = \{\dots, -2, -1, 0, 1, \dots, n\}$  b)  $A_1 = \{\dots, -2, -1, 0, 1\}$

50. a) As  $i$  increases, the sets get smaller:  $\dots \subset A_3 \subset A_2 \subset A_1$ . All the sets are subsets of  $A_1$ , which is the set of positive integers,  $\mathbb{Z}^+$ . It follows that  $\bigcap_{i=1}^{\infty} A_i = \mathbb{Z}^+$ . Every positive integer is excluded from at least one

of the sets (in fact from infinitely many), so  $\bigcap_{i=1}^{\infty} A_i = \emptyset$ .

b) All the sets are subsets of the set of natural numbers  $\mathbb{N}$  (the nonnegative integers). The number 0 is in

each of the sets, and every positive integer is in exactly one of the sets, so  $\bigcup_{i=1}^{\infty} A_i = \mathbb{N}$  and  $\bigcap_{i=1}^{\infty} A_i = \{0\}$ .

c) As  $i$  increases, the sets get larger:  $A_1 \subset A_2 \subset A_3 \subset \dots$ . All the sets are subsets of the set of positive real numbers  $\mathbb{R}^+$ , and every positive real number is included eventually, so  $\bigcup_{i=1}^{\infty} A_i = \mathbb{R}^+$ . Because  $A_1$  is a subset

of each of the others,  $\bigcap_{i=1}^{\infty} A_i = A_1 = (0, 1)$  (the interval of all real numbers between 0 and 1, exclusive). d)

This time, as in part (a), the sets are getting smaller as  $i$  increases:  $\dots \subset A_3 \subset A_2 \subset A_1$ . Because  $A_1$

includes all the others,  $\bigcap_{i=1}^{\infty} A_i = (1, \infty)$  (all real numbers greater than 1). Every number eventually

gets excluded as  $i$  increases, so  $\bigcup_{i=1}^{\infty} A_i = \emptyset$ . Notice that  $\infty$  is not a real number, so we cannot write

$\bigcap_{i=1}^{\infty} A_i = \{\infty\}$ .

$\bigcup_{i=1}^{\infty} A_i = \emptyset$ . Notice that  $\infty$  is not a real number, so we cannot write

52. a) 00 1110 0000 b) 10 1001 0001 c) 01 1100 1110

54. a) No elements are included, so this is the empty set.

b) All elements are included, so this is the universal set.

56. The bit string for the symmetric difference is obtained by taking the bitwise exclusive OR of the two bit strings for the two sets, since we want to include those elements that are in one set or the other but not both.

58. We can take the bitwise OR (for union) or AND (for intersection) of all the bit strings for these sets.

60. The successor set has one more element than the original set, namely the original set itself. Therefore the answer is  $n + 1$ .

## Section 2.3 Functions 45

62. a) If the departments share the equipment, then the maximum number of each type is all that is required, so we want to take the union of the multisets,  $A \cup B$ .

b) Both departments will use the minimum number of each type, so we want to take the intersection of the multisets,  $A \cap B$ .

c) This will be the difference  $B - A$  of the multisets.

d) If no sharing is allowed, then the university needs to purchase a quantity of each type of equipment that is the sum of the quantities used by the departments; this is the sum of the multisets,  $A + B$ .

64. Taking the maximum for each person, we have  $S \cup T = \{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$ .

## SECTION 2.3 Functions

2. a) This is not a function because the rule is not well-defined. We do not know whether  $f(3) = 3$  or  $f(3) = -3$ . For a function, it cannot be both at the same time.
- b) This is a function. For all integers  $n$ ,  $\sqrt{n^2 + 1}$  is a well-defined real number.
- c) This is not a function with domain  $\mathbb{Z}$ , since for  $n = 2$  (and also for  $n = -2$ ) the value of  $f(n)$  is not defined by the given rule. In other words,  $f(2)$  and  $f(-2)$  are not specified since division by 0 makes no sense.
4. a) The domain is the set of nonnegative integers, and the range is the set of digits (0 through 9). b) The domain is the set of positive integers, and the range is the set of integers greater than 1. c) The domain is the set of all bit strings, and the range is the set of nonnegative integers. d) The domain is the set of all bit strings, and the range is the set of nonnegative integers (a bit string can have length 0).
6. a) The domain is  $\mathbb{Z}^+ \times \mathbb{Z}^+$  and the range is  $\mathbb{Z}^+$ .
- b) Since the largest decimal digit of a strictly positive integer cannot be 0, we have domain  $\mathbb{Z}^+$  and range  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ .
- c) The domain is the set of all bit strings. The number of 1's minus number of 0's can be any positive or negative integer or 0, so the range is  $\mathbb{Z}$ .
- d) The domain is given as  $\mathbb{Z}^+$ . Clearly the range is  $\mathbb{Z}^+$  as well.
- e) The domain is the set of bit strings. The range is the set of strings of 1's, i.e.,  $\{\lambda, 1, 11, 111, \dots\}$ , where  $\lambda$  is the empty string (containing no symbols).
8. We simply round up or down in each case.
- a) 1 b) 2 c) -1 d) 0 e) 3 f) -2 g)  $1 \frac{1}{2} + 12 = 1 \frac{3}{2} = 1$
- h)  $30 + 1 + \frac{1}{2} \cdot 4 = 3 \frac{3}{2} = 2$
10. a) This is one-to-one. b) This is not one-to-one, since b is the image of both a and b. c) This is not one-to-one, since d is the image of both a and d.
12. a) This is one-to-one, since if  $n_1 - 1 = n_2 - 1$ , then  $n_1 = n_2$ .
- b) This is not one-to-one, since, for example,  $f(3) = f(-3) = 10$ .
- c) This is one-to-one, since if  $n_1^3 = n_2^3$ , then  $n_1 = n_2$  (take the cube root of each side).
- d) This is not one-to-one, since, for example,  $f(3) = f(4) = 2$ .
- 46 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices
14. a) This is clearly onto, since  $f(0, -n) = n$  for every integer  $n$ .
- b) This is not onto, since, for example, 2 is not in the range. To see this, if  $m^2 - n^2 = (m - n)(m + n) = 2$ , then  $m$  and  $n$  must have same parity (both even or both odd). In either case, both  $m - n$  and  $m + n$  are then even, so this expression is divisible by 4 and hence cannot equal 2.
- c) This is clearly onto, since  $f(0, n - 1) = n$  for every integer  $n$ .
- d) This is onto. To achieve negative values we set  $m = 0$ , and to achieve nonnegative values we set  $n = 0$ .
- e) This is not onto, for the same reason as in part (b). In fact, the range here is clearly a subset of the range in that part.
16. a) This would normally be one-to-one, unless somehow two students in the class had a strange mobile

phone service in which they shared the same phone number.

b) This is surely one-to-one; otherwise the student identification number would not “identify” students very well!

c) This is almost surely not one-to-one; unless the class is very small, it is very likely that two students will receive the same grade.

d) This function will be one-to-one as long as no two students in the class hail from the same town (which is rather unlikely, so the function is probably not one-to-one).

18. Student answers may vary, depending on the choice of codomain.

a) A codomain could be all ten-digit positive integers; the function is not onto because there are many possible phone numbers assigned to people not in the class.

b) Under some student record systems, the student number consists of eight digits, so the codomain could be all natural numbers less than 100,000,000. The class does not have 100,000,000 students in it, so this function is not onto.

c) A codomain might be {A, B, C, D, F} (the answer depends on the grading system used at that school). If there were people at all five performance levels in this class, then the function would be onto. If not (for example, if no one failed the course), then it would not be onto.

d) The codomain could be the set of all cities and towns in the world. The function is clearly not onto. Alternatively, the codomain could be just the set of cities and towns from which the students in that class hail, in which case the function would be onto.

20. a)  $f(n) = n + 17$  b)  $f(n) = 3n/24$

c) We let  $f(n) = n - 1$  for even values of  $n$ , and  $f(n) = n + 1$  for odd values of  $n$ . Thus we have  $f(1) = 2$ ,  $f(2) = 1$ ,  $f(3) = 4$ ,  $f(4) = 3$ , and so on. Note that this is just one function, even though its definition used two formulae, depending on the parity of  $n$ .

d)  $f(n) = 17$

22. If we can find an inverse, the function is a bijection. Otherwise we must explain why the function is not on-to-one or not onto.

a) This is a bijection since the inverse function is  $f^{-1}(x) = (4 - x)/3$ .

b) This is not one-to-one since  $f(17) = f(-17)$ , for instance. It is also not onto, since the range is the interval  $(-\infty, 7]$ . For example, 42548 is not in the range.

c) This function is a bijection, but not from  $\mathbb{R}$  to  $\mathbb{R}$ . To see that the domain and range are not  $\mathbb{R}$ , note that  $x = -2$  is not in the domain, and  $x = 1$  is not in the range. On the other hand,  $f$  is a bijection from  $\mathbb{R} - \{-2\}$  to  $\mathbb{R} - \{1\}$ , since its inverse is  $f^{-1}(x) = (1 - 2x)/(x - 1)$ .

d) It is clear that this continuous function is increasing throughout its entire domain ( $\mathbb{R}$ ) and it takes on both arbitrarily large values and arbitrarily small (large negative) ones. So it is a bijection. Its inverse is clearly  $f^{-1}(x) = \sqrt[5]{x} - 1$ .

$$f^{-1}(x) = \sqrt[5]{x} - 1.$$

#### Section 2.3 Functions 47

24. The key here is that larger denominators make smaller fractions, and smaller denominators make larger fractions. We have two things to prove, since this is an “if and only if” statement. First, suppose that  $f$  is strictly increasing. This means that  $f(x) < f(y)$  whenever  $x < y$ . To show that  $g$  is strictly decreasing, suppose that  $x < y$ . Then  $g(x) = 1/f(x) > 1/f(y) = g(y)$ . Conversely, suppose that  $g$  is strictly decreasing. This means that  $g(x) > g(y)$  whenever  $x < y$ . To show that  $f$  is strictly increasing, suppose that  $x < y$ . Then  $f(x) = 1/g(x) < 1/g(y) = f(y)$ .

26. a) Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be the given function. We are told that  $f(x_1) < f(x_2)$  whenever  $x_1 < x_2$ . We need to show that  $f(x_1) = f(x_2)$  whenever  $x_1 = x_2$ . This follows immediately from the given conditions, because without loss of generality, we may assume that  $x_1 < x_2$ .

b) We need to make the function increasing, but not strictly increasing, so, for example, we could take the trivial function  $f(x) = 17$ . If we want the range to be all of  $\mathbb{R}$ , we could define  $f$  in parts this way:  $f(x) = x$  for  $x < 0$ ;  $f(x) = 0$  for  $0 \leq x \leq 1$ ; and  $f(x) = x - 1$  for  $x > 1$ .

28. For the function to be invertible, it must be a one-to-one correspondence. This means that it has to be one-to-one, which it is, and onto, which it is not, because, its range is the set of positive real numbers, rather than the set of all real numbers. When we restrict the codomain to be the set of positive real numbers, we get an invertible function. In fact, there is a well-known name for the inverse function in this case—the natural logarithm function ( $g(x) = \ln x$ ).

30. In all parts, we simply need to compute the values  $f(-1)$ ,  $f(0)$ ,  $f(2)$ ,  $f(4)$ , and  $f(7)$  and collect the values into a set.

a)  $\{1\}$  (all five values are the same) b)  $\{-1, 1, 5, 8, 15\}$  c)  $\{0, 1, 2\}$  d)  $\{0, 1, 5, 16\}$  32. a) the set of

even integers b) the set of positive even integers c) the set of real numbers

34. To clarify the setting, suppose that  $g : A \rightarrow B$  and  $f : B \rightarrow C$ , so that  $f \circ g : A \rightarrow C$ . We will prove that if  $f \circ g$  is one-to-one, then  $g$  is also one-to-one, so not only is the answer to the question “yes,” but part of the hypothesis is not even needed. Suppose that  $g$  were not one-to-one. By definition this means that there are distinct elements  $a_1$  and  $a_2$  in  $A$  such that  $g(a_1) = g(a_2)$ . Then certainly  $f(g(a_1)) = f(g(a_2))$ , which is the same statement as  $(f \circ g)(a_1) = (f \circ g)(a_2)$ . By definition this means that  $f \circ g$  is not one-to-one, and our proof is complete.

36. We have  $(f \circ g)(x) = f(g(x)) = f(x + 2) = (x + 2)^2 + 1 = x^2 + 4x + 5$ , whereas  $(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = x^2 + 1 + 2 = x^2 + 3$ . Note that they are not equal.

38. Forming the compositions we have  $(f \circ g)(x) = acx + ad + b$  and  $(g \circ f)(x) = cax + cb + d$ . These are equal if and only if  $ad + b = cb + d$ . In other words, equality holds for all 4-tuples  $(a, b, c, d)$  for which  $ad + b = cb + d$ .

40. a) This really has two parts. First suppose that  $b$  is in  $f(S \cup T)$ . Thus  $b = f(a)$  for some  $a \in S \cup T$ . Either  $a \in S$ , in which case  $b \in f(S)$ , or  $a \in T$ , in which case  $b \in f(T)$ . Thus in either case  $b \in f(S) \cup f(T)$ . This shows that  $f(S \cup T) \subseteq f(S) \cup f(T)$ . Conversely, suppose  $b \in f(S) \cup f(T)$ . Then either  $b \in f(S)$  or  $b \in f(T)$ . This means either that  $b = f(a)$  for some  $a \in S$  or that  $b = f(a)$  for some  $a \in T$ . In either case,  $b = f(a)$  for some  $a \in S \cup T$ , so  $b \in f(S \cup T)$ . This shows that  $f(S) \cup f(T) \subseteq f(S \cup T)$ , and our proof is complete.

b) Suppose  $b \in f(S \cap T)$ . Then  $b = f(a)$  for some  $a \in S \cap T$ . This implies that  $a \in S$  and  $a \in T$ , so we have  $b \in f(S)$  and  $b \in f(T)$ . Therefore  $b \in f(S) \cap f(T)$ , as desired.

48 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

42. a) The answer is the set of all solutions to  $x^2 = 1$ , namely  $\{1, -1\}$ .

b) In order for  $x^2$  to be strictly between 0 and 1, we need  $x$  to be either strictly between 0 and 1 or strictly between  $-1$  and 0. Therefore the answer is  $\{x \mid -1 < x < 0 \vee 0 < x < 1\}$ .

c) In order for  $x^2$  to be greater than 4, we need either  $x > 2$  or  $x < -2$ . Therefore the answer is  $\{x \mid x > 2 \vee x < -2\}$ .

44. a) We need to prove two things. First suppose  $x \in f^{-1}(S \cup T)$ . This means that  $f(x) \in S \cup T$ . Therefore either  $f(x) \in S$  or  $f(x) \in T$ . In the first case  $x \in f^{-1}(S)$ , and in the second case  $x \in f^{-1}(T)$ . In either case, then,  $x \in f^{-1}(S) \cup f^{-1}(T)$ . Thus we have shown that  $f^{-1}(S \cup T) \subseteq f^{-1}(S) \cup f^{-1}(T)$ . Conversely, suppose that  $x \in f^{-1}(S) \cup f^{-1}(T)$ . Then either  $x \in f^{-1}(S)$  or  $x \in f^{-1}(T)$ , so either  $f(x) \in S$  or  $f(x) \in T$ . Thus we know that  $f(x) \in S \cup T$ , so by definition  $x \in f^{-1}(S \cup T)$ . This shows that  $f^{-1}(S) \cup f^{-1}(T) \subseteq f^{-1}(S \cup T)$ , as desired.
- b) This is similar to part (a). We have  $x \in f^{-1}(S \cap T)$  if and only if  $f(x) \in S \cap T$ , if and only if  $f(x) \in S$  and  $f(x) \in T$ , if and only if  $x \in f^{-1}(S)$  and  $x \in f^{-1}(T)$ , if and only if  $x \in f^{-1}(S) \cap f^{-1}(T)$ .
46. There are three cases. Define the "fractional part" of  $x$  to be  $f(x) = x - 1x2$ . Clearly  $f(x)$  is always between 0 and 1 (inclusive at 0, exclusive at 1), and  $x = 1x2 + f(x)$ . If  $f(x)$  is less than  $\frac{1}{2}$ , then  $x + \frac{1}{2}$  will have a value slightly less than  $1x2 + 1$ , so when we round down, we get  $1x2$ . In other words, in this case  $1x + \frac{1}{2}2 = 1x2$ , and indeed that is the integer closest to  $x$ . If  $f(x)$  is greater than  $\frac{1}{2}$ , then  $x + \frac{1}{2}$  will have a value slightly greater than  $1x2 + 1$ , so when we round down, we get  $1x2 + 1$ . In other words, in this case  $1x + \frac{1}{2}2 = 1x2 + 1$ , and indeed that is the integer closest to  $x$  in this case. Finally, if the fractional part is exactly  $\frac{1}{2}$ , then  $x$  is midway between two integers, and  $1x + \frac{1}{2}2 = 1x2 + 1$ , which is the larger of these two integers.
48. If  $x$  is not an integer, then  $3x4$  is the integer just larger than  $x$ , and  $1x2$  is the integer just smaller than  $x$ . Clearly they differ by 1. If  $x$  is an integer, then  $3x4 - 1x2 = x - x = 0$ .
50. Write  $x = n - "$ , where  $n$  is an integer and  $0 \leq " < 1$ ; thus  $3x4 = n$ . Then  $3x + m4 = 3n - " + m4 = n + m = 3x4 + m$ . Alternatively, we could proceed along the lines of the proof of property 4a of Table 1, shown in the text.
52. a) The "if" direction is trivial, since  $x \leq 3x4$ . For the other direction, suppose that  $x \leq n$ . Since  $n$  is an integer no smaller than  $x$ , and  $3x4$  is by definition the smallest such integer, clearly  $3x4 \leq n$ . b) The "if" direction is trivial, since  $1x2 \leq x$ . For the other direction, suppose that  $n \leq x$ . Since  $n$  is an integer not exceeding  $x$ , and  $1x2$  is by definition the largest such integer, clearly  $n \leq 1x2$ .
54. To prove the first equality, write  $x = n - "$ , where  $n$  is an integer and  $0 \leq " < 1$ ; thus  $3x4 = n$ . Therefore,  $1-x2 = 1-n + "2 = -n + "2 = -3x4$ . The second equality is proved in the same manner, writing  $x = n + "$ , where  $n$  is an integer and  $0 \leq " < 1$ . This time  $1x2 = n$ , and  $3-x4 = 3-n - "4 = -n - "4 = -1x2$ .
56. In some sense this question is its own answer—the number of integers between  $a$  and  $b$ , inclusive, is the number of integers between  $a$  and  $b$ , inclusive. Presumably we seek an expression involving  $a$ ,  $b$ , and the floor and/or ceiling function to answer this question. If we round  $a$  up and round  $b$  down to integers, then we will be looking at the smallest and largest integers just inside the range of integers we want to count, respectively. These values are of course  $3a4$  and  $1b2$ , respectively. Then the answer is  $1b2 - 3a4 + 1$  (just think of counting all the integers between these two values, including both ends—if a row of fenceposts one foot apart extends for  $k$  feet, then there are  $k + 1$  fenceposts). Note that this even works when, for example,  $a = 0.3$  and  $b = 0.7$ .

## Section 2.3 Functions 49

58. Since a byte is eight bits, all we are asking for in each case is  $3n/84$ , where  $n$  is the number of bits.
- a)  $34/84 = 1$  b)  $310/84 = 2$  c)  $3500/84 = 63$  d)  $33000/84 = 375$
60. From Example 28 we know that one ATM cell is 53 bytes, or  $53 \cdot 8 = 424$  bits long. Thus in each case we

need to divide the number of bits transmitted in 10 seconds by 424 and round down.

a) In 10 seconds, this link can transmit  $128,000 \cdot 10 = 1,280,000$  bits. Therefore the answer is  $11,280,000/4242 = 3018$ .

b) In 10 seconds, this link can transmit  $300,000 \cdot 10 = 3,000,000$  bits. So the answer is  $13,000,000/4242 = 7075$ . c) In 10 seconds, this link can transmit  $1,000,000 \cdot 10 = 10,000,000$  bits. So the answer is  $110,000,000/4242 = 23,584$ .

62. The graph consists of the points  $(n, 1 - n^2)$  for all  $n \in \mathbb{Z}$ . The picture shows part of the graph on the usual coordinate axes.

64. The graph is similar to the graph of  $f(x) = 1x^2$ ; the only difference is a change in the scale of the x-axis.

66. The function values for this step function change only at integer values of  $x$ , and different things happen for odd  $x$  and for even  $x$  because of the  $x/2$  term. Whatever jump pattern is established on the closed interval  $[0, 2]$  must repeat indefinitely in both directions. A thoughtful analysis then yields the following graph.

68. a) We can rewrite this as  $f(x) = 33(x - \frac{2}{3})^4$ . The graph will therefore look exactly like the graph of the function  $f(x) = 33x^4$ , except that the picture will be shifted to the right by  $\frac{2}{3}$  unit, since  $x$  has been replaced by  $x - \frac{2}{3}$ . The graph of  $f(x) = 33x^4$  is just like the graph shown in Figure 10b, except that the x-axis needs to be rescaled by a factor of 3 (the first jump on the positive x-axis occurs at  $x = \frac{1}{3}$  here). Putting this all together yields the following picture. (Alternatively, we can think of this as the graph of  $f(x) = 33x^4$  shifted down 2 units, since  $33x - 24 = 33x^4 - 2$ .)

b) The graph will look exactly like the graph shown in Figure 10b, except that the x-axis needs to be rescaled by a factor of 5 (the first jump on the positive x-axis occurs at  $x = 5$  here).

c) Since  $1 - 1/x^2 = -31/x^4$  (see Exercise 54), the picture is just the picture for Exercise 67d flipped upside down.

d) The basic shape is the parabola,  $y = x^2$ . However, because of the greatest integer function, the curve is broken into steps, with jumps at  $x = \pm 1, \pm \sqrt{2}, \pm \sqrt{3}, \dots$ . Note the symmetry around the y-axis.

e) The basic shape is the parabola,  $y = x^2/4$ . However, because of the step functions, the curve is broken into steps. For  $x$  an even integer,  $f(x) = x^4/4$ , since the terms inside the floor and ceiling function symbols

are integers. Note how these are isolated point, as in Exercise 67f.  
 Section 2.3 Functions 51

f) When  $x$  is an even integer, this is just  $x$ . When  $x$  is between two even integers, however, this has the value of the odd integer between them. The graph is therefore as shown here.

g) Despite the complicated-looking formula, this is not too hard. Note that the expression inside the outer floor function symbols is always going to be an integer plus  $\frac{1}{2}$ ; therefore we can tell exactly what its rounded-down value will be, namely  $23x/24$ . This is just the graph in Figure 10b, rescaled on both axes.

70. This follows immediately from the definition. We want to show that  $\#(f \circ g) \circ (g^{-1} \circ f^{-1})^{\$}(z) = z$  for all  $z \in Z$  and that  $\#(g^{-1} \circ f^{-1}) \circ (f \circ g)^{\$}(x) = x$  for all  $x \in X$ . For the first we have

$$\begin{aligned} \#(f \circ g) \circ (g^{-1} \circ f^{-1})^{\$}(z) &= (f \circ g)((g^{-1} \circ f^{-1})(z)) \\ &= (f \circ g)(g^{-1}(f^{-1}(z))) \\ &= f(g(g^{-1}(f^{-1}(z)))) \\ &= f(f^{-1}(z)) = z. \end{aligned}$$

The second equality is similar.

72. If  $f$  is one-to-one, then every element of  $A$  gets sent to a different element of  $B$ . If in addition to the range of  $A$  there were another element in  $B$ , then  $|B|$  would be at least one greater than  $|A|$ . This cannot happen, so we conclude that  $f$  is onto. Conversely, suppose that  $f$  is onto, so that every element of  $B$  is the image of some element of  $A$ . In particular, there is an element of  $A$  for each element of  $B$ . If two or more elements of

A were sent to the same element of B, then  $|A|$  would be at least one greater than the  $|B|$ . This cannot happen, so we conclude that  $f$  is one-to-one.

## 52 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

74. a) This is true. Since  $3 \times 4$  is already an integer,  $13 \times 42 = 3 \times 4$ .

b) A little experimentation shows that this is not always true. To disprove it we need only produce a counterexample, such as  $x = y = \frac{3}{4}$ . In this case the left-hand side is  $13/22 = 1$ , while the right-hand side is  $0 + 0 = 0$ .

c) A little trial and error fails to produce a counterexample, so maybe this is true. We look for a proof. Since we are dividing by 4, let us write  $x = 4n + k$ , where  $0 \leq k < 4$ . In other words, write  $x$  in terms of how much it exceeds the largest multiple of 4 not exceeding it. There are three cases. If  $k = 0$ , then  $x$  is already a multiple of 4, so both sides equal  $n$ . If  $0 < k \leq 2$ , then  $3x/24 = 2n + 1$ , so the left-hand side is  $3n + \frac{1}{2} \cdot 4 = n + 1$ . Of course the right-hand side is  $n + 1$  as well, so again the two sides agree. Finally, suppose that  $2 < k < 4$ . Then  $3x/24 = 2n + 2$ , and the left-hand side is  $3n + 14 = n + 1$ ; of course the right-hand side is still  $n + 1$ , as well. Since we proved that the two sides are equal in all cases, the proof is complete.

d) For  $x = 8.5$ , the left-hand side is 3, whereas the right-hand side is 2.

e) This is true. Write  $x = n + \epsilon$  and  $y = m + \delta$ , where  $n$  and  $m$  are integers and  $\epsilon$  and  $\delta$  are nonnegative real numbers less than 1. The left-hand side is  $n + m + (n + m)$  or  $n + m + (n + m + 1)$ , the latter occurring if and only if  $\epsilon + \delta \geq 1$ . The right-hand side is the sum of two quantities. The first is either  $2n$  (if  $\epsilon < \frac{1}{2}$ ) or  $2n + 1$  (if  $\epsilon \geq \frac{1}{2}$ ). The second is either  $2m$  (if  $\delta < \frac{1}{2}$ ) or  $2m + 1$  (if  $\delta \geq \frac{1}{2}$ ). The only way, then, for the left-hand side to exceed the right-hand side is to have the left-hand side be  $2n + 2m + 1$  and the right-hand side be  $2n + 2m$ . This can occur only if  $\epsilon + \delta \geq 1$  while  $\epsilon < \frac{1}{2}$  and  $\delta < \frac{1}{2}$ . But that is an impossibility, since the sum of two numbers less than  $\frac{1}{2}$  cannot be as large as 1. Therefore the right-hand side is always at least as large as the left-hand side.

76. A straightforward way to do this problem is to consider the three cases determined by where in the interval between two consecutive integers the real number  $x$  lies. Certainly every real number  $x$  lies in an interval  $[n, n + 1)$  for some integer  $n$ ; indeed,  $n = \lfloor x \rfloor$ . (Recall that  $[s, t)$  is the notation for the set of real numbers greater than or equal to  $s$  and less than  $t$ .) If  $x \in [n, n + \frac{1}{3})$ , then  $3x$  lies in the interval  $[3n, 3n + 1)$ , so  $13x2 = 3n$ . Moreover in this case  $x + \frac{1}{3}$  is still less than  $n + 1$ , and  $x + \frac{2}{3}$  is still less than  $n + 1$ , so  $1x2 + 1x + \frac{1}{3} \cdot 2 + 1x + \frac{2}{3} \cdot 2 = n + n + n = 3n$  as well. For the second case, we assume that  $x \in [n + \frac{1}{3}, n + \frac{2}{3})$ . This time  $3x \in [3n + 1, 3n + 2)$ , so  $13x2 = 3n + 1$ . Moreover in this case  $x + \frac{1}{3}$  is in  $[n + \frac{2}{3}, n + 1)$ , and  $x + \frac{2}{3}$  is in  $[n + 1, n + \frac{4}{3})$ , so  $1x2 + 1x + \frac{1}{3} \cdot 2 + 1x + \frac{2}{3} \cdot 2 = n + n + (n + 1) = 3n + 1$  as well. The third case,  $x \in [n + \frac{2}{3}, n + 1)$ , is similar, with both sides equaling  $3n + 2$ .

78. a) We merely have to remark that  $f^*$  is well-defined by the rule given here. For each  $a \in A$ , either  $a$  is in the domain of definition of  $f$  or it is not. If it is, then  $f^*(a)$  is the well-defined element  $f(a) \in B$ , and otherwise  $f^*(a) = u$ . In either case  $f^*(a)$  is a well-defined element of  $B \cup \{u\}$ .

b) We simply need to set  $f^*(a) = u$  for each  $a$  not in the domain of definition of  $f$ . In part (a), then,  $f^*(n) = 1/n$  for  $n \neq 0$ , and  $f^*(0) = u$ . In part (b) we have a total function already, so  $f^*(n) = 3n/24$  for all  $n \in \mathbb{Z}$ . In part (c)  $f^*(m, n) = m/n$  if  $n \neq 0$ , and  $f^*(m, 0) = u$  for all  $m \in \mathbb{Z}$ . In part (d) we have a total function already, so  $f^*(m, n) = mn$  for all values of  $m$  and  $n$ . In part (e) the rule only applies if  $m > n$ , so  $f^*(m, n) = m - n$  if  $m > n$ , and  $f^*(m, n) = u$  if  $m \leq n$ .

80. For the "if" direction, we simply need to note that if  $S$  is a finite set, with cardinality  $m$ , then every proper subset of  $S$  has cardinality strictly smaller than  $m$ , so there is no possible one-to-one correspondence between the elements of  $S$  and the elements of the proper subset. (This is essentially the pigeonhole principle, to be discussed in Section 6.2.)

The "only if" direction is much deeper. Let  $S$  be the given infinite set. Clearly  $S$  is not empty, because

by definition, the empty set has cardinality 0, a nonnegative integer. Let  $a_0$  be one element of  $S$ , and let  $A = S - \{a_0\}$ . Clearly  $A$  is also infinite (because if it were finite, then we would have  $|S| = |A| + 1$ , making  $S$  finite). We will now construct a one-to-one correspondence between  $S$  and  $A$ ; think of this as a one-to-one and onto function  $f$  from  $S$  to  $A$ . (This construction is an infinite process; technically we are using something called the Axiom of Choice.) In order to define  $f(a_0)$ , we choose an arbitrary element  $a_1$  in  $A$  (which is possible because  $A$  is infinite) and set  $f(a_0) = a_1$ . Next we define  $f$  at  $a_1$ . To do so, we choose an arbitrary element  $a_2$  in  $A - \{a_1\}$  (which is possible because  $A - \{a_1\}$  is necessarily infinite) and set  $f(a_1) = a_2$ . Next we define  $f$  at  $a_2$ . To do so, we choose an arbitrary element  $a_3$  in  $A - \{a_1, a_2\}$  (which is possible because  $A - \{a_1, a_2\}$  is necessarily infinite) and set  $f(a_2) = a_3$ . We continue this process forever. Finally, we let  $f$  be the identity function on  $S - \{a_0, a_1, a_2, \dots\}$ . The function thus defined has  $f(a_i) = a_{i+1}$  for all natural numbers  $i$  and  $f(x) = x$  for all  $x \in S - \{a_0, a_1, a_2, \dots\}$ . Our construction forced  $f$  to be one-to-one and onto.

## SECTION 2.4 Sequences and Summations

2. In each case we just plug  $n = 8$  into the formula.

a)  $2^{8-1} = 128$  b)  $7$  c)  $1 + (-1)^8 = 0$  d)  $-(-2)^8 = -256$

4. a)  $a_0 = (-2)^0 = 1$ ,  $a_1 = (-2)^1 = -2$ ,  $a_2 = (-2)^2 = 4$ ,  $a_3 = (-2)^3 = -8$

b)  $a_0 = a_1 = a_2 = a_3 = 3$

c)  $a_0 = 7 + 4^0 = 8$ ,  $a_1 = 7 + 4^1 = 11$ ,  $a_2 = 7 + 4^2 = 23$ ,  $a_3 = 7 + 4^3 = 71$

d)  $a_0 = 2^0 + (-2)^0 = 2$ ,  $a_1 = 2^1 + (-2)^1 = 0$ ,  $a_2 = 2^2 + (-2)^2 = 8$ ,  $a_3 = 2^3 + (-2)^3 = 0$

6. These are easy to compute by hand, calculator, or computer.

a)  $10, 7, 4, 1, -2, -5, -8, -11, -14, -17$

b) We can use the formula in Table 2, or we can just keep adding to the previous term ( $1 + 2 = 3$ ,  $3 + 3 = 6$ ,  $6 + 4 = 10$ , and so on):  $1, 3, 6, 10, 15, 21, 28, 36, 45, 55$ . These are called the triangular numbers. c)  $1, 5, 19, 65, 211, 665, 2059, 6305, 19171, 58025$

d)  $1, 1, 1, 2, 2, 2, 2, 2, 3, 3$  (there will be  $2k + 1$  copies of  $k$ ) e)  $1, 5, 6, 11, 17, 28, 45, 73, 118, 191$  f) The largest number whose binary expansion has  $n$  bits is  $(11 \dots 1)_2$ , which is  $2^n - 1$ . So the sequence is  $1, 3, 7, 15, 31, 63, 127, 255, 511, 1023$ .

g)  $1, 2, 2, 4, 8, 11, 33, 37, 148, 153$  h)  $1, 2, 2, 2, 2, 3, 3, 3, 3, 3$

8. One rule could be that each term is 2 greater than the previous term; the sequence would be  $3, 5, 7, 9, 11, 13, \dots$ . Another rule could be that the  $n^{\text{th}}$  term is the  $n^{\text{th}}$  odd prime; the sequence would be  $3, 5, 7, 11, 13, 17, \dots$ . Actually, we could choose any number we want for the fourth term (say 12) and find a third degree polynomial whose value at  $n$  would be the  $n^{\text{th}}$  term; in this case we need to solve for  $A, B, C$ , and  $D$  in the equations  $y = Ax^3 + Bx^2 + Cx + D$  where  $(1, 3), (2, 5), (3, 7), (4, 12)$  have been plugged in for  $x$  and  $y$ . Doing so yields  $(x^3 - 6x^2 + 15x - 4)/2$ . With this formula, the sequence is  $3, 5, 7, 12, 23, 43, 75, 122, 187, 273$ . Obviously many other answers are possible.

10. In each case we simply plug  $n = 0, 1, 2, 3, 4, 5$ , using the initial conditions for the first few and then the recurrence relation.

a)  $a_0 = -1$ ,  $a_1 = -2a_0 = 2$ ,  $a_2 = -2a_1 = -4$ ,  $a_3 = -2a_2 = 8$ ,  $a_4 = -2a_3 = -16$ ,  $a_5 = -2a_4 = 32$  b)  $a_0 = 2$ ,  $a_1 = -1$ ,  $a_2 = a_1 - a_0 = -3$ ,  $a_3 = a_2 - a_1 = -2$ ,  $a_4 = a_3 - a_2 = 1$ ,  $a_5 = a_4 - a_3 = 3$  c)  $a_0 = 1$ ,  $a_1 = 3a_0^2 = 3$ ,  $a_2 = 3a_1^2 = 27 = 3^3$ ,  $a_3 = 3a_2^2 = 2187 = 3^7$ ,  $a_4 = 3a_3^2 = 14348907 = 3^{15}$ ,  $a_5 = 3a_4^2 = 617673396283947 = 3^{31}$

d)  $a_0 = -1$ ,  $a_1 = 0$ ,  $a_2 = 2a_1 + a_0^2 = 1$ ,  $a_3 = 3a_2 + a_1^2 = 3$ ,  $a_4 = 4a_3 + a_2^2 = 13$ ,  $a_5 = 5a_4 + a_3^2 = 74$  e)  $a_0 = 1$ ,  $a_1 = 1$ ,  $a_2 = 2$ ,  $a_3 = a_2 - a_1 + a_0 = 2$ ,  $a_4 = a_3 - a_2 + a_1 = 1$ ,  $a_5 = a_4 - a_3 + a_2 = 1$

12. a)  $-3a_{n-1} + 4a_{n-2} = -3 \cdot 0 + 4 \cdot 0 = 0 = a_n$  b)  $-3a_{n-1} + 4a_{n-2} = -3 \cdot 1 + 4 \cdot 1 = 1 = a_n$  c)  $-3a_{n-1} + 4a_{n-2} = -3$

$$\cdot (-4)^{n-1} + 4 \cdot (-4)^{n-2} = (-4)^{n-2} \cdot (-3)(-4) + 4 = (-4)^{n-2} \cdot 16 = (-4)^{n-2}(-4)^2 = (-4)^n = a_n$$

d)  $-3a_{n-1} + 4a_{n-2} = -3 \cdot 2(-4)^{n-1} + 3 + 4 \cdot 2(-4)^{n-2} + 3 = (-4)^{n-2} \cdot (-6)(-4) + 4 \cdot 2 \cdot (-9 + 12) = (-4)^{n-2} \cdot 32 + 3 = (-4)^{n-2}(-4)^2 \cdot 2 + 3 = 2 \cdot (-4)^n + 3 = a_n$

14. In each case, one possible answer is just the equation as presented (it is a recurrence relation of degree 0). We will give an alternate answer.

a) One possible answer is  $a_n = a_{n-1}$ .

b) Note that  $a_n - a_{n-1} = 2n - (2n - 2) = 2$ . Therefore we have  $a_n = a_{n-1} + 2$  as one possible answer. c)

Just as in part (b), we have  $a_n = a_{n-1} + 2$ .

d) Probably the simplest answer is  $a_n = 5a_{n-1}$ .

e) Since  $a_n - a_{n-1} = n^2 - (n-1)^2 = 2n - 1$ , we have  $a_n = a_{n-1} + 2n - 1$ .

f) This is similar to part (e). One answer is  $a_n = a_{n-1} + 2n$ .

g) Note that  $a_n - a_{n-1} = n + (-1)^n - (n-1) - (-1)^{n-1} = 1 + 2(-1)^n$ . Thus we have  $a_n = a_{n-1} + 1 + 2(-1)^n$ . h)  $a_n = na_{n-1}$

16. In the iterative approach, we write  $a_n$  in terms of  $a_{n-1}$ , then write  $a_{n-1}$  in terms of  $a_{n-2}$  (using the recurrence relation with  $n-1$  plugged in for  $n$ ), and so on. When we reach the end of this procedure, we use the given initial value of  $a_0$ . This will give us an explicit formula for the answer or it will give us a finite series, which we then sum to obtain an explicit formula for the answer.

a)  $a_n = -a_{n-1} = (-1)^2 a_{n-2} = \dots = (-1)^n a_{n-n} = (-1)^n a_0 = 5 \cdot (-1)^n$

b)  $a_n = 3 + a_{n-1} = 3 + 3 + a_{n-2} = 2 \cdot 3 + a_{n-2} = 3 \cdot 3 + a_{n-3} = \dots = n \cdot 3 + a_{n-n} = n \cdot 3 + a_0 = 3n + 1$  c)  $a_n = -n + a_{n-1}$

$$\begin{aligned} &= -n + \overset{\#}{-(n-1)} + \overset{\$}{a_{n-2}} = -n + (n-1) + \overset{\#}{a_{n-2}} \\ &= -n + (n-1) + \overset{\$}{-(n-2)} + \overset{\#}{a_{n-3}} = -n + (n-1) + (n-2) + \overset{\$}{a_{n-3}} \\ &\quad \ddots \\ &= -n + (n-1) + (n-2) + \dots + (n-(n-1)) + \overset{\$}{a_{n-n}} \\ &= -n + (n-1) + (n-2) + \dots + 1 + \overset{\$}{a_0} \\ &= 2 + 4 = -n^2 - n + 8 \\ &= -n(n+1) \end{aligned}$$

2

d)  $a_n = -3 + 2a_{n-1}$

$$= -3 + 2(-3 + 2a_{n-2}) = -3 + 2(-3) + 4a_{n-2}$$

$$= -3 + 2(-3) + 4(-3 + 2a_{n-3}) = -3 + 2(-3) + 4(-3) + 8a_{n-3}$$

$$= -3 + 2(-3) + 4(-3) + 8(-3 + 2a_{n-4}) = -3 + 2(-3) + 4(-3) + 8(-3) + 16a_{n-4} \dots$$

$$= -3(1 + 2 + 4 + \dots + 2^{n-1}) + 2^n a_{n-n} = -3(2^n - 1) + 2^n(-1) = -2^{n+2} + 3$$
 e)  $a_n$

$$= (n+1)a_{n-1} = (n+1)na_{n-2}$$

$$= (n+1)n(n-1)a_{n-3} = (n+1)n(n-1)(n-2)a_{n-4}$$

$\ddots$

$$= (n+1)n(n-1)(n-2)(n-3) \dots (n-(n-2)) a_{n-n}$$

$$= (n+1)n(n-1)(n-2)(n-3) \dots 2 \cdot a_0$$

$$= (n+1)! \cdot 2 = 2(n+1)!$$

## Section 2.4 Sequences and Summations 55

f)  $a_n = 2na_{n-1}$

$$\begin{aligned} &= 2n \cdot 2(n-1)a_{n-2} = 2^{n(n-1)} a_{n-2} \\ &= 2^{n(n-1)} \cdot 2(n-2)a_{n-3} = 2^{n(n-1)(n-2)} a_{n-3} \\ &\quad \vdots \\ &= 2^n n(n-1)(n-2)(n-3) \cdots n - (n-1) a_{n-n} \\ &= 2^n n(n-1)(n-2)(n-3) \cdots 1 \cdot a_0 \\ &= 3 \cdot 2^n n! \end{aligned}$$

g)  $a_n = n - 1 - a_{n-1}$

$$\begin{aligned} &= n - 1 - (n - 1 - 1) - a_{n-2} = (n - 1) - (n - 2) + a_{n-2} \\ &= (n - 1) - (n - 2) + (n - 2 - 1) - a_{n-3} = (n - 1) - (n - 2) + (n - 3) - a_{n-3} \\ &\quad \vdots \\ &= (n - 1) - (n - 2) + \cdots + (-1)^{n-1}(n - n) + (-1)^n a_{n-n} \\ &= 2n - 1 + (-1)^n \\ &\quad 4 + (-1)^n \cdot 7 \end{aligned}$$

18. a) The amount after  $n-1$  years is multiplied by 1.09 to give the amount after  $n$  years, since 9% of the value must be added to account for the interest. Thus we have  $a_n = 1.09a_{n-1}$ . The initial condition is  $a_0 = 1000$ . b) Since we multiply by 1.09 for each year, the solution is  $a_n = 1000(1.09)^n$ . c)  $a_{100} = 1000(1.09)^{100} \approx \$5,529,041$

20. This is just like Exercise 18. We are letting  $a_n$  be the population, in billions of people,  $n$  years after 2010. a)  $a_n = 1.011a_{n-1}$ , with  $a_0 = 6.9$  b)  $a_n = 6.9 \cdot (1.011)^n$  c)  $a_{20} = 6.9 \cdot (1.011)^{20} \approx 8.6$  billion people

22. We let  $a_n$  be the salary, in thousands of dollars,  $n$  years after 2009.

a)  $a_n = 1 + 1.05a_{n-1}$ , with  $a_0 = 50$

- b) Here  $n = 8$ . We can either iterate the recurrence relation 8 times, or we can use the result of part (c). The answer turns out to be approximately  $a_8 = 83.4$ , i.e., a salary of approximately \$83,400. c) We use the iterative approach.

$$\begin{aligned} a_n &= 1 + 1.05a_{n-1} \\ &= 1 + 1.05(1 + 1.05a_{n-2}) \\ &= 1 + 1.05 + (1.05)^2 a_{n-2} \\ &\quad \vdots \\ &= 1 + 1.05 + (1.05)^2 + \cdots + (1.05)^{n-1} + (1.05)^n a_0 \\ &= (1.05)^n - 1 \\ &\quad 1.05 - 1 + 50 \cdot (1.05)^n \\ &= 70 \cdot (1.05)^n - 20 \end{aligned}$$

24. a) Each month our account accrues some interest that must be paid. Since the balance the previous month is  $B(k - 1)$ , the amount of interest we owe is  $(r/12)B(k - 1)$ . After paying this interest, the rest of the  $P$  dollar payment we make each month goes toward reducing the principle. Therefore we have  $B(k) =$
- 56 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

$B(k - 1) - (P - (r/12)B(k - 1))$ . This can be simplified to  $B(k) = (1 + (r/12))B(k - 1) - P$ . The initial condition is that  $B(0) =$  the amount borrowed.

b) Solving this by iteration yields

$$B(k) = (1 + (r/12))^k(B(0) - 12P/r) + 12P/r.$$

Setting  $B(k) = 0$  and solving this for  $k$  yields the desired value of  $T$  after some messy algebra, namely

$$T = \frac{\log(-12P/(B(0)r - 12P))}{\log(1 + (r/12))}.$$

26. a) The first term is 3, and the  $n^{\text{th}}$  term is obtained by adding  $2n - 1$  to the previous term. In other words, we successively add 3, then 5, then 7, and so on. Alternatively, we see that the  $n^{\text{th}}$  term is  $n^2 + 2$ ; we can see this by inspection if we happen to notice how close each term is to a perfect square, or we can fit a quadratic polynomial to the data. The next three terms are 123, 146, 171.
- b) This is an arithmetic sequence whose first term is 7 and whose difference is 4. Thus the  $n^{\text{th}}$  term is  $7 + 4(n - 1) = 4n + 3$ . Thus the next three terms are 47, 51, 55.
- c) The  $n^{\text{th}}$  term is clearly the binary expansion of  $n$ . Thus the next three terms are 1100, 1101, 1110. d) The sequence consists of one 1, followed by three 2's, followed by five 3's, followed by seven 5's, and so on, with the number of copies of the next value increasing by 2 each time, and the values themselves following the rule that the first two values are 1 and 2 and each subsequent value is the sum of the previous two values. Obviously other answers are possible as well. By our rule, the next three terms would be 8, 8, 8. e) If we stare at this sequence long enough and compare it with Table 1, then we notice that the  $n^{\text{th}}$  term is  $3^n - 1$ . Thus the next three terms are 59048, 177146, 531440.
- f) We notice that each term evenly divides the next, and the multipliers are successively 3, 5, 7, 9, 11, and so on. That must be the intended pattern. One notation for this is to use  $n!!$  to mean  $n(n - 2)(n - 4) \cdots$ ; thus the  $n^{\text{th}}$  term is  $(2n - 1)!!$ . Thus the next three terms are 654729075, 13749310575, 316234143225. g) The sequence consists of one 1, followed by two 0s, then three 1s, four 0s, five 1s, and so on, alternating between 0s and 1s and having one more item in each group than in the previous group. Thus six 0's will follow next, so the next three terms are 0, 0, 0.
- h) It doesn't take long to notice that each term is the square of its predecessor. The next three terms get very big very fast: 18446744073709551616, 340282366920938463463374607431768211456, and then

1157920892373161954235709850086879078532699846656405640394575840079131

29639936. (These were computed using Maple.)

28. Let us ask ourselves which is the last term in the sequence whose value is  $k$ ? Clearly it is  $1 + 2 + 3 + \cdots + k$ , which equals  $k(k + 1)/2$ . We can rephrase this by saying that  $a_n \leq k$  if and only if  $k(k + 1)/2 \geq n$ . Thus, to find  $k$  as a function of  $n$ , we must find the smallest  $k$  such that  $k(k + 1)/2 \geq n$ . This is equivalent to  $k^2 + k - 2n \geq 0$ . By the quadratic formula, this tells us that  $k$  has to be at least  $(-1 + \sqrt{1 + 8n})/2$ . Therefore we have  $k$

$= 3(-1 + \sqrt{1 + 8n})/24 = \frac{1}{24}(-1 + \sqrt{1 + 8n})$ . By Exercise 47 in Section 2.3, this is the same as the integer closest to  $\frac{1}{24}(-1 + \sqrt{1 + 8n})$ , where we choose the smaller of the two closest integers if

is a half integer. The desired answer is  $\frac{1}{24}(-1 + \sqrt{1 + 8n})$ , which by Exercise 46 in Section 2.3 is the integer closest to  $\sqrt{2n}$  (note that  $\sqrt{2n}$  can never be a half integer). To see that these are the same, note that it can never happen that  $\sqrt{2n} \leq m + \frac{1}{2}$  while  $\frac{1}{24}(-1 + \sqrt{1 + 8n}) > m + \frac{1}{2}$  for some positive integer  $m$ , since this would imply that  $2n \leq m^2 + m + \frac{1}{4}$  and  $2n > m^2 + m$ , an impossibility. Therefore the integer closest to  $\sqrt{2n}$  and the (smaller) integer closest to  $\frac{1}{24}(-1 + \sqrt{1 + 8n})$  are done.

#### Section 2.4 Sequences and Summations 57

30. a)  $1 + 3 + 5 + 7 = 16$  b)  $1^2 + 3^2 + 5^2 + 7^2 = 84$

c)  $(1/1) + (1/3) + (1/5) + (1/7) = 176/105$  d)  $1 + 1 + 1 + 1 = 4$

32. a) The terms of this sequence alternate between 2 (if  $j$  is even) and 0 (if  $j$  is odd). Thus the sum is  $2 + 0 + 2 + 0 + 2 + 0 + 2 + 0 + 2 = 10$ .

b) We can break this into two parts and compute  $\sum_{j=0}^8 3^j - \sum_{j=0}^8 2^j$ . Each summation can be computed from the formula for the sum of a geometric progression. Thus the answer is

$$3^9 - 1$$

$$3 - 1 - 2^9 - 1$$

$$2 - 1 = 9841 - 511 = 9330.$$

c) As in part (b) we can break this into two parts and compute  $\sum_{j=0}^8 2 \cdot 3^j + \sum_{j=0}^8 3 \cdot 2^j$ . Each summation can be computed from the formula for the sum of a geometric progression. Thus the answer is

$$2 \cdot 3^9 - 2$$

$$3 - 1 + 3 \cdot 2^9 - 3$$

$$2 - 1 = 19682 + 1533 = 21215.$$

d) This could be worked as in part (b), but it is easier to note that the sum telescopes (see Exercise 35). Each power of 2 cancels except for the  $-2^0$  when  $j = 0$  and the  $2^9$  when  $j = 8$ . Therefore the answer is  $2^9 - 2^0 = 511$ . (Alternatively, note that  $2^{j+1} - 2^j = 2^j$ .)

34. We will just write out the sums explicitly in each case.

a)  $(1 - 1) + (1 - 2) + (2 - 1) + (2 - 2) + (3 - 1) + (3 - 2) = 3$

b)  $(0+0)+(0+2)+(0+4)+(3+0)+(3+2)+(3+4)+(6+0)+(6+2)+(6+4)+(9+0)+(9+2)+(9+4) = 78$  c)  $(0 + 1 + 2) + (0 + 1 + 2) + (0 + 1 + 2) = 9$

d)  $(0 + 0 + 0 + 0) + (0 + 1 + 8 + 27) + (0 + 4 + 32 + 108) = 180$

36. We use the suggestion (simple algebra shows that this is indeed an identity) and note that all the terms in the summation cancel out except for the  $1/k$  when  $k = 1$  and the  $1/(k + 1)$  when  $k = n$ :

$$\begin{array}{ccccccc}
 +^n_{k=1} & & 1 & & , & 1 & & k+1 \\
 & & & & & & & - \\
 & & & & & & & 1 \\
 k(k+1) = +^n_{k=1} & & k-1 & & & & & = 1 - 1
 \end{array}$$

$$n + 1$$

$$n + 1 = n$$

38. First we note that  $k^3 - (k - 1)^3 = 3k^2 - 3k + 1$ . Then we sum this equation for all values of  $k$  from 1 to  $n$ . On the left, because of telescoping, we have just  $n^3$ ; on the right we have

$$\sum_{k=1}^n (k^3 - (k-1)^3) = \sum_{k=1}^n (3k^2 - 3k + 1)$$

$$n^3 = 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + \sum_{k=1}^n 1$$

Equating the two sides and solving for  $\sum_{k=1}^n k^2$ , we obtain the desired formula.

$$\sum_{k=1}^n k^2 = \frac{1}{3} n^3 + \sum_{k=1}^n (3k - 3) + n$$

$$= \frac{1}{3} n^3 + 3 \sum_{k=1}^n k - 3n + n$$

$$= \frac{1}{3} n^3 + 3 \cdot \frac{n(n+1)}{2} - 2n$$

$$= \frac{n^3}{3} + \frac{3n^2 + 3n}{2} - 2n$$

$$= \frac{2n^3 + 3n^2 + 3n - 4n}{6} = \frac{n(n+1)(2n+1)}{6}$$

40. This exercise is like Example 23. From Table 2 we know that  $\sum_{k=1}^{200} k^3 = 200^2 \cdot 201^2 / 4 = 404,010,000$ , and  $\sum_{k=1}^{98} k^3 = 98^2 \cdot 99^2 / 4 = 23,532,201$ . Therefore the desired sum is  $404,010,000 - 23,532,201 = 380,477,799$ .

58 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

42. If we write down the first few terms of this sum we notice a pattern. It starts  $(1 + 1 + 1 + 1 + 1 + 1 + 1) + (2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2) + (3 + 3 + 3 + 3 + \dots + 3) + \dots$ . There are seven 1s, then 19 2s, then 37 3s, and so on; in general, the number of  $i$ 's is  $(i + 1)^3 - i^3 = 3i^2 + 3i + 1$ . So we need to sum  $i(3i^2 + 3i + 1)$  for an appropriate range of values for  $i$ . We must find this range. It gets a little messy at the end if  $m$  is such that the sequence stops before a complete range of the last value is present.

Let  $n = \sqrt[3]{m^2} - 1$ . Then there are  $n + 1$  blocks, and  $(n + 1)^3 - 1$  is where the next-to-last block ends. The

sum of those complete blocks is  $\sum_{i=1}^n i(3i^2 + 3i + 1) = \sum_{i=1}^n (3i^3 + 3i^2 + i) = n(3n+4)(n+1)^2/4$  (using Table 2 and algebra). The remaining terms in our summation all have the value  $n + 1$  and the number of them present is  $m - ((n + 1)^3 - 1)$ . Our final answer is therefore  $n(3n + 4)(n + 1)^2/4 + (n + 1)(m - (n + 1)^3 + 1)$ , where, once again,  $n = \sqrt[3]{m^2} - 1$ .

44.  $n! = \prod_{i=1}^n i$

46.  $(0!)(1!)(2!)(3!)(4!) = 1 \cdot 1 \cdot 2 \cdot 6 \cdot 24 = 288$

## SECTION 2.5 Cardinality of Sets

2. a) This set is countably infinite. The integers in the set are 11, 12, 13, 14, and so on. We can list these numbers in that order, thereby establishing the desired correspondence. In other words, the

correspondence is given by  $1 \leftrightarrow 11, 2 \leftrightarrow 12, 3 \leftrightarrow 13$ , and so on; in general  $n \leftrightarrow (n + 10)$ .

b) This set is countably infinite. The integers in the set are  $-1, -3, -5, -7$ , and so on. We can list these numbers in that order, thereby establishing the desired correspondence. In other words, the correspondence is given by  $1 \leftrightarrow -1, 2 \leftrightarrow -3, 3 \leftrightarrow -5$ , and so on; in general  $n \leftrightarrow -(2n - 1)$ .

c) This set is  $\{-999,999, -999,998, \dots, -1, 0, 1, \dots, 999,999\}$ . It is finite, with cardinality 1,999,999. d) This set is uncountable. We can prove it by the same diagonalization argument as was used to prove that the set of all reals is uncountable in Example 5.

e) This set is countable. We can list its elements in the order  $(2, 1), (3, 1), (2, 2), (3, 2), (2, 3), (3, 3), \dots$ , giving us the one-to-one correspondence  $1 \leftrightarrow (2, 1), 2 \leftrightarrow (3, 1), 3 \leftrightarrow (2, 2), 4 \leftrightarrow (3, 2), 5 \leftrightarrow (2, 3), 6 \leftrightarrow (3, 3), \dots$

f) This set is countable. The integers in the set are  $0, \pm 10, \pm 20, \pm 30$ , and so on. We can list these numbers in the order  $0, 10, -10, 20, -20, 30, \dots$ , thereby establishing the desired correspondence. In other words, the correspondence is given by  $1 \leftrightarrow 0, 2 \leftrightarrow 10, 3 \leftrightarrow -10, 4 \leftrightarrow 20, 5 \leftrightarrow -20, 6 \leftrightarrow 30$ , and so on.

4. a) This set is countable. The integers in the set are  $\pm 1, \pm 2, \pm 4, \pm 5, \pm 7$ , and so on. We can list these numbers in the order  $1, -1, 2, -2, 4, -4, 5, -5, 7, -7, \dots$ , thereby establishing the desired correspondence. In other words, the correspondence is given by  $1 \leftrightarrow 1, 2 \leftrightarrow -1, 3 \leftrightarrow 2, 4 \leftrightarrow -2, 5 \leftrightarrow 4$ , and so on. b) This is similar to part (a); we can simply list the elements of the set in order of increasing absolute value, listing each positive term before its corresponding negative:  $5, -5, 10, -10, 15, -15, 20, -20, 25, -25, 30, -30, 40, -40, 45, -45, 50, -50, \dots$

c) This set is countable but a little tricky. We can arrange the numbers in a 2-dimensional table as follows:

|      |    |       |        |         |          |           |            |     |
|------|----|-------|--------|---------|----------|-----------|------------|-----|
| .1   | .1 | .11   | .111   | .1111   | .11111   | .111111   | .1111111   | ... |
| 1.1  | 1  | 1.1   | 1.11   | 1.111   | 1.1111   | 1.11111   | 1.111111   | ... |
| 11.1 | 11 | 11.1  | 11.11  | 11.111  | 11.1111  | 11.11111  | 11.111111  | ... |
| 111. | 1  | 111.1 | 111.11 | 111.111 | 111.1111 | 111.11111 | 111.111111 | ... |

Thus we have shown that our set is the countable union of countable sets (each of the countable sets is one row of this table). Therefore by Exercise 27, the entire set is countable. For an explicit correspondence with

## Section 2.5 Cardinality of Sets 59

the positive integers, we can zigzag along the positive-sloping diagonals as in Figure 3:  $1 \leftrightarrow .1, 2 \leftrightarrow 1.1, 3 \leftrightarrow .1, 4 \leftrightarrow .11, 5 \leftrightarrow 1$ , and so on.

d) This set is not countable. We can prove it by the same diagonalization argument as was used to prove that the set of all reals is uncountable in Example 5. All we need to do is choose  $d_i = 1$  when  $d_{ii} = 9$  and choose  $d_i = 9$  when  $d_{ii} = 1$  or  $d_{ii}$  is blank (if the decimal expansion is finite).

6. We want a one-to-one function from the set of positive integers to the set of odd positive integers. The simplest one to use is  $f(n) = 2n - 1$ . We put the guest currently in Room  $n$  into Room  $(2n - 1)$ . Thus the guest in Room 1 stays put, the guest in Room 2 moves to Room 3, the guest in Room 3 moves to Room 5, and so on.
8. First we can make the move explained in Exercise 6, which frees up all the even-numbered rooms. The new guests can go into those rooms (the first into Room 2, the second into Room 4, and so on).
10. In each case, let us take  $A$  to be the set of real numbers.
- a) We can let  $B$  be the set of real numbers as well; then  $A - B = \emptyset$ , which is finite.
- b) We can let  $B$  be the set of real numbers that are not positive integers; in symbols,  $B = A - \mathbb{Z}^+$ . Then  $A - B = \mathbb{Z}^+$ , which is countably infinite.

c) We can let  $B$  be the set of positive real numbers. Then  $A - B$  is the set of negative real numbers and 0, which is certainly uncountable.

12. The definition of  $|A| \leq |B|$  is that there is a one-to-one function from  $A$  to  $B$ . In this case the desired function is just  $f(x) = x$  for each  $x \in A$ .
14. If  $A$  and  $B$  have the same cardinality, then we have a one-to-one correspondence  $f : A \rightarrow B$ . The function  $f$  meets the requirement of the definition that  $|A| \leq |B|$ , and  $f^{-1}$  meets the requirement of the definition that  $|B| \leq |A|$ .
16. If a set  $A$  is countable, then we can list its elements,  $a_1, a_2, a_3, \dots, a_n, \dots$  (possibly ending after a finite number of terms). Every subset of  $A$  consists of some (or none or all) of the items in this sequence, and we can list them in the same order in which they appear in the sequence. This gives us a sequence (again, infinite or finite) listing all the elements of the subset. Thus the subset is also countable.
18. The hypothesis gives us a one-to-one and onto function  $f$  from  $A$  to  $B$ . By Exercise 16e in the supplementary exercises for this chapter, the function  $S_f$  from  $P(A)$  to  $P(B)$  defined by  $S_f(X) = f(X)$  for all  $X \subseteq A$  is one-to-one and onto. Therefore  $P(A)$  and  $P(B)$  have the same cardinality.
20. By definition, we have one-to-one onto functions  $f : A \rightarrow B$  and  $g : B \rightarrow C$ . Then  $g \circ f$  is a one-to-one onto function from  $A$  to  $C$ , so  $|A| = |C|$ .
22. If  $A = \emptyset$ , then the only way for the conditions to be met are that  $B = \emptyset$  as well, and we are done. So assume that  $A$  is nonempty. Let  $f$  be the given onto function from  $A$  to  $B$ , and let  $g : \mathbb{Z}^+ \rightarrow A$  be an onto function that establishes the countability of  $A$ . (If  $A$  is finite rather than countably infinite, say of cardinality  $k$ , then the function  $g$  will be defined so that  $g(1), g(2), \dots, g(k)$  will list the elements of  $A$ , and  $g(n) = g(1)$  for  $n > k$ .) We need to find an onto function from  $\mathbb{Z}^+$  to  $B$ . The function  $f \circ g$  does the trick, because the composition of two onto functions is onto (Exercise 33b in Section 2.3).
24. Because  $|A| < |\mathbb{Z}^+|$ , there is a one-to-one function  $f : A \rightarrow \mathbb{Z}^+$ . We are also given that  $A$  is infinite, so the range of  $f$  has to be infinite. We will construct a bijection  $g$  from  $\mathbb{Z}^+$  to  $A$ . For each  $n \in \mathbb{Z}^+$ , let  $m$  be the  $n^{\text{th}}$  smallest element in the range of  $f$ . Then  $g(n) = f^{-1}(m)$ . The existence of  $g$  contradicts the definition of  $|A| < |\mathbb{Z}^+|$ , and our proof is complete.

## 60 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

26. We can label the rational numbers with strings from the set  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, /, -\}$  by writing down the string that represents that rational number in its simplest form (no leading 0's, denominator not 0, no common factors greater than 1 between numerator and denominator, and the minus sign in front if the number is negative). The labels are unique. It follows immediately from Exercise 25 that the set of rational numbers is countable.
28. We can think of  $\mathbb{Z}^+ \times \mathbb{Z}^+$  as the countable union of countable sets, where the  $i^{\text{th}}$  set in the collection, for  $i \in \mathbb{Z}^+$ , is  $\{(i, n) \mid n \in \mathbb{Z}^+\}$ . The statement now follows from Exercise 27.
30. There are at most two real solutions of each quadratic equation, so the number of solutions is countable as long as the number of triples  $(a, b, c)$ , with  $a, b$ , and  $c$  integers, is countable. But this follows from Exercise 27 in the following way. There are a countable number of pairs  $(b, c)$ , since for each  $b$  (and there are countably many  $b$ 's) there are only a countable number of pairs with that  $b$  as its first coordinate. Now for each  $a$  (and there are countably many  $a$ 's) there are only a countable number of triples with that  $a$  as its first coordinate (since we just showed that there are only a countable number of pairs  $(b, c)$ ). Thus again by Exercise 27 there are only countably many triples.

32. We saw in Exercise 31 that

$$f(m, n) = (m + n - 2)(m + n - 1)2 + m$$

is a one-to-one function with domain  $Z^+ \times Z^+$ . We want to expand the domain to be  $Z \times Z$ , so things need to be spread out a little if we are to keep it one-to-one. If we can find a one-to-one function  $g$  from  $Z \times Z$  to  $Z^+ \times Z^+$ , then composing these two functions will be our desired one-to-one function from  $Z \times Z$  to  $Z$  (we know from Exercise 33a in Section 2.3 that the composition of one-to-one functions is one-to-one). The function suggested here is  $g(m, n) = ((3m + 1)^2, (3n + 1)^2)$ , so that the composed function is  $(f \circ g)(m, n) = ((3m + 1)^2 + (3n + 1)^2 - 2)((3m + 1)^2 + (3n + 1)^2 - 1)/2 + (3m + 1)^2$ . To see that  $g$  is one-to-one, first note that it is enough to show that the behavior in each coordinate is one-to-one; that is, the function that sends integer  $k$  to positive integer  $(3k + 1)^2$  is one-to-one. To see this, first note that if  $k_1 \neq k_2$  and  $k_1$  and  $k_2$  are both positive or both negative, then  $(3k_1 + 1)^2 \neq (3k_2 + 1)^2$ . And if one is nonnegative and the other is negative, then they cannot have the same images under this function because the nonnegative integers are sent to squares of numbers that leave a remainder of 1 when divided by 3 ( $0 \rightarrow 1^2$ ,  $1 \rightarrow 4^2$ ,  $2 \rightarrow 7^2$ , . . .), but negative integers are sent to squares of numbers that leave a remainder of 2 when divided by 3 ( $-1 \rightarrow 2^2$ ,  $-2 \rightarrow 5^2$ ,  $-3 \rightarrow 8^2$ , . . .).

34. It suffices to find one-to-one functions  $f : (0, 1) \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow (0, 1)$ . We can obviously use the function  $f(x) = x$  in the first case. For the second, we can compress  $\mathbb{R}$  onto  $(0, 1)$  by using the arctangent function, which is known to be injective; let  $g(x) = 2 \arctan(x)/\pi$ . It then follows from the Schröder-Bernstein theorem that  $|(0, 1)| = |\mathbb{R}|$ .

36. We can encode subsets of the set of positive integers as strings of, say, 5's and 6's, where the  $i^{\text{th}}$  symbol is a 5 if  $i$  is in the subset and a 6 otherwise. If we interpret this string as a real number by putting a 0 and a decimal point in front, then we have constructed a one-to-one function from  $P(Z^+)$  to  $(0, 1)$ . Also, we can construct a one-to-one function from  $(0, 1)$  to  $P(Z^+)$  by sending the number whose binary expansion is  $0.d_1d_2d_3 \dots$  to the set  $\{i \mid d_i = 1\}$ . Therefore by the Schröder-Bernstein theorem we have  $|P(Z^+)| = |(0, 1)|$ . By Exercise 34,  $|(0, 1)| = |\mathbb{R}|$ , so we have shown that  $|P(Z^+)| = |\mathbb{R}|$ . (We already know from Cantor's diagonal argument that  $\aleph_0 < |\mathbb{R}|$ .) There is one technical point here. In order for our function from  $(0, 1)$  to  $P(Z^+)$  to be well-defined, we must choose which of two equivalent expressions to represent numbers that have terminating binary expansions to use (for example,  $0.100101$  versus  $0.100110$ ); we can decide to always use the terminating form, i.e., the one ending in all 0's.)

## Section 2.6 Matrices 61

38. We know from Example 5 that the set of real numbers between 0 and 1 is uncountable. Let us associate to each real number in this range (including 0 but excluding 1) a function from the set of positive integers to the set  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$  as follows: If  $x$  is a real number whose decimal representation is  $0.d_1d_2d_3 \dots$  (with ambiguity resolved by forbidding the decimal to end with an infinite string of 9's), then we associate to  $x$  the function whose rule is given by  $f(n) = d_n$ . Clearly this is a one-to-one function from the set of real numbers between 0 and 1 and a subset of the set of all functions from the set of positive integers to the set  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ . Two different real numbers must have different decimal representations, so the corresponding functions are different. (A few functions are left out, because of forbidding representations such as  $0.239999 \dots$ .) Since the set of real numbers between 0 and 1 is uncountable, the subset of functions we have associated with them must be uncountable. But the set of all such functions has at least this cardinality, so it, too, must be uncountable (by Exercise 15).

40. We follow the hint. Suppose that  $f$  is a function from  $S$  to  $P(S)$ . We must show that  $f$  is not onto. Let  $T = \{s \in S \mid s \notin f(s)\}$ . We will show that  $T$  is not in the range of  $f$ . If it were, then we would have  $f(t) = T$  for some  $t$

$\in S$ . Now suppose that  $t \in T$ . Then because  $t \in f(t)$ , it follows from the definition of  $T$  that  $t \notin T$ ; this is a contradiction. On the other hand, suppose that  $t \notin T$ . Then because  $t \in f(t)$ , it follows from the definition of  $T$  that  $t \in T$ ; this is again a contradiction. This completes our proof by contradiction that  $f$  is not onto. On the other hand, the function sending  $x$  to  $\{x\}$  for each  $x \in S$  is a one-to-one function from  $S$  to  $P(S)$ , so by Definition 2  $|S| \leq |P(S)|$ . By the same definition, since  $|S| = |P(S)|$  (from what we have just proved and Definition 1), it follows that  $|S| < |P(S)|$ .

## SECTION 2.6 Matrices

2. We just add entry by entry.

$$\begin{array}{r} \square \\ \text{a) } \square \begin{array}{ccc} 0 & 3 & 9 \\ 1 & 4 & -1 \\ 2 & -5 & -3 \end{array} \end{array} \quad \begin{array}{c} 4 \\ -4 \ -5 \ 4 \ 0 \end{array}$$

$$\begin{array}{r} \square \\ \text{b) } \square \begin{array}{ccc} 3 & -4 & 9 \\ 2 & 10 & \end{array} \end{array}$$

4. To multiply matrices  $A$  and  $B$ , we compute the  $(i, j)^{\text{th}}$  entry of the product  $AB$  by adding all the products of elements from the  $i^{\text{th}}$  row of  $A$  with the corresponding element in the  $j^{\text{th}}$  column of  $B$ , that is  $\sum_{k=1}^n a_{ik}b_{kj}$ . This can only be done, of course, when the number of columns of  $A$  equals the number of rows of  $B$  (called  $n$  in the formula shown here).

$$\begin{array}{r} \square \\ \text{a) } \square \begin{array}{ccc} -1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & -2 & 1 \end{array} \end{array} \quad \begin{array}{c} \square \\ -7 \ -5 \ 8 \ 5 \\ 4 \ 0 \ 7 \ 3 \\ \square \end{array} \quad \begin{array}{c} \square \\ 24 \ -7 \ 20 \ 29 \ 2 \\ -10 \ 4 \ -17 \ -24 \ -3 \end{array}$$

$$\begin{array}{r} \square \\ \text{b) } \square \begin{array}{ccc} 4 & -1 & -7 \\ 6 & \end{array} \end{array} \quad \begin{array}{c} \square \\ \text{c) } \square \begin{array}{ccc} 2 & 0 & -3 \\ -4 & -1 & \end{array} \end{array}$$

6. First note that  $A$  must be a  $3 \times 3$  matrix in order for the sizes to work out as shown. If we name the elements of  $A$  in the usual way as  $[a_{ij}]$ , then the given equation is really nine equations in the nine unknowns  $a_{ij}$ ,

62 Chapter 2 Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

obtained simply by writing down what the matrix multiplication on the left means:

$$\begin{aligned} 1 \cdot a_{11} + 3 \cdot a_{21} + 2 \cdot a_{31} &= 7 \\ 1 \cdot a_{12} + 3 \cdot a_{22} + 2 \cdot a_{32} &= 1 \\ 1 \cdot a_{13} + 3 \cdot a_{23} + 2 \cdot a_{33} &= 3 \\ 2 \cdot a_{11} + 1 \cdot a_{21} + 1 \cdot a_{31} &= 1 \\ 2 \cdot a_{12} + 1 \cdot a_{22} + 1 \cdot a_{32} &= 0 \end{aligned}$$

$$2 \cdot a_{13} + 1 \cdot a_{23} + 1 \cdot a_{33} = 3$$

$$4 \cdot a_{11} + 0 \cdot a_{21} + 3 \cdot a_{31} = -1$$

$$4 \cdot a_{12} + 0 \cdot a_{22} + 3 \cdot a_{32} = -3$$

$$4 \cdot a_{13} + 0 \cdot a_{23} + 3 \cdot a_{33} = 7$$

This is really not as bad as it looks, since each variable only appears in three equations. For example, the first, fourth, and seventh equations are a system of three equations in the three variables  $a_{11}$ ,  $a_{21}$ , and  $a_{31}$ . We can solve them using standard algebraic techniques to obtain  $a_{11} = -1$ ,  $a_{21} = 2$  and  $a_{31} = 1$ . By similar reasoning we also obtain  $a_{12} = 0$ ,  $a_{22} = 1$  and  $a_{32} = -1$ ; and  $a_{13} = 1$ ,  $a_{23} = 0$  and  $a_{33} = 1$ . Thus our answer is

$$A = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}.$$

As a check we can carry out the matrix multiplication and verify that we obtain the given right-hand side.

8. Since the entries of  $A + B$  are  $a_{ij} + b_{ij}$  and the entries of  $B + A$  are  $b_{ij} + a_{ij}$ , that  $A + B = B + A$  follows from the commutativity of addition of real numbers.

10. a) This product is a  $3 \times 5$  matrix.

b) This is not defined since the number of columns of  $B$  does not equal the number of rows of  $A$ .

c) This product is a  $3 \times 4$  matrix.

d) This is not defined since the number of columns of  $C$  does not equal the number of rows of  $A$ .

e) This is not defined since the number of columns of  $B$  does not equal the number of rows of  $C$ .

f) This product is a  $4 \times 5$  matrix.

12. We use the definition of matrix addition and multiplication. All summations here are from 1 to  $k$ . a)

$$(A + B)C = \sum_{q=1}^6 (a_{iq} + b_{iq})c_{qj} = \sum_{q=1}^6 a_{iq}c_{qj} + \sum_{q=1}^6 b_{iq}c_{qj} = AC + BC$$

$$b) C(A + B) = \sum_{q=1}^6 c_{iq}(a_{qj} + b_{qj}) = \sum_{q=1}^6 c_{iq}a_{qj} + \sum_{q=1}^6 c_{iq}b_{qj} = CA + CB$$

14. Let  $A$  and  $B$  be two diagonal  $n \times n$  matrices. Let  $C = [c_{ij}]$  be the product  $AB$ . From the definition of matrix multiplication,  $c_{ij} = \sum_{q=1}^n a_{iq}b_{qj}$ . Now all the terms  $a_{iq}$  in this expression are 0 except for  $q = i$ , so  $c_{ij} = a_{ii}b_{ij}$ . But  $b_{ij} = 0$  unless  $i = j$ , so the only nonzero entries of  $C$  are the diagonal entries  $c_{ii} = a_{ii}b_{ii}$ .

16. The  $(i, j)^{\text{th}}$  entry of  $(A^t)^t$  is the  $(j, i)^{\text{th}}$  entry of  $A^t$ , which is the  $(i, j)^{\text{th}}$  entry of  $A$ .

18. We need to multiply these two matrices together in both directions and check that both products are  $I_3$ . Indeed, they are.

20. a) Using Exercise 19, noting that  $ad - bc = -5$ , we write down the inverse immediately:

$$\begin{bmatrix} 3 & 4 \\ -3/5 & 1/5 \end{bmatrix}$$

Supplementary Exercises 63

b) We multiply to obtain  $A^2 = \begin{bmatrix} 3 & 4 & 2 & 11 \\ 4 & 3 & 1 & 18 & 9 & 37 \end{bmatrix}$  and then  $A^3 = 4 \begin{bmatrix} 3 & 1 & 18 & 9 & 37 \end{bmatrix}$ .

c) We multiply to obtain  $(A^{-1})^2 = \begin{bmatrix} 3 & 11/25 & -4/25 & -2/25 & 3/25 & 4 \end{bmatrix}$ .

and then  $(A^{-1})^3 = \begin{pmatrix} 3 & -37/125 & 18/125 \\ 9/125 & -1/125 & 4 \end{pmatrix}$ .

d) Applying the method of Exercise 19 for obtaining inverses to the answer in part (b), we obtain the answer in part (c). Therefore  $(A^3)^{-1} = (A^{-1})^3$ .

22. A matrix is symmetric if and only if it equals its transpose. So let us compute the transpose of  $AA^t$  and see if we get this matrix back. Using Exercise 17b and then Exercise 16, we have  $(AA^t)^t = (A^t)^t A^t = AA^t$ , as desired.

24. a) We simply note that under the given definitions of  $A$ ,  $X$ , and  $B$ , the definition of matrix multiplication is exactly the system of equations shown.

b) The given system is the matrix equation  $AX = B$ . If  $A$  is invertible with inverse  $A^{-1}$ , then we can multiply both sides of this equation by  $A^{-1}$  to obtain  $A^{-1}AX = A^{-1}B$ . The left-hand side simplifies to  $IX$ , however, by the definition of inverse, and this is simply  $X$ . Thus the given system is equivalent to the system  $X = A^{-1}B$ , which obviously tells us exactly what  $X$  is (and therefore what all the values  $x_i$  are).

26. We follow the definitions.

a)  $\begin{pmatrix} 3 & 1 & 1 & 1 & 1 \\ 4 & 3 & 0 & 1 & 0 & 0 \\ 4 & 3 & 1 & 1 & 1 \\ 4 & 1 & 0 & 1 & 0 \end{pmatrix}$   $\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ \square \\ \square \\ \square \end{pmatrix}$   
 b)  $\begin{pmatrix} 3 & 0 & 1 & 0 & 0 \\ 4 & 3 & 1 & 1 & 1 \\ 4 & 1 & 0 & 1 & 0 \end{pmatrix}$  28. We follow the definition and obtain  $\begin{pmatrix} \square \\ \square \\ \square \end{pmatrix}$ .  
 c)  $\begin{pmatrix} 3 & 1 & 1 & 1 & 1 \\ 4 & 3 & 0 & 1 & 0 & 0 \\ 4 & 3 & 1 & 1 & 1 \\ 4 & 1 & 0 & 1 & 0 \end{pmatrix}$   $\begin{pmatrix} 1 & 0 \\ \square & 1 & 0 \end{pmatrix}$

30. a)  $A \vee A = [a_{ij} \vee a_{ij}] = [a_{ij}] = A$  b)  $A \wedge A = [a_{ij} \wedge a_{ij}] = [a_{ij}] = A$

32. a)  $(A \vee B) \vee C = [(a_{ij} \vee b_{ij}) \vee c_{ij}] = [a_{ij} \vee (b_{ij} \vee c_{ij})] = A \vee (B \vee C)$   
 b) This is identical to part (a), with  $\wedge$  replacing  $\vee$ .

34. Since the  $i^{\text{th}}$  row of  $I$  consists of all 0's except for a 1 in the  $(i,i)^{\text{th}}$  position, we have  $I \vee A = [(0 \wedge a_{1j}) \vee \cdots \vee (1 \wedge a_{ij}) \vee \cdots \vee (0 \wedge a_{nj})] = [a_{ij}] = A$ . Similarly, since the  $j^{\text{th}}$  column of  $I$  consists of all 0's except for a 1 in the  $(j,j)^{\text{th}}$  position, we have  $A \vee I = [(a_{i1} \wedge 0) \vee \cdots \vee (a_{ij} \wedge 1) \vee \cdots \vee (a_{in} \wedge 0)] = [a_{ij}] = A$ .

## SUPPLEMENTARY EXERCISES FOR CHAPTER 2

2. We are given that  $A \subseteq B$ . We want to prove that the power set of  $A$  is a subset of the power set of  $B$ , which means that if  $C \subseteq A$  then  $C \subseteq B$ . But this follows directly from Exercise 17 in Section 2.1.

4. a)  $Z$  b)  $\emptyset$  c)  $O$  d)  $E$

6. If  $A \subseteq B$ , then every element in  $A$  is also in  $B$ , so clearly  $A \cap B = A$ . Conversely, if  $A \cap B = A$ , then every element of  $A$  must also be in  $A \cap B$ , and hence in  $B$ . Therefore  $A \subseteq B$ .

8. This identity is true, so we must show that every element in the left-hand side is also an element in the right-hand side and conversely. Let  $x \in (A - B) - C$ . Then  $x \in A - B$  but  $x \notin C$ . This means that  $x \in A$ , but  $x \notin B$  and  $x \notin C$ . Therefore  $x \in A - C$ , and therefore  $x \in (A - C) - B$ . The converse is proved in exactly the same way.

10. The inequality follows from the obvious fact that  $A \cap B \subseteq A \cup B$ . Equality can hold only if there are no elements in either  $A$  or  $B$  that are not in both  $A$  and  $B$ , and this can happen only if  $A = B$ .
12. Since  $A \cap B = (A \cup B) - (A \cup B) \setminus (A \cap B)$ , we are asked to show that  $|A \cup B| = |A| + |B| - |A \cap B|$ . This follows immediately from the facts that  $|X| = |U| - |X|$  (which is clear from the definitions) and (see the discussion following Example 5 in Section 2.2) that  $|A \cup B| = |A| + |B| - |A \cap B|$ .
14. Define a function  $g : f(S) \rightarrow S$  by choosing, for each element  $x$  in  $f(S)$ , an element  $g(x) \in S$  such that  $f(g(x)) = x$ . Clearly  $g$  is one-to-one, so  $|f(S)| \leq |S|$ . Note that we do not need the hypothesis that  $A$  and  $B$  are finite.
16. a) We are given that  $f$  is one-to-one, and we must show that  $S_f$  is one-to-one. So suppose that  $X_1 \neq X_2$ , where these are subsets of  $A$ . We have to show that  $S_f(X_1) \neq S_f(X_2)$ . Without loss of generality there is an element  $a \in X_1 - X_2$ . This means that  $f(a) \in S_f(X_1)$ . If  $f(a)$  were also an element of  $S_f(X_2)$ , then we would need an element  $a' \in X_2$  such that  $f(a') = f(a)$ . But since  $f$  is one-to-one, this forces  $a' = a$ , which is impossible, because  $a \notin X_2$ . Therefore  $f(a) \in S_f(X_1) - S_f(X_2)$ , so  $S_f(X_1) \neq S_f(X_2)$ .
- b) We are given that  $f$  is onto, and we must show that  $S_f$  is onto. So suppose that  $Y \subseteq B$ . We have to find  $X \subseteq A$  such that  $S_f(X) = Y$ . Let  $X = \{x \in A \mid f(x) \in Y\}$ . We claim that  $S_f(X) = Y$ . Clearly  $S_f(X) \subseteq Y$ . To see that  $Y \subseteq S_f(X)$ , suppose that  $b \in Y$ . Then because  $f$  is onto, there is some  $a \in A$  such that  $f(a) = b$ . By our definition of  $X$ ,  $a \in X$ . Therefore by definition  $b \in S_f(X)$ .
- c) We are given that  $f$  is onto, and we must show that  $S_{f^{-1}}$  is one-to-one. So suppose that  $Y_1 \neq Y_2$ , where these are subsets of  $B$ . We have to show that  $S_{f^{-1}}(Y_1) \neq S_{f^{-1}}(Y_2)$ . Without loss of generality there is an element  $b \in Y_1 - Y_2$ . Because  $f$  is onto, there is an  $a \in A$  such that  $f(a) = b$ . Therefore  $a \in S_{f^{-1}}(Y_1)$ . But we also know that  $a \notin S_{f^{-1}}(Y_2)$ , because if  $a$  were an element of  $S_{f^{-1}}(Y_2)$ , then we would have  $b = f(a) \in Y_2$ , contrary to our choice of  $b$ . The existence of this  $a$  shows that  $S_{f^{-1}}(Y_1) \neq S_{f^{-1}}(Y_2)$ .
- d) We are given that  $f$  is one-to-one, and we must show that  $S_{f^{-1}}$  is onto. So suppose that  $X \subseteq A$ . We have to find  $Y \subseteq B$  such that  $S_{f^{-1}}(Y) = X$ . Let  $Y = S_f(X)$ . In other words,  $Y = \{f(x) \mid x \in X\}$ . We must show that  $S_{f^{-1}}(Y) = X$ , which means that we must show that  $\{u \in A \mid f(u) \in \{f(x) \mid x \in X\}\} = X$  (we changed the dummy variable to  $u$  for clarity). That the right-hand side is a subset of the left-hand side is immediate, because if  $u \in X$ , then  $f(u)$  is an  $f(x)$  for some  $x \in X$ . Conversely, suppose that  $u$  is in the left-hand side. Thus  $f(u) = f(x_0)$  for some  $x_0 \in X$ . But because  $f$  is one-to-one, we know that  $u = x_0$ ; that is  $u \in X$ .
- e) This follows immediately from the earlier parts, because to be a one-to-one correspondence means to be one-to-one and onto.
18. If  $n$  is even, then  $n/2$  is an integer, so  $3n/24 + 1n/22 = (n/2) + (n/2) = n$ . If  $n$  is odd, then  $3n/24 = (n+1)/2$  and  $1n/22 = (n-1)/2$ , so again the sum is  $n$ .
20. This is certainly true if either  $x$  or  $y$  is an integer, since then this equation is equivalent to the identity (4b) in Table 1 of Section 2.3. Otherwise, write  $x$  and  $y$  in terms of their integer and fractional parts:  $x = n + \alpha$  and  $y = m + \delta$ , where  $n = \lfloor x \rfloor$ ,  $0 < \alpha < 1$ ,  $m = \lfloor y \rfloor$ , and  $0 < \delta < 1$ . If  $\delta + \alpha > 1$ , then the equation is true, since both sides equal  $m + n + 2$ ; if  $\delta + \alpha \leq 1$ , then the equation is false, since the left-hand side equals  $m + n + 1$ , but the right-hand side equals  $m + n + 2$ . To summarize: the equation is true if and only if either at least one of  $x$  and  $y$  is an integer or the sum of the fractional parts of  $x$  and  $y$  exceeds 1.
22. The values of the floor and ceiling function will depend on whether their arguments are integral or not. So there seem to be two cases here. First let us suppose that  $n$  is even. Then  $n/2$  is an integer, and  $n^2/4$  is also an integer, so the equation is a simple algebraic fact. The second case is harder. Suppose that  $n$  is

odd, say  $n = 2k + 1$ . Then  $n/2 = k + \frac{1}{2}$ . Therefore the left-hand side gives us  $k(k + 1) = k^2 + k$ , since we have to round down for the first factor and round up for the second. What about the right-hand side?  $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$ , so  $n^2/4 = k^2 + k + \frac{1}{4}$ . Therefore the floor function gives us  $k^2 + k$ , and the proof is completed.

24. Since we are dividing by 4, let us write  $x = 4n - k$ , where  $0 \leq k < 4$ . In other words, write  $x$  in terms of how much it is less than the smallest multiple of 4 not less than it. There are three cases. If  $k = 0$ , then  $x$  is already a multiple of 4, so both sides equal  $n$ . If  $0 < k \leq 2$ , then  $1x/22 = 2n - 1$ , so the left-hand side is  $1n - \frac{1}{2} = n - 1$ . Of course the right-hand side is  $n - 1$  as well, so again the two sides agree. Finally, suppose that  $2 < k < 4$ . Then  $1x/22 = 2n - 2$ , and the left-hand side is  $1n - 12 = n - 1$ ; of course the right-hand side is still  $n - 1$ , as well. Since we proved that the two sides are equal in all cases, the proof is complete.

26. If  $x$  is an integer, then of course the two sides are identical. So suppose that  $x = k + \theta$ , where  $k$  is an integer and  $\theta$  is a real number with  $0 < \theta < 1$ . Then the values of the left-hand side, which is  $1(k + n)/m2$ , and the right-hand side, which is  $1(k + n + \theta)/m2$ , are the same, since adding a number strictly between 0 and 1 to the numerator of a fraction whose numerator and denominator are integers cannot cause the fraction to reach the next higher integer value (the numerator cannot reach the next multiple of  $m$ ).

28. a) 1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28, 36, 38, 47, 48, 53, 57, 62, 69

b) Suppose there were only a finite set of Ulam numbers, say  $u_1 < u_2 < \dots < u_n$ . Then it is clear that  $u_{n-1} + u_n$  can be written uniquely as the sum of two distinct Ulam numbers, so this is an Ulam number larger than  $u_n$ , a contradiction. Therefore there are an infinite number of Ulam numbers.

30. If we work at this long enough, we might notice that each term after the first three is the sum of the previous three terms. With this rule the next four terms will be 169, 311, 572, 1052. One way to use the power of technology here is to submit the given sequence to The On-Line Encyclopedia of Integer Sequences (oeis.org).

32. We know that the set of rational numbers is countable. If the set of irrational numbers were also countable, then the union of these two sets would also be countable by Theorem 1 in Section 2.5. But their union, the set of real numbers, is known to be uncountable. This contradiction tells us that the set of irrational numbers is not countable.

34. A finite subset of  $\mathbb{Z}^+$  has a largest element and therefore is a subset of  $\{1, 2, 3, \dots, n\}$  for some positive integer  $n$ . Let  $S_n$  be the set of subsets of  $\{1, 2, 3, \dots, n\}$ . It is finite and therefore countable; in fact  $|S_n| = 2^n$ . The set of all finite subsets of  $\mathbb{Z}^+$  is the union  $\bigcup_{n=1}^{\infty} S_n$ .

Being a countable union of countable sets, it is countable by Exercise 27 in Section 2.5.

36. This follows immediately from Exercise 35, because  $\mathbb{C}$  can be identified with  $\mathbb{R} \times \mathbb{R}$  by sending the complex number  $a + bi$ , where  $a$  and  $b$  are real numbers, to the ordered pair  $(a, b)$ .

38. Since  $A$  is the matrix defined by  $a_{ij} = c$  and  $a_{ij} = 0$  for  $i \neq j$ , it is easy to see from the definition of multiplication that  $AB$  and  $BA$  are both the same as  $B$  except that every entry has been multiplied by  $c$ . Therefore these two matrices are equal.

40. We simply need to show that the alleged inverse of  $AB$  has the correct defining property—that its product with  $AB$  (on either side) is the identity. Thus we compute

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

and similarly  $(B^{-1}A^{-1})(AB) = I$ . Therefore  $(AB)^{-1} = B^{-1}A^{-1}$ . (Note that the indicated matrix multiplications

were all defined, since the hypotheses implied that both A and B were  $n \times n$  matrices for some (and the same)  $n$ .)

## CHAPTER 3

### Algorithms

#### SECTION 3.1 Algorithms

2. a) This procedure is not finite, since execution of the while loop continues forever. b) This procedure is not effective, because the step  $m := 1/n$  cannot be performed when  $n = 0$ , which will eventually be the case. c) This procedure lacks definiteness, since the value of  $i$  is never set. d) This procedure lacks definiteness, since the statement does not tell whether  $x$  is to be set equal to  $a$  or to  $b$ .

4. Set the answer to be  $-\infty$ . For  $i$  going from 1 through  $n - 1$ , compute the value of the  $(i + 1)^{\text{st}}$  element in the list minus the  $i^{\text{th}}$  element in the list. If this is larger than the answer, reset the answer to be this value.

6. We need to go through the list and count the negative entries.

```
procedure negatives( $a_1, a_2, \dots, a_n$  : integers)
 $k := 0$ 
for  $i := 1$  to  $n$ 
    if  $a_i < 0$  then  $k := k + 1$ 
return  $k$  {the number of negative integers in the list}
```

8. This is similar to Exercise 7, modified to keep track of the largest even integer we encounter.

```
procedure largest even location( $a_1, a_2, \dots, a_n$  : integers)
 $k := 0$ 
largest :=  $-\infty$ 
for  $i := 1$  to  $n$ 
    if ( $a_i$  is even and  $a_i > \text{largest}$ ) then
         $k := i$ 
        largest :=  $a_i$ 
return  $k$  {the desired location (or 0 if there are no evens)}
```

10. We assume that if the input  $x = 0$ , then  $n > 0$ , since otherwise  $x^n$  is not defined. In our procedure, we let  $m = |n|$  and compute  $x^m$  in the obvious way. Then if  $n$  is negative, we replace the answer by its reciprocal.

```
procedure power ( $x$  : real number,  $n$  : integer)
 $m := |n|$ 
power := 1
for  $i := 1$  to  $m$ 
    power := power  $\cdot x$ 
if  $n < 0$  then power :=  $1/\text{power}$ 
return power { power =  $x^n$  }
```

12. Four assignment statements are needed, one for each of the variables and a temporary assignment to get started so that we do not lose one of the original values.

Section 3.1 Algorithms 67

```
temp := x
x := y
y := z
z := temp
```

14. a) With linear search we start at the beginning of the list, and compare 7 successively with 1, 3, 4, 5, 6, 8, 9, and 11. When we come to the end of the list and still have not found 7, we conclude that it is not in the list.
- b) We begin the search on the entire list, with  $i = 1$  and  $j = n = 8$ . We set  $m := 4$  and compare 7 to the fourth element of the list. Since  $7 > 5$ , we next restrict the search to the second half of the list, with  $i = 5$  and  $j = 8$ . This time we set  $m := 6$  and compare 7 to the sixth element of the list. Since  $7 \neq 8$ , we next restrict ourselves to the first half of the second half of the list, with  $i = 5$  and  $j = 6$ . This time we set  $m := 5$ , and compare 7 to the fifth element. Since  $7 > 6$ , we now restrict ourselves to the portion of the list between  $i = 6$  and  $j = 6$ . Since at this point  $i \neq j$ , we exit the loop. Since the sixth element of the list is not equal to 7, we conclude that 7 is not in the list.

16. We let min be the smallest element found so far. At the end, it is the smallest element, since we update it as necessary as we scan through the list.

```
procedure smallest( $a_1, a_2, \dots, a_n$  : natural numbers)
min :=  $a_1$ 
for i := 2 to n
    if  $a_i < \text{min}$  then min :=  $a_i$ 
return min {the smallest integer among the input}
```

18. This is similar to Exercise 17.

```
procedure last smallest( $a_1, a_2, \dots, a_n$  : integers)
min :=  $a_1$ 
location := 1
for i := 2 to n
    if min  $\geq a_i$  then
        min :=  $a_i$ 
        location := i
return location {the location of the last occurrence of the smallest element in the list}
```

20. We just combine procedures for finding the largest and smallest elements.

```
procedure smallest and largest( $a_1, a_2, \dots, a_n$  : integers)
min :=  $a_1$ 
max :=  $a_1$ 
for i := 2 to n
    if  $a_i < \text{min}$  then min :=  $a_i$ 
    if  $a_i > \text{max}$  then max :=  $a_i$ 
```

{ min is the smallest integer among the input, and max is the largest}

22. We assume that the input is a sequence of symbols,  $a_1, a_2, \dots, a_n$ , each of which is either a letter or a blank. We build up the longest word in word; its length is length. We denote the empty word by  $\lambda$ .

68 Chapter 3 Algorithms

```
procedure longest word( $a_1, a_2, \dots, a_n$  : symbols)
  maxlength := 0
  maxword :=  $\lambda$ 
  i := 1
  while i  $\leq$  n
    word :=  $\lambda$ 
    length := 0
    while  $a_i \neq$  blank and i  $\leq$  n
      length := length + 1
      word := concatenation of word and  $a_i$ 
      i := i + 1
    if length > max then
      maxlength := length
      maxword := word
    i := i + 1
  return maxword {the longest word in the sentence}
```

24. This is similar to Exercise 23. We let the array hit keep track of which elements of the codomain B have already been found to be images of elements of the domain A. When we find an element that has already been hit being hit again, we conclude that the function is not one-to-one.

```
procedure one one(f : function,  $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_m$  : integers)
  for i := 1 to m
    hit( $b_i$ ) := 0
  one one := true
  for j := 1 to n
    if hit( $f(a_j)$ ) = 0 then hit( $f(a_j)$ ) := 1
    else one one := false
  return one one
```

26. There are two changes. First, we need to test  $x = a_m$  (right after the computation of m) and take appropriate action if equality holds (what we do is set i and j both to be m). Second, if  $x \neq a_m$ , then instead of setting j equal to m, we can set j equal to  $m - 1$ . The advantages are that this allows the size of the “half” of the list being looked at to shrink slightly faster, and it allows us to stop essentially as soon as we have found the element we are looking for.

28. This could be thought of as just doing two iterations of binary search at once. We compare the sought-after element to the middle element in the still-active portion of the list, and then to the middle element of either the top half or the bottom half. This will restrict the subsequent search to one of four sublists, each about one-quarter the size of the previous list. We need to stop when the list has length three or less and make explicit checks. Here is the pseudocode.

Section 3.1 Algorithms 69

```

procedure tetrary search(x : integer, a1, a2, . . . , an : increasing integers)
  i := 1
  j := n
  while i < j - 2
    l := &(i + j)/4'
    m := &(i + j)/2'
    u := &3(i + j)/4'
    if x > am then if x ≤ au then
      i := m + 1
      j := u
    else i := u + 1
  else if x > al then
    i := l + 1
    j := m
  else j := l
  if x = ai then location := i
  else if x = aj then location := j
  else if x = al(i+j)/2' then location := &(i + j)/2'
  else location := 0
  return location {the subscript of the term equal to x (0 if not found)}

```

30. The following algorithm will find all modes in the sequence and put them into a list L. At each point in the execution of this algorithm, modecount is the number of occurrences of the elements found to occur most often so far (the elements in L). Whenever a more frequently occurring element is found (the main inner loop), modecount and L are updated; whenever an element is found with this same count, it is added to L.

```

procedure find all modes(a1, a2, . . . , an : nondecreasing integers)
  modecount := 0
  i := 1
  while i ≤ n
    value := ai
    count := 1
    while i ≤ n and ai = value
      count := count + 1
      i := i + 1
    if count > modecount then
      modecount := count
      set L to consist just of value
    else if count = modecount then add value to L
  return L {the list of all the values occurring most often, namely modecount times}

```

32. The following algorithm will find all terms of a finite sequence of integers that are greater than the sum of all

the previous terms. We put them into a list L, but one could just as easily have them printed out, if that were desired. It might be more useful to put the indices of these terms into L, rather than the terms themselves (i.e., their values), but we take the former approach for variety. As usual, the empty list is considered to have sum 0, so the first term in the sequence is included in L if and only if it positive.

```
procedure find all biggies( $a_1, a_2, \dots, a_n$  : integers)
```

```
set L to be the empty list
```

```
sum := 0
```

```
i := 1
```

```
while  $i \leq n$ 
```

```
    if  $a_i > \text{sum}$  then append  $a_i$  to L
```

```
    sum := sum +  $a_i$ 
```

```
    i := i + 1
```

```
return L {the list of all the values that exceed the sum of all the previous terms in the sequence}
```

## 70 Chapter 3 Algorithms

34. There are five passes through the list. After one pass the list reads 2, 3, 1, 5, 4, 6, since the 6 is compared and moved at each stage. During the next pass, the 2 and the 3 are not interchanged, but the 3 and the 1 are, as are the 5 and the 4, yielding 2, 1, 3, 4, 5, 6. On the third pass, the 2 and the 1 are interchanged, yielding 1, 2, 3, 4, 5, 6. There are two more passes, but no further interchanges are made, since the list is now in order.

36. The procedure is the same as that given in the solution to Exercise 35. We will exhibit the lists obtained after each step, with all the lists obtained during one pass on the same line.

```
dfkmab, dfkmab, dfkmab, dfkamb, dfkabm
```

```
dfkabm, dfkabm, dfakbm, dfabkm
```

```
dfabkm, dafbkm, dabfkm
```

```
adbfkm, abdfkm
```

```
abdfkm
```

38. We start with 6, 2, 3, 1, 5, 4. The first step inserts 2 correctly into the sorted list 6, producing 2, 6, 3, 1, 5, 4. Next 3 is inserted into 2, 6, and the list reads 2, 3, 6, 1, 5, 4. Next 1 is inserted into 2, 3, 6, and the list reads 1, 2, 3, 6, 5, 4. Next 5 is inserted into 1, 2, 3, 6, and the list reads 1, 2, 3, 5, 6, 4. Finally 4 is inserted into 1, 2, 3, 5, 6, and the list reads 1, 2, 3, 4, 5, 6. At each insertion, the element to be inserted is compared with the elements already sorted, starting from the beginning, until its correct spot is found, and then the previously sorted elements beyond that spot are each moved one position toward the back of the list.

40. We start with d, f, k, m, a, b. The first step inserts f correctly into the sorted list d, producing no change. Similarly, no change results when k and m are inserted into the sorted lists d, f and d, f, k, respectively. Next a is inserted into d, f, k, m, and the list reads a, d, f, k, m, b. Finally b is inserted into a, d, f, k, m, and the list reads a, b, d, f, k, m. At each insertion, the element to be inserted is compared with the elements already sorted, starting from the beginning, until its correct spot is found, and then the previously sorted elements beyond that spot are each moved one position toward the back of the list.

42. We let minspot be the place at which the minimum remaining element is found. After we find it on the  $i^{\text{th}}$  pass, we just have to interchange the elements in location minspot and location i.

```
procedure selection( $a_1, a_2, \dots, a_n$ )
```

```
for i := 1 to  $n - 1$ 
```

```
    minspot := i
```

```
    for j := i + 1 to n
```

```

        if  $a_j < a_{\text{minspot}}$  then minspot := j
    interchange  $a_{\text{minspot}}$  and  $a_i$ 
    {the list is now in order}

```

44. We carry out the binary search algorithm given as Algorithm 3 in this section, except that we replace the final check with if  $x < a_i$  then location := i else location := i + 1.

46. We are counting just the comparisons of the numbers in the list, not any comparisons needed for the book keeping in the for loop. The second element in the list must be compared only with the first (in other words, when  $j = 2$  in Algorithm 5,  $i$  takes the values 1 before we drop out of the while loop). Similarly, the third element must be compared only with the first. We continue in this way, until finally the  $n^{\text{th}}$  element must be compared only with the first. So the total number of comparisons is  $n - 1$ . This is the best case for insertion sort in terms of the number of comparisons, but moving the elements to do the insertions requires much more effort.

### Section 3.1 Algorithms 71

48. For the insertion sort, one comparison is needed to find the correct location of the 4, one for the 3, four for the 8, one for the 1, four for the 5, and two for the 2. This is a total of 13 comparisons. For the binary insertion sort, one comparison is needed to find the correct location of the 4, two for the 3, two for the 8, three for the 1, three for the 5, and four for the 2. This is a total of 15 comparisons. If the list were long (and not almost in decreasing order to begin with), we would use many fewer comparisons using binary insertion sort. The reason that the answer came out “wrong” here is that the list is so short that the binary search was not efficient.

50. a) This is essentially the same as Algorithm 5, but working from the other end. However, we can do the moving while we do the searching for the correct insertion spot, so the pseudocode has only one section.

```

procedure backward insertion sort( $a_1, a_2, \dots, a_n$  : real numbers with  $n \geq 2$ )
for j := 2 to n
    m :=  $a_j$ 
    i := j - 1
    while (m <  $a_i$  and i > 0)
         $a_{i+1}$  :=  $a_i$ 
        i := i - 1
     $a_{i+1}$  := m
{  $a_1, a_2, \dots, a_n$  are sorted}

```

b) On the first pass the 2 is compared to the 3 and found to be less, so the 3 moves to the right. We have reached the beginning of the list, so the loop terminates ( $i = 0$ ), and the 2 is inserted, yielding 2, 3, 4, 5, 1, 6. On the second pass the 4 is compared to the 3, and since  $4 > 3$ , the while loop terminates and nothing changes. Similarly, no changes are made as the 5 is inserted. On the fourth pass, the 1 is compared all the way to the front of the list, with each element moving toward the back of the list as the comparisons go on, and finally the 1 is inserted in its correct position, yielding 1, 2, 3, 4, 5, 6. The final pass produces no change. c) Only one comparison is used during each pass, since the condition  $m < a_i$  is immediately false. Therefore a total of  $n - 1$  comparisons are used.

d) The  $j^{\text{th}}$  pass requires  $j - 1$  comparisons of elements, so the total number of comparisons is  $1 + 2 + \dots + (n - 1) = n(n - 1)/2$ .

52. In each case we use as many quarters as we can, then as many dimes to achieve the remaining amount, then as many nickels, then as many pennies.

- a) The algorithm uses the maximum number of quarters, three, leaving 12 cents. It then uses the maximum number of dimes (one) and nickels (none), before using two pennies.
- b) one quarter, leaving 24 cents, then two dimes, leaving 4 cents, then four pennies
- c) three quarters, leaving 24 cents, then two dimes, leaving 4 cents, then four pennies
- d) one quarter, leaving 8 cents, then one nickel and three pennies

54. a) The algorithm uses the maximum number of quarters, three, leaving 12 cents. It then uses the maximum number of dimes (one), and then two pennies. The greedy algorithm worked, since we got the same answer as in Exercise 52.
- b) one quarter, leaving 24 cents, then two dimes, leaving 4 cents, then four pennies (the greedy algorithm worked, since we got the same answer as in Exercise 52)
- c) three quarters, leaving 24 cents, then two dimes, leaving 4 cents, then four pennies (the greedy algorithm worked, since we got the same answer as in Exercise 52)
- d) The greedy algorithm would have us use one quarter, leaving 8 cents, then eight pennies, a total of nine coins. However, we could have used three dimes and three pennies, a total of six coins. Thus the greedy algorithm is not correct for this set of coins.

## 72 Chapter 3 Algorithms

56. One approach is to come up with an example in which using the 12-cent coin before using dimes or nickels would be inefficient. A dime and a nickel together are worth 15 cents, but the greedy algorithm would have us use four coins (a 12-cent coin and three pennies) rather than two. An alternative example would be 29 cents, in which case the greedy algorithm would use a quarter and four pennies, but we could have done better using two 12-cent coins and a nickel.
58. Here is one counterexample, using 11 talks. Suppose the start and end times are as follows: A 1–3, B 3–5, C 5–7, D 7–9, E 2–4, F 2–4, G 2–4, H 4–6, J 6–8, K 6–8, L 6–8. The optimal schedule is talks A, B, C, and D. However, the talk with the fewest overlaps with other talks is H, which overlaps only with B and C (all the other talks overlap with three or four other talks). However, once we have decided to include talk H, we can no longer schedule four talks, so this algorithm will not produce an optimum solution.
60. If all the men get their first choices, then the matching will be stable, because no man will be part of an unstable pair, preferring another woman to his assigned partner. Thus the pairing  $(m_1w_3, m_2w_1, m_3w_2)$  is stable. Similarly, if all the women get their first choices, then the matching will be stable, because no woman will be part of an unstable pair, preferring another man to her assigned partner. Thus the matching  $(m_1w_1, m_2w_2, m_3w_3)$  is stable. Two of the other four matchings pair  $m_1$  with  $w_2$ , and this cannot be stable, because  $m_1$  prefers  $w_1$  to  $w_2$ , his assigned partner, and  $w_1$  prefers  $m_1$  to her assigned partner, whoever it is, because  $m_1$  is her favorite. In a similar way, the matching  $(m_1w_3, m_2w_2, m_3w_1)$  is unstable because of the unhappy unmatched pair  $m_3w_3$  (each preferring the other to his or her assigned partner). Finally, the matching  $(m_1w_1, m_2w_3, m_3w_2)$  is stable, because each couple has a reason not to break up:  $w_1$  got her favorite and so is content,  $m_3$  got his favorite and so is content, and  $w_3$  only prefers  $m_3$  to her assigned partner but he doesn't prefer her to his assigned partner.
62. The algorithm given in the solution to Exercise 61 will terminate if at some point at the conclusion of the while loop, no man is rejected. If this happens, then that must mean that each man has one and only one proposal pending with some woman, because he proposed to only one in that round, and since he was not rejected, his proposal is the only one pending with that woman. It follows that at that point there are  $s$  pending proposals, one from each man, so each woman will be matched with a unique man. Finally, we argue that there are at most  $s^2$  iterations of the while loop, so the algorithm must terminate. Indeed, if at the

conclusion of the while loop rejected men remain, then some man must have been rejected, because no man is marked as rejected at the conclusion of the proposal phase (first for loop inside the while loop). If a man is rejected, then his rejection list grows. Thus each pass through the while loop, at least one more of the  $s^2$  possible rejections will have been recorded, unless the loop is about to terminate. (Actually there will be fewer than  $s^2$  iterations, because no man is rejected by the woman with whom he is eventually matched.) There is one more subtlety we need to address. Is it possible that at the end of some round, some man has been rejected by every woman and therefore the algorithm cannot continue? We claim not. If at the end of some round some man has been rejected by every woman, then every woman has one pending proposal at the completion of that round (from someone she likes better—otherwise she never would have rejected that poor man), and of course these proposals are all from different men because a man proposes only once in each round. That means  $s$  men have pending proposals, so in fact our poor universally-rejected man does not exist.

64. Suppose we had a program  $S$  that could tell whether a program with its given input ever prints the digit 1. Here is an algorithm for solving the halting problem: Given a program  $P$  and its input  $I$ , construct a program  $P^\#$ , which is just like  $P$  but never prints anything (even if  $P$  did print something) except that if and when it is about to halt, it prints a 1 and halts. Then  $P$  halts on an input  $I$  if and only if  $P^\#$  ever prints a 1 on that same input. Feed  $P^\#$  and  $I$  to  $S$ , and that will tell us whether or not  $P$  halts on input  $I$ . Since we know that the halting problem is in fact not solvable, we have a contradiction. Therefore no such program  $S$  exists.

### Section 3.2 The Growth of Functions 73

66. The decision problem has no input. The answer is either always yes or always no, depending on whether or not the specific program with its specific input halts or not. In the former case, the decision procedure is “say yes,” and in the latter case it is “say no.”

## SECTION 3.2 The Growth of Functions

2. Note that the choices of  $C$  and  $k$  witnesses are not unique.

a) Yes, since  $17x + 11 \leq 17x + x = 18x \leq 18x^2$  for all  $x > 11$ . The witnesses are  $C = 18$  and  $k = 11$ . b) Yes, since  $x^2 + 1000 \leq x^2 + x^2 = 2x^2$  for all  $x > \sqrt{1000}$ . The witnesses are  $C = 2$  and  $k = \sqrt{1000}$ . c) Yes, since  $x \log x \leq x \cdot x = x^2$  for all  $x$  in the domain of the function. (The fact that  $\log x < x$  for all  $x$  follows from the fact that  $x < 2^x$  for all  $x$ , which can be seen by looking at the graphs of these two functions.) The witnesses are  $C = 1$  and  $k = 0$ . d) No. If there were a constant  $C$  such that  $x^4/2 \leq Cx^2$  for sufficiently large  $x$ , then we would have  $C \geq x^2/2$ . This is clearly impossible for a constant to satisfy. e) No. If  $2^x$  were  $O(x^2)$ , then the fraction  $2^x/x^2$  would have to be bounded above by some constant  $C$ . It can be shown that in fact  $2^x > x^3$  for all  $x \geq 10$  (using mathematical induction—see Section 5.1—or calculus), so  $2^x/x^2 \geq x^3/x^2 = x$  for large  $x$ , which is certainly not less than or equal to  $C$ . f) Yes, since  $x(x+1) \leq x \cdot 2x = 2x^2$  for all  $x > 1$ . The witnesses are  $C = 2$  and  $k = 1$ .

4. If  $x > 5$ , then  $2^x + 17 \leq 2^x + 2^x = 2 \cdot 2^x \leq 2 \cdot 3^x$ . This shows that  $2^x + 17$  is  $O(3^x)$  (the witnesses are  $C = 2$  and  $k = 5$ ).

6. We can use the following inequalities, valid for all  $x > 1$  (note that making the denominator of a fraction smaller makes the fraction larger).

$$\begin{aligned} 2x + 1 &\leq x^3 + 2x^3 \\ 2x &= \frac{3}{2}x^2 \\ x^3 + 2x &\end{aligned}$$

This proves the desired statement, with witnesses  $k = 1$  and  $C = 3/2$ .

8. a) Since  $x^3 \log x$  is not  $O(x^3)$  (because the  $\log x$  factor grows without bound as  $x$  increases),  $n = 3$  is too small. On the other hand, certainly  $\log x$  grows more slowly than  $x$ , so  $2x^2 + x^3 \log x \leq 2x^4 + x^4 = 3x^4$ . Therefore  $n = 4$  is the answer, with  $C = 3$  and  $k = 0$ .
- b) The  $(\log x)^4$  is insignificant compared to the  $x^5$  term, so the answer is  $n = 5$ . Formally we can take  $C = 4$  and  $k = 1$  as witnesses.
- c) For large  $x$ , this fraction is fairly close to 1. (This can be seen by dividing numerator and denominator by  $x^4$ .) Therefore we can take  $n = 0$ ; in other words, this function is  $O(x^0) = O(1)$ . Note that  $n = -1$  will not do, since a number close to 1 is not less than a constant times  $n^{-1}$  for large  $n$ . Formally we can write  $f(x) \leq 3x^4/x^4 = 3$  for all  $x > 1$ , so witnesses are  $C = 3$  and  $k = 1$ .
- d) This is similar to the previous part, but this time  $n = -1$  will do, since for large  $x$ ,  $f(x) \approx 1/x$ . Formally we can write  $f(x) \leq 6x^3/x^3 = 6$  for all  $x > 1$ , so witnesses are  $C = 6$  and  $k = 1$ .

10. Since  $x^3 \leq x^4$  for all  $x > 1$ , we know that  $x^3$  is  $O(x^4)$  (witnesses  $C = 1$  and  $k = 1$ ). On the other hand, if  $x^4 \leq Cx^3$ , then (dividing by  $x^3$ )  $x \leq C$ . Since this latter condition cannot hold for all large  $x$ , no matter what the value of the constant  $C$ , we conclude that  $x^4$  is not  $O(x^3)$ .

12. We showed that  $x \log x$  is  $O(x^2)$  in Exercise 2c. To show that  $x^2$  is not  $O(x \log x)$  it is enough to show that  $x^2/(x \log x)$  is unbounded. This is the same as showing that  $x/\log x$  is unbounded. First let us note that  $\log x < \sqrt{x}$  for all  $x > 16$ . This can be seen by looking at the graphs of these functions, or by calculus. Therefore the fraction  $x/\log x$  is greater than  $x/\sqrt{x} = \sqrt{x}$  for all  $x > 16$ , and this clearly is not bounded.

#### 74 Chapter 3 Algorithms

14. a) No, by an argument similar to Exercise 10.  
 b) Yes, since  $x^3 \leq x^3$  for all  $x$  (witnesses  $C = 1$ ,  $k = 0$ ).  
 c) Yes, since  $x^3 \leq x^2 + x^3$  for all  $x$  (witnesses  $C = 1$ ,  $k = 0$ ).  
 d) Yes, since  $x^3 \leq x^2 + x^4$  for all  $x$  (witnesses  $C = 1$ ,  $k = 0$ ).  
 e) Yes, since  $x^3 \leq 2^x \leq 3^x$  for all  $x > 10$  (see Exercise 2e). Thus we have witnesses  $C = 1$  and  $k = 10$ . f) Yes, since  $x^3 \leq 2 \cdot (x^3/2)$  for all  $x$  (witnesses  $C = 2$ ,  $k = 0$ ).
16. The given information says that  $|f(x)| \leq C|x|$  for all  $x > k$ , where  $C$  and  $k$  are particular constants. Let  $k^\#$  be the larger of  $k$  and 1. Then since  $|x| \leq |x^2|$  for all  $x > 1$ , we have  $|f(x)| \leq C|x^2|$  for all  $x > k^\#$ , as desired.
18.  $1^k + 2^k + \dots + n^k \leq n^k + n^k + \dots + n^k = n \cdot n^k = n^{k+1}$
20. They both are. For the first we have  $\log(n+1) < \log(2n) = \log n + \log 2 < 2 \log n$  for  $n > 2$ . For the second one we have  $\log(n^2+1) < \log(2n^2) = 2 \log n + \log 2 < 3 \log n$  for  $n > 2$ .
22. The ordering is straightforward when we remember that exponential functions grow faster than polynomial functions, that factorial functions grow faster still, and that logarithmic functions grow very slowly. The order is  $(\log n)^3$ ,  $\sqrt{n} \log n$ ,  $n^{99} + n^{98}$ ,  $n^{100}$ ,  $1.5^n$ ,  $10^n$ ,  $(n!)^2$ .
24. The first algorithm uses fewer operations because  $n^2 2^n$  is  $O(n!)$  but  $n!$  is not  $O(n^2 2^n)$ . In fact, the second function overtakes the first function for good at  $n = 8$ , when  $8^2 \cdot 2^8 = 16,384$  and  $8! = 40,320$ .
26. The approach in these problems is to pick out the most rapidly growing term in each sum and discard the rest (including the multiplicative constants).  
 a) This is  $O(n^3 \cdot \log n + \log n \cdot n^3)$ , which is the same as  $O(n^3 \cdot \log n)$ .

b) Since  $2^n$  dominates  $n^2$ , and  $3^n$  dominates  $n^3$ , this is  $O(2^n \cdot 3^n) = O(6^n)$ .

c) The dominant terms in the two factors are  $n^n$  and  $n!$ , respectively. Therefore this is  $O(n^n n!)$ .

28. We can use the following rule of thumb to determine what simple big-Theta function to use: throw away all the lower order terms (those that don't grow as fast as other terms) and all constant coefficients. a) This function is  $\Theta(1)$ , so it is not  $\Theta(x)$ , since 1 (or 10) grows more slowly than  $x$ . To be precise,  $x$  is not  $O(10)$ . For the same reason, this function is not  $\Omega(x)$ .

b) This function is  $\Theta(x)$ ; we can ignore the "+ 7" since it is a lower order term, and we can ignore the coefficient. Of course, since  $f(x)$  is  $\Theta(x)$ , it is also  $\Omega(x)$ .

c) This function grows faster than  $x$ . Therefore  $f(x)$  is not  $\Theta(x)$  but it is  $\Omega(x)$ .

d) This function grows more slowly than  $x$ . Therefore  $f(x)$  is not  $\Theta(x)$  or  $\Omega(x)$ .

e) This function has values that are, for all practical purposes, equal to  $x$  (certainly  $x/2$  is always between  $x/2$  and  $x$ , for  $x > 2$ ), so it is  $\Theta(x)$  and therefore also  $\Omega(x)$ .

f) As in part (e) this function has values that are, for all practical purposes, equal to  $x/2$ , so it is  $\Theta(x)$  and therefore also  $\Omega(x)$ .

30. a) This follows from the fact that for all  $x > 7$ ,  $x \leq 3x + 7 \leq 4x$ .

b) For large  $x$ , clearly  $x^2 \leq 2x^2 + x - 7$ . On the other hand, for  $x \geq 1$  we have  $2x^2 + x - 7 \leq 3x^2$ . c)

For  $x > 2$  we certainly have  $x + \frac{1}{2} \leq 2x$  and also  $x \leq 2x + \frac{1}{2}$ .

d) For  $x > 2$ ,  $\log(x^2 + 1) \leq \log(2x^2) = 1 + 2 \log x \leq 3 \log x$  (recall that  $\log$  means  $\log_2$ ). On the other hand, since  $x < x^2 + 1$  for all positive  $x$ , we have  $\log x \leq \log(x^2 + 1)$ .

e) This follows from the fact that  $\log_{10} x = C(\log_2 x)$ , where  $C = 1/\log_2 10$ .

#### Section 3.2 The Growth of Functions 75

32. We just need to look at the definitions. To say that  $f(x)$  is  $O(g(x))$  means that there are constants  $C$  and  $k$  such that  $|f(x)| \leq C|g(x)|$  for all  $x > k$ . Note that without loss of generality we may take  $C$  and  $k$  to be positive. To say that  $g(x)$  is  $\Omega(f(x))$  is to say that there are positive constants  $C^\#$  and  $k^\#$  such that  $|g(x)| \geq C^\#|f(x)|$  for all  $x > k$ . These are saying exactly the same thing if we set  $C^\# = 1/C$  and  $k^\# = k$ .

34. a) By Exercise 31 we have to show that  $3x^2 + x + 1$  is  $O(3x^2)$  and that  $3x^2$  is  $O(3x^2 + x + 1)$ . The latter is trivial, since  $3x^2 \leq 3x^2 + x + 1$  for  $x > 0$ . The former is almost as trivial, since  $3x^2 + x + 1 \leq 3x^2 + 3x^2 = 2 \cdot 3x^2$  for all  $x > 1$ . What we have shown is that  $1 \cdot 3x^2 \leq 3x^2 + x + 1 \leq 2 \cdot 3x^2$  for all  $x > 1$ ; in other words,  $C_1 = 1$  and  $C_2 = 2$  in Exercise 33.

b) The following picture shows that graph of  $3x^2 + x + 1$  falls in the shaded region between the graph of  $3x^2$  and the graph of  $2 \cdot 3x^2$  for all  $x > 1$ .

36. Looking at the definition, we see that to say that  $f(x)$  is  $\Omega(1)$  means that  $|f(x)| \geq C$  when  $x > k$ , for some positive constants  $k$  and  $C$ . In other words,  $f(x)$  keeps at least a certain distance away from 0 for large enough  $x$ . For example,  $1/x$  is not  $\Omega(1)$ , since it gets arbitrary close to 0; but  $(x - 2)(x - 10)$  is  $\Omega(1)$ , since  $f(x) \geq 9$  for  $x > 11$ .

38. The  $n^{\text{th}}$  odd positive integer is  $2n-1$ . Thus each of the first  $n$  odd positive integers is at most  $2n$ . Therefore their product is at most  $(2n)^n$ , so one answer is  $O((2n)^n)$ . Of course other answers are possible as well.

40. This follows from the fact that  $\log_b x$  and  $\log_a x$  are the same except for a multiplicative constant, namely  $d = \log_b a$ . Thus if  $f(x) \leq C \log_b x$ , then  $f(x) \leq Cd \log_a x$ .

42. This does not follow. Let  $f(x) = 2x$  and  $g(x) = x$ . Then  $f(x)$  is  $O(g(x))$ . Now  $2^{f(x)} = 2^{2x} = 4^x$ , and  $2^{g(x)} = 2^x$ , and  $4^x$  is not  $O(2^x)$ . Indeed,  $4^x/2^x = 2^x$ , so the ratio grows without bound as  $x$  grows—it is not bounded by a constant.

44. The definition of " $f(x)$  is  $\Theta(g(x))$ " is that  $f(x)$  is both  $O(g(x))$  and  $\Omega(g(x))$ . That means that there are positive constants  $C_1$ ,  $k_1$ ,  $C_2$ , and  $k_2$  such that  $|f(x)| \leq C_2|g(x)|$  for all  $x > k_2$  and  $|f(x)| \geq C_1|g(x)|$  for all  $x > k_1$ . Similarly, we have that there are positive constants  $C_1^\#$ ,  $k_1^\#$ ,  $C_2^\#$ , and  $k_2^\#$  such that  $|g(x)| \leq C_2^\#|h(x)|$  for all  $x > k_2^\#$  and  $|g(x)| \geq C_1^\#|h(x)|$  for all  $x > k_1^\#$ . We can combine these inequalities to obtain  $|f(x)| \leq C_2C_2^\#|h(x)|$  for all  $x > \max(k_2, k_2^\#)$  and  $|f(x)| \geq C_1C_1^\#|h(x)|$  for all  $x > \max(k_1, k_1^\#)$ . This means that  $f(x)$  is  $\Theta(h(x))$ .

46. The definitions tell us that there are positive constants  $C_1$ ,  $k_1$ ,  $C_2$ , and  $k_2$  such that  $|f_1(x)| \leq C_2|g_1(x)|$  for all  $x > k_2$  and  $|f_1(x)| \geq C_1|g_1(x)|$  for all  $x > k_1$ , and that there are positive constants  $C_1^\#$ ,  $k_1^\#$ ,  $C_2^\#$ , and  $k_2^\#$  such that  $|f_2(x)| \leq C_2^\#|g_2(x)|$  for all  $x > k_2^\#$  and  $|f_2(x)| \geq C_1^\#|g_2(x)|$  for all  $x > k_1^\#$ . We can multiply these inequalities to obtain  $|f_1(x)f_2(x)| \leq C_2C_2^\#|g_1(x)g_2(x)|$  for all  $x > \max(k_2, k_2^\#)$  and  $|f_1(x)f_2(x)| \geq C_1C_1^\#|g_1(x)g_2(x)|$  for all  $x > \max(k_1, k_1^\#)$ . This means that  $f_1(x)f_2(x)$  is  $\Theta(g_1(x)g_2(x))$ .

#### 76 Chapter 3 Algorithms

48. Typically  $C$  will be less than 1. From some point onward to the right ( $x > k$ ), the graph of  $f(x)$  must be above the graph of  $g(x)$  after the latter has been scaled down by the factor  $C$ . Note that  $f(x)$  does not have to be larger than  $g(x)$  itself.

50. We need to show inequalities both ways. First, we show that  $|f(x)| \leq Cx^n$  for all  $x \geq 1$ , as follows, noting that  $x^i \leq x^n$  for such values of  $x$  whenever  $i < n$ . We have the following inequalities, where  $M$  is the largest of the absolute values of the coefficients and  $C$  is  $M(n+1)$ :

$$\begin{aligned} |f(x)| &= |a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0| \\ &\leq |a_n| x^n + |a_{n-1}| x^{n-1} + \cdots + |a_1| x + |a_0| \\ &\leq |a_n| x^n + |a_{n-1}| x^n + \cdots + |a_1| x^n + |a_0| x^n \\ &\leq Mx^n + Mx^n + \cdots + Mx^n + Mx^n = Cx^n \end{aligned}$$

For the other direction, which is a little messier, let  $k$  be chosen larger than 1 and larger than  $2nm/|a_n|$ , where  $m$  is the largest of the absolute values of the  $a_i$ 's for  $i < n$ . Then each  $a_{n-i}/x^i$  will be smaller than  $|a_n|/2n$  in absolute value for all  $x > k$ . Now we have for all  $x > k$ ,

$$|f(x)| = |a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0|$$

$$= x^n |a_n + \frac{a_{n-1}}{x} + \dots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n}|$$

$$\geq x^n |a_n/2|, \quad \text{as desired.} \quad x^n$$

52. We just make the analogous change in the definition of big-Omega that was made in the definition of big-O: there exist positive constants  $C$ ,  $k_1$ , and  $k_2$  such that  $|f(x, y)| \geq C|g(x, y)|$  for all  $x > k_1$  and  $y > k_2$ .

54. For all values of  $x$  and  $y$  greater than 1, each term of the given expression is greater than  $x^3 y^3$ , so the entire expression is greater than  $x^3 y^3$ . In other words, we take  $C = k_1 = k_2 = 1$  in the definition given in Exercise 52.

56. For all positive values of  $x$  and  $y$ , we know that  $\lceil xy \rceil \geq xy$  by definition (since the ceiling function value cannot be less than the argument). Thus  $\lceil xy \rceil$  is  $\Omega(xy)$  from the definition, taking  $C = 1$  and  $k_1 = k_2 = 0$ . In fact,  $\lceil xy \rceil$  is also  $O(xy)$  (and therefore  $\Theta(xy)$ ); this is easy to see since  $\lceil xy \rceil \leq (x+1)(y+1) \leq (2x)(2y) = 4xy$  for all  $x$  and  $y$  greater than 1.

58. It suffices to show that  $\frac{d}{n} = 0$ ,

$\lim_{n \rightarrow \infty} (\log_b n)^c$   
where we think of  $n$  as a continuous variable. Because both numerator and denominator approach  $\infty$ , we apply L'Hôpital's rule and evaluate

$$\lim_{n \rightarrow \infty} c(\log_b n)^{c-1} d \cdot n^d \cdot \ln b.$$

Section 3.2 The Growth of Functions 77 At this point, if  $c \leq 1$ , then the limit is 0. Otherwise we again have an

expression of type  $\infty/\infty$ , so we apply

L'Hôpital's rule once more, obtaining  $(\ln b)^2$ .

$$\lim_{n \rightarrow \infty} c(c-1)(\log_b n)^{c-2} d^2 \cdot n^d.$$

If  $c \leq 2$ , then the limit is 0; if not, we repeat. Eventually the exponent on  $\log_b n$  becomes nonpositive and we conclude that the limit is 0, as desired.

60. It suffices to look at  $\lim_{n \rightarrow \infty} b^n/c^n = (b/c)^n$  and  $\lim_{n \rightarrow \infty} c^n/b^n = (c/b)^n$ . Because  $c > b > 1$ , we have  $0 < b/c < 1$  and  $c/b > 1$ , so the former limit is clearly 0 and the latter limit is clearly  $\infty$ .

62. a) Under the hypotheses,

$$\lim_{x \rightarrow \infty} cf(x)$$

$$g(x) = c \lim_{x \rightarrow \infty} f(x)$$

$$g(x) = c \cdot 0 = 0.$$

b) Under the hypotheses,

$$\lim_{x \rightarrow \infty} f_1(x) + f_2(x)$$

$$g(x) = \lim_{x \rightarrow \infty} f_1(x)$$

$$g(x) = \lim_{x \rightarrow \infty} f_2(x)$$

$$g(x) = 0 + 0 = 0.$$

64. The behaviors of  $f$  and  $g$  alone are not really at issue; what is important is whether  $f(x)/g(x)$  approaches 0 as  $x \rightarrow \infty$ . Thus, as shown in the picture, it might happen that the graphs of  $f$  and  $g$  rise, but  $f$  increases enough more rapidly than  $g$  so that the ratio gets small. In the picture, we see that  $f(x)/g(x)$  is asymptotic to the  $x$ -axis.

66. No. Let  $f(x) = x$  and  $g(x) = x^2$ . Then clearly  $f(x)$  is  $o(g(x))$ , but the ratio of the logs of the absolute values is the constant 2, and 2 does not approach 0. Therefore it is not the case in this example that  $\log |f(x)|$  is  $o(\log |g(x)|)$ .

68. This follows from the fact that the limit of  $f(x)/g(x)$  is 0 in this case, as can be most easily seen by dividing numerator and denominator by  $x^n$  (the numerator then is bounded and the absolute value of the denominator grows without bound as  $x \rightarrow \infty$ ).

70. Since  $f(x) = 1/x$  is a decreasing function which has the value  $1/x$  at  $x = j$ , it is clear that  $1/j < 1/x$  throughout the interval from  $j - 1$  to  $j$ . Summing over all the intervals for  $j = 2, 3, \dots, n$ , and noting that the definite integral is the area under the curve, we obtain the inequality in the hint. Therefore

$$H_n = 1 + \sum_{j=2}^n \frac{1}{j} < 1 + \int_1^n \frac{1}{x} dx = 1 + \ln n = 1 + C$$

for  $n > 2$ , where  $C = \log n \leq 2C \log n$

78 Chapter 3 Algorithms

72. By Example 6,  $\log n!$  is  $O(n \log n)$ . By Exercise 71,  $n \log n$  is  $O(\log n!)$ . Thus by Exercise 31,  $\log n!$  is  $\Theta(n \log n)$ .

74. In each case we need to evaluate the limit of  $f(x)/g(x)$  as  $x \rightarrow \infty$ . If it equals 1, then  $f$  and  $g$  are asymptotic; otherwise (including the case in which the limit does not exist) they are not. Most of these are straightforward

applications of algebra, elementary notions about limits, or L'Hôpital's rule.

a)  $\lim_{x \rightarrow \infty} x^2 + 3x + 7$

$$x^2 + 10 = \lim_{x \rightarrow \infty} 1 + 3/x + 7/x^2$$

b)  $\lim_{x \rightarrow \infty} x^2 \log x$

$$\frac{3}{x} = \lim_{x \rightarrow \infty} \log x$$

$1 + 10/x^2 = 1$ , so  $f$  and  $g$  are asymptotic.

$g$  are not asymptotic.

$$x = \lim_{x \rightarrow \infty} 1$$

$x \cdot \ln 2 = 0$  (we used L'Hôpital's rule for the last equivalence), so  $f$  and

c) Here  $f(x)$  is dominated by its leading term,  $x^4$ , and  $g(x)$  is a polynomial of degree 4, so the ratio approaches 1, the ratio of the leading coefficients, as in part (a). Therefore  $f$  and  $g$  are asymptotic. d) Here  $f$  and  $g$  are polynomials of degree 12, so the ratio approaches 1, the ratio of the leading coefficients, as in part (a). Therefore  $f$  and  $g$  are asymptotic.

### SECTION 3.3 Complexity of Algorithms

2. The statement  $t := t + i + j$  is executed  $n^2$  times, so the number of operations is  $O(n^2)$ . (Specifically,  $2n^2$  additions are used, not counting any arithmetic needed for bookkeeping in the loops.)
4. The value of  $i$  keeps doubling, so the loop terminates after  $k$  iterations as soon as  $2^k > n$ . The value of  $k$  that makes this happen is  $O(\log n)$ , because  $2^{\log n} = n$ . Within the loop there are two additions or multiplications, so the answer to the question is  $O(\log n)$ .
6. a) We can sort the first four elements by copying the steps in Algorithm 5 but only up to  $j = 4$ .

procedure sort four ( $a_1, a_2, \dots, a_n$  : real numbers)

for  $j := 2$  to 4

$i := 1$

    while  $a_j > a_i$

$i := i + 1$

$m := a_j$

    for  $k := 0$  to  $j - i - 1$

$a_{j-k} := a_{j-k-1}$

$a_i := m$

b) Only a (small) finite number of steps are performed here, regardless of the length of the list, so this algorithm has complexity  $O(1)$ .

8. If we successively square  $k$  times, then we have computed  $x^{2^k}$ . Thus we can compute  $x^{2^k}$  with only  $k$  multiplications, rather than the  $2^k - 1$  multiplications that the naive algorithm would require, so this method is much more efficient.
10. a) By the way that  $S - 1$  is defined, it is clear that  $S \wedge (S - 1)$  is the same as  $S$  except that the rightmost 1 bit has been changed to a 0. Thus we add 1 to count for every one bit (since we stop as soon as  $S = 0$ , i.e., as soon as  $S$  consists of just 0 bits).  
b) Obviously the number of bitwise AND operations is equal to the final value of count, i.e., the number of one bits in  $S$ .

### Section 3.3 Complexity of Algorithms 79

12. a) There are three loops, each nested inside the next. The outer loop is executed  $n$  times, the middle loop is executed at most  $n$  times, and the inner loop is executed at most  $n$  times. Therefore the number of times the one statement inside the inner loop is executed is at most  $n^3$ . This statement requires one comparison,

so the total number of comparisons is  $O(n^3)$ .

b) We follow the hint, not worrying about the fractions that might result from roundoff when dividing by 2 or 4 (these don't affect the final answer in big-Omega terms). The outer loop is executed at least  $n/4$  times, once for each value of  $i$  from 1 to  $n/4$  (we ignore the rest of the values of  $i$ ). The middle loop is executed at least  $n/4$  times, once for each value of  $j$  from  $3n/4$  to  $n$ . The inner loop for these values of  $i$  and  $j$  is executed at least  $(3n/4) - (n/4) = n/2$  times. Therefore the statement within the inner loop, which requires one comparison, is executed at least  $(n/4)(n/4)(n/2) = n^3/32$  times, which is  $\Omega(n^3)$ . The second statement follows by definition.

14. a) Initially  $y := 3$ . For  $i = 1$  we set  $y$  to  $3 \cdot 2 + 1 = 7$ . For  $i = 2$  we set  $y$  to  $7 \cdot 2 + 1 = 15$ , and we are done. b) There is one multiplication and one addition for each of the  $n$  passes through the loop, so there are  $n$  multiplications and  $n$  additions in all.

16. If each bit operation takes  $10^{-11}$  second, then we can carry out  $10^{11}$  bit operations per second, and therefore  $60 \cdot 60 \cdot 24 \cdot 10^{11} = 864 \cdot 10^{13}$  bit operations per day. Therefore in each case we want to solve the inequality  $f(n) = 864 \cdot 10^{13}$  for  $n$  and round down to an integer. Obviously a calculator or computer software will come in handy here.

a) If  $\log n = 864 \cdot 10^{13}$ , then  $n = 2^{864 \cdot 10^{13}}$ , which is an unfathomably huge number.

b) If  $1000n = 864 \cdot 10^{13}$ , then  $n = 864 \cdot 10^{10}$ , which is still a very large number.

c) If  $n^2 = 864 \cdot 10^{13}$ , then  $n = \sqrt{864 \cdot 10^{13}}$ , which works out to about  $9.3 \cdot 10^7$ .

d) If  $1000n^2 = 864 \cdot 10^{13}$ , then  $n = \sqrt{864 \cdot 10^{10}}$ , which works out to about  $2.9 \cdot 10^6$ .

e) If  $n^3 = 864 \cdot 10^{13}$ , then  $n = (864 \cdot 10^{13})^{1/3}$ , which works out to about  $2.1 \cdot 10^5$ .

f) If  $2^n = 864 \cdot 10^{13}$ , then  $n = \lceil \log(864 \cdot 10^{13}) \rceil = 52$ . (Remember, we are taking log to the base 2.)

g) If  $2^{2n} = 864 \cdot 10^{13}$ , then  $n = \lceil \log(864 \cdot 10^{13})/2 \rceil = 26$ .

h) If  $2^{2^n} = 864 \cdot 10^{13}$ , then  $n = \lceil \log(\log(864 \cdot 10^{13})) \rceil = 5$ .

18. We are asked to compute  $(2n^2 + 2^n) \cdot 10^{-9}$  for each of these values of  $n$ . When appropriate, we change the units from seconds to some larger unit of time.

a)  $1.224 \times 10^{-6}$  seconds b) approximately  $1.05 \times 10^{-3}$  seconds

c) approximately  $1.13 \times 10^6$  seconds, which is about 13 days (nonstop)

d) approximately  $1.27 \times 10^{21}$  seconds, which is about  $4 \times 10^{13}$  years (nonstop)

20. In each case we want to compare the function evaluated at  $2n$  to the function evaluated at  $n$ . The most desirable form of the comparison (subtraction or division) will vary.

a) Notice that

$$\log \log 2n - \log \log n = \log \log 2 + \log n$$

$$\log n = \log 1 + \log n$$

$$\log n \cdot$$

If  $n$  is large, the fraction in this expression is approximately equal to 1, and therefore the expression is approximately equal to 0. In other words, hardly any extra time is required. For example, in going from  $n = 1024$  to  $n = 2048$ , the number of extra milliseconds is  $\log 11/10 \approx 0.14$ .

b) Here we have  $\log 2n - \log n = \log 2^n/n = \log 2 = 1$ . One extra millisecond is required, independent of  $n$ . c)

This time it makes more sense to use a ratio comparison, rather than a difference comparison. Because  $100(2n)/(100n) = 2$ , we conclude that twice as much time is needed for the larger problem.

d) The controlling factor here is  $n$ , rather than  $\log n$ , so again we look at the ratio:

$$2n \log(2n)$$

$$n \log n = 2 \cdot \frac{1 + \log n}{\log n}$$

For large  $n$ , the final fraction is approximately 1, so we can say that the time required for  $2n$  is a bit more than twice what it is for  $n$ .

e) Because  $(2n)^2/n^2 = 4$ , we see that four times as much time is required for the larger problem. f) Because  $(3n)^2/n^2 = 9$ , we see that nine times as much time is required for the larger problem. g) The relevant ratio is  $2^{2n}/2^n$ , which equals  $2^n$ . If  $n$  is large, then this is a huge number. For example, in going from  $n = 10$  to  $n = 20$ , the number of milliseconds increases over 1000-fold.

22. a) The number of comparisons does not depend on the values of  $a_1$  through  $a_n$ . Exactly  $2n - 1$  comparisons are used, as was determined in Example 1. In other words, the best case performance is  $O(n)$ . b) In the best case  $x = a_1$ . We saw in Example 4 that three comparisons are used in that case. The best case performance, then, is  $O(1)$ . c) It is hard to give an exact answer, since it depends on the binary representation of the number  $n$ , among other things. In any case, the best case performance is really not much different from the worst case performance, namely  $O(\log n)$ , since the list is essentially cut in half at each iteration, and the algorithm does not stop until the list has only one element left in it.
24. a) In order to find the maximum element of a list of  $n$  elements, we need to make at least  $n - 1$  comparisons, one to rule out each of the other elements. Since Algorithm 1 in Section 3.1 used just this number (not counting bookkeeping), it is optimal. b) Linear search is not optimal, since we found that binary search was more efficient. This assumes that we can be given the list already sorted into increasing order.
26. We will count comparisons of elements in the list to  $x$ . (This ignores comparisons of subscripts, but since we are only interested in a big- $O$  analysis, no harm is done.) Furthermore, we will assume that the number of elements in the list is a power of 4, say  $n = 4^k$ . Just as in the case of binary search, we need to determine the maximum number of times the while loop is iterated. Each pass through the loop cuts the number of elements still being considered (those whose subscripts are from  $i$  to  $j$ ) by a factor of 4. Therefore after  $k$  iterations, the active portion of the list will have length 1; that is, we will have  $i = j$ . The loop terminates at this point. Now each iteration of the loop requires two comparisons in the worst case (one with  $a_m$  and one with either  $a_l$  or  $a_u$ ). Three more comparisons are needed at the end. Therefore the number of comparisons is  $2k + 3$ , which is  $O(k)$ . But  $k = \log_4 n$ , which is  $O(\log n)$  since logarithms to different bases differ only by multiplicative constants, so the time complexity of this algorithm (in all cases, not just the worst case) is  $O(\log n)$ .
28. The algorithm we gave for finding all the modes essentially just goes through the list once, doing a little bookkeeping at each step. In particular, between any two successive executions of the statement  $i := i + 1$  there are at most about eight operations (such as comparing count with modecount, or reinitializing value). Therefore at most about  $8n$  steps are done in all, so the time complexity in all cases is  $O(n)$ .
30. The algorithm we gave is clearly of linear time complexity, i.e.,  $O(n)$ , since we were able to keep updating the sum of previous terms, rather than recomputing it each time. This applies in all cases, not just the worst case.
32. The algorithm read through the list once and did a bounded amount of work on each term. Looked at another way, only a bounded amount of work was done between increments of  $j$  in the algorithm given in

the solution. Thus the complexity is  $O(n)$ .